

BQ79600-Q1 Automotive SPI/UART Communication Interface Functional-Safety Compliant With Automatic Host Wakeup

1 Features

- Qualified for automotive applications
- AEC-Q100 Qualified with the following results:
 - Device temperature grade 1: –40°C to +125°C ambient operating temperature range
 - Device HBM ESD classification level 2
 - Device CDM ESD classification level C4B
- [Functional Safety-Compliant](#)
 - Developed for functional safety applications
 - Documentation available to aid ISO 26262 system design
 - [Safety Manual](#)
 - [Functional Safety Analysis Report](#)
 - Systematic capability up to ASIL-D
 - Hardware capability up to ASIL-D
- Automatic wake up of BMS/BMU system when fault is detected in ring architecture
- Support supply from 4.75-V to 40-V
- UART/SPI Host interface
- Compatible with 3.3-V/5-V logic
- Isolated differential daisy chain
 - Support ring architecture with one device
 - Support transformer/ capacitor isolation
- Designed for BCI/EMI/EMC robustness
- Supports BQ7961X-Q1 family, BQ7965X, BQ7963X and future products

2 Applications

- [Battery management system \(BMS\)](#)
- [Other HEV/EV](#)
- Fuel cell
- Energy storage

3 Description

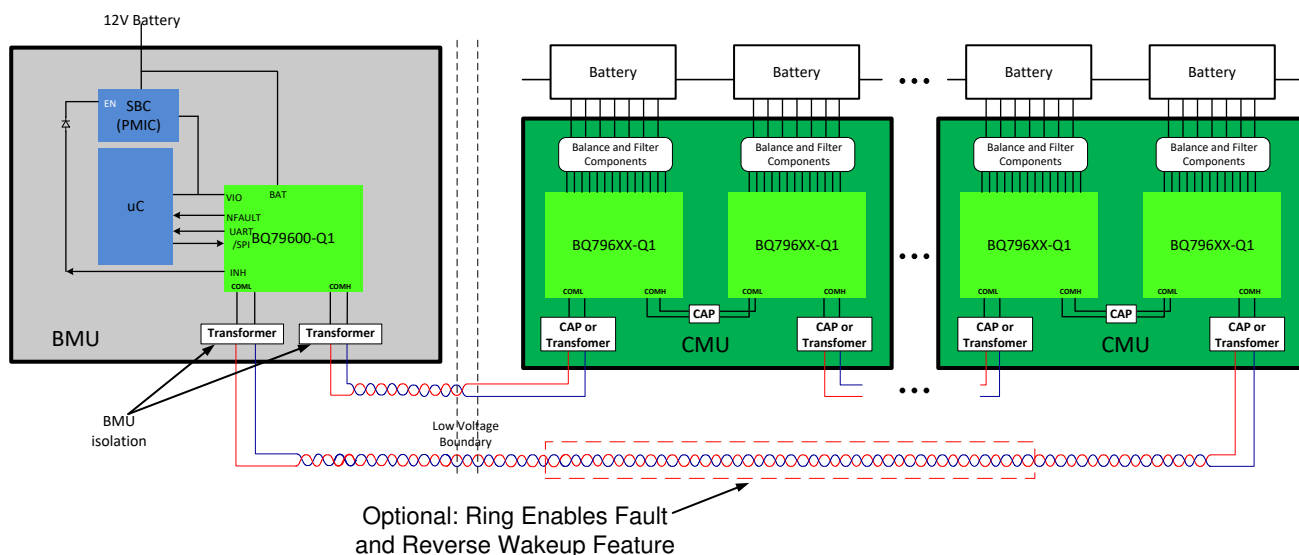
The BQ79600-Q1 is a communication (bridge) IC designed to interface between a microcontroller (MCU) and TI battery monitoring ICs, for example the BQ7961X-Q1. The information from the MCU is translated by the device to signals recognized by TI's battery management daisy chain protocol, and transmitted out. And signals from daisy chain are decoded to bit stream and then sent back to MCU.

When the MCU and PMIC are in SHUTDOWN/ SLEEP, the BQ79600-Q1 can wake them up if any unmasked fault is detected when using ring architecture.

Device Information

PART NUMBER ⁽¹⁾	PACKAGE	BODY SIZE (NOM)
BQ79600-Q1	TSSOP (16-pin)	6.6 mm × 5.1 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.



Simplified System Diagram



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4 Revision History

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• Changed from Advance Information to Production Data.....	1

5 Pin Configuration and Functions

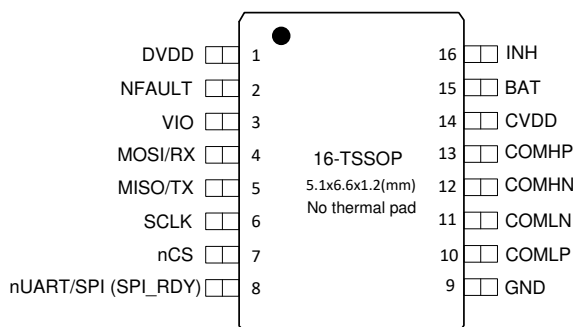


Figure 5-1. PW Package 16-Pin TSSOP Top View

Pin Functions

PIN		TYPE ⁽¹⁾	DESCRIPTION
NAME	NO.		
VIO	3	P	Power supply input for UART/SPI input/output pins. Decouple with a 0.1µF capacitor to GND. VIO should be powered before SCLK, nCS, RX/MOSI, TX/MISO, NFAULT, nUART/SPI (SPI_RDY) is driven.
SCLK	6	DI	SPI clock input. If SPI interface is used, this pin is connected to SPI master controller. Connect to GND with a 10-100kohm pull-down resistor. If not used, connect to GND.
nCS	7	DI	Active low chip select pin for SPI interface. Connect to VIO with 10-100kohm pull-up resistor in SPI mode. Cannot hardwire to GND in SPI mode. Connect to GND in UART mode.
RX/MOSI	4	DI	UART receiver input or SPI master out slave in. Connect to VIO with a 10-100kohm pull-up resistor. Don't leave it unconnected.
TX/MISO	5	DO	UART transmitter output or SPI master in slave out. Pull up to VIO with a 10-100kohm pull-up resistor at MCU side. MISO pin drives high in SPI idle mode, cannot directly support SPI multidrop, if multidrop is needed, add a tri-state buffer between BQ79600-Q1 and MCU.
NFAULT	2	DO	Fault indication output. Open drain active low. Pull up to VIO with 100kohm resistor. Connect NFAULT to host MCU GPIO. If not used, connect to GND.
nUART/SPI (SPI_RDY)	8	DI/O	If used as UART mode, connect this pin to GND. If used as SPI mode, connect to VIO through 10-100kohm pull up resistor and connect this pin to MCU GPIO. This pin is used as an input pin to select SPI or UART interface before device finishes wakeup/reset initialization (SPI communication is ready). If SPI mode is selected, SPI_RDY has to be used by host to decide if read/write can be initiated or needing further wait. Refer to Section 7.3.2.1.2.2.1 details.
DVDD	1	P	1.8V regulated output. DVDD supplies the internal digital circuits. Bypass DVDD with a 0.22µF (recommended) or a 0.47µF capacitor to GND.
GND	9	P	Ground.
CVDD	14	P	Dedicated 5V supply used for the daisy chain communications. Decouple with a 0.22µF (recommended) or a 0.47µF to GND.
COMHN	12	AC-I/O	This is an AC coupled bi-directional I/O pin for daisy chain (VIF) communication. Do not apply external DC voltage to this pin. Shall connect to COMLN/P of adjacent device through proper isolation, see Section 7.3.2.1.1 . Leave unconnected if (Ring Architecture) not used.
COMHP	13	AC-I/O	
COMLN	11	AC-I/O	This is an AC coupled bi-directional I/O pin for daisy chain (VIF) communication. Do not apply external DC voltage to this pin. Shall connect to COMHN/P of adjacent device through proper isolation, see Section 7.3.2.1.1 . Leave unconnected if (Ring Architecture) not used.
COMLP	10	AC-I/O	
INH	16	HV-O	Inhibit pin (PMOS open drain) to control system voltage regulators, connect 100kohm resistor to GND. If reverse wake up feature is not used, connect this pin to BAT pin. Don't leave this pin floating.
BAT	15	HV-P	Battery supply Input. Supply internal LDOs and wakeup circuit. Connect to external supply through 10ohm resistor. Bypass to GND with 0.1µF.

(1) DI = digital input, DO = digital output, HV = high voltage

6 Specifications

6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)⁽¹⁾

		MIN	MAX	UNIT
Input Voltage	BAT to GND	−0.3	40	V
	INH to GND (INH is lower than BAT)	−0.3	40	V
	VIO, CVDD to GND	−0.3	5.5	V
	DVDD to GND	−0.3	1.98	V
	MISO/TX, MOSI/RX, nCS, SCLK, NFAULT, nUART/SPI(SPI_RDY) to GND	−0.3	VIO+0.3	V
	COMHP, COMHN, COMLP, COMLN to GND	−10	10	V
	COMHP to COMHN, COMLP to COMLN	−6.5	6.5	V
INH current			4	mA
I/O current	MISO/TX current (100pF load, VIO=5V, 10ns transition-time)		8	mA
T _A	Ambient temperature	−40	125	°C
T _J	Junction temperature	−40	150	°C
T _{stg}	Storage temperature	−65	150	°C

- (1) Stresses beyond those listed under Absolute Maximum Rating may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under Recommended Operating Condition. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

6.2 ESD Ratings

			VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human body model (HBM), per AEC Q100-002 ⁽¹⁾	±2000	V
		Charged device model (CDM), per AEC Q100-011	Corner pins (1, 8, 9 and 16)	
			Other pins	
			±750	
			±500	

- (1) AEC Q100-002 indicates that HBM stressing shall be in accordance with the ANSI/ESDA/JEDEC JS-001 specification.

6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

		MIN	TYP	MAX	UNIT
V _{BAT, INH}	BAT, INH (INH is equal to or lower than BAT) (Powered by 12V Battery)	5.5		24	V
V _{VIO_RANGE}	VIO input (applies to nCS, MOSI/RX, MISO/TX, SCLK, nUART/SPI (SPI_RDY), NFAULT)	3.135		5.25	V
I _{MISO/TX}	MISO/TX current			3	mA
I _{INH}	INH output current			2	mA
T _A	Operation temperature	−40		125	°C

6.4 Thermal Information

THERMAL METRIC ⁽¹⁾		BQ79600-Q1	UNIT
		PW(TSSOP)	
		16 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	102	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	28.9	°C/W
R _{θJB}	Junction-to-board thermal resistance	49	°C/W
Ψ _{JT}	Junction-to-top characterization parameter	14	°C/W

THERMAL METRIC ⁽¹⁾		BQ79600-Q1	UNIT
		PW(TSSOP)	
		16 PINS	
Ψ_{JB}	Junction-to-board characterization parameter	48.2	°C/W

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

6.5 Electrical Characteristics

VIO = 3.3V, over operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
OPERATION VOLTAGE						
V_{BAT}		When reverse wakeup feature is used, BAT pin is powered by "12V battery"	5.5		24	V
V_{BAT}		When reverse wakeup feature is not used, BAT/CVDD are powered by regulated 5V	4.75		5.25	V
THERMAL SHUTDOWN						
T_{SHUT_R}	Thermal shutdown (rising direction)	DieTemp sensor	126	138	150	°C
T_{SHUT_F}	Thermal shutdown (falling direction)	DieTemp sensor	116		141	°C
SUPPLY CURRENTS						
I_{SHDN_1}	Supply current in SHUTDOWN mode device powered by regulated 5V supply	VBAT shorted to CVDD, both equal to 5V, measured through GND pin			7	μA
I_{SHDN_2}	Supply current in SHUTDOWN mode powered by "12V" battery directly	VBAT= 17V, CVDD self powered, measured through GND pin			9	μA
$I_{VALIDATE}$	Supply current in VALIDATE mode	Current on BAT pin			168	μA
$I_{SLP(IDLE)}$	Baseline supply current in SLEEP mode	Current on BAT pin, no fault, COMH and COML RX disabled, no HB TX			110	μA
$I_{SLP_RX_O_N}$	Additional supply current to SLEEP mode base line	When COML OR COMH RX is on		35		μA
$I_{SLP_TX_O_N}$	Additional supply current to SLEEP mode base line	When COML or COMH tone transmitter is on (HB tone)			8	μA
$I_{ACT(IDLE)}$	Baseline supply current in ACTIVE mode	Current on BAT pin, no fault, no communication, Tone RX/TX is off		3	4	mA
I_{COMT}	Additional average current for one of BQ79600 daisy chain transmitters is on	Average current into VBAT when BQ79600 transmits 14 bytes of data (brdcast write 8 bytes of 0x00 into address 0X1B)		10		mA
Supplies (AVAO_REF, always on internal supply)						
$V_{AVAOREG}$	AVAOREG voltage	VBAT > min VBAT		2.45		V
$V_{AVDDREF_OV}$	AVDDREF OV threshold	VBAT > min VBAT, hys = 130mV	2.8		3.1	V
Supplies (CVDD)						
V_{CVDD}	CVDD output voltage	No external load, $C_{SUPPLIES} = 0.22\mu F$, ACTIVE mode	4.9	5	5.1	V
V_{CVDD_OV}	CVDD OV rising threshold	Hys = 140mV	5.3	5.5	5.65	V
$V_{CVDD_UV_F}$	CVDD UV falling threshold		4.35	4.5	4.65	V
$V_{CVDD_UV_R}$	CVDD UV rising threshold		4.45	4.6	4.75	V
V_{CVDD_ILIMIT}	CVDD current limit		53		81	mA
C_{CVDD}	Capacitance range on CVDD pin	Not capacitor value	0.1		0.8	μF

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VIO = 3.3V, over operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
Supplies (DVDD)						
V _{DVDD}	DVDD output voltage	No external load, C _{SUPPLIES} = 0.22μF, ACTIVE mode	1.75	1.8	1.85	V
V _{DVDD_OV}	DVDD OV rising threshold	Hys = 65mV	1.9		2.1	V
V _{DVDD_UV_F}	DVDD UV falling threshold		1.63		1.69	V
V _{DVDD_UV_R}	DVDD UV rising threshold		1.68		1.75	V
V _{DVDD_ILIMIT}	DVDD current limit		20		57	mA
D _{DVDD}	Load capacitance on DVDD pin	Not capacitor value	0.1		0.8	μF
SNIFF DETECTOR						
V _{VAL_THR_P}	Sniffer detector threshold, rising swing on COMHP has to be larger than value	Sniffer is enabled, and device is in SHUTDOWN mode	3.2		3.6	V
INH Driver						
V _{DROP_INH}	When INH is pulled up, voltage drop from BAT to INH	I _{INH} = -0.5mA		0.5	1	V
V _{INH_DET}	Threshold to set [INH_STAT] to '1'		2.2			V
Reference Voltages						
Digital I/Os (TX, RX, NFAULT)						
V _{VIO_UV_R}	VIO UV rising	Hys = 200mV	2.5		3.1	V
V _{OH}	Output as logic level high (TX)	FET pull up, I _{out} =1mA, VIO = 3.3V or 5V	V _{VIO} -0.1			V
V _{OL}	Output as logic level low (TX)	FET pull down, I _{out} =1mA, VIO = 3.3V or 5V			0.1	V
V _{IH}	Input as logic level high (RX), requirement for user	VIO = 3.3V or 5V	0.75 x V _{VIO}			V
V _{IL}	Input as logic level low (RX), requirement for user	VIO = 3.3V or 5V			0.25 x V _{VIO}	V
R _{NFAULT}	NFAULT pull down impedance	Use 100kohm external pull up			1000	Ω
Daisy-chain Communication Bus						
V _{DCCM_1}	Common mode voltage (COML and COMH)	ACTIVE mode		2.2		V
V _{DCCM_2}	Common mode voltage (COML and COMH)	SLEEP or VALIDATE mode		1		V
V _{COMM_DATA}	COMM port data receiver threshold range (V _{COMP} -V _{COMN})		1.04		1.75	V
V _{COMM_TONE}	COMM port HB/FAULT tone receiver threshold range (V _{COMP} -V _{COMN})		1.13		1.94	V

6.6 Timing Requirements

over operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
POWER STATE TIMING						
t _{POR2COMM}		From V _{BAT} (rising) > V _{POR} to device ready to receive WAKE ping, ramp up VBAT and VIO in 10μs			1	ms

over operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
$t_{SU(WAKE_SHUT)}$	Startup from SHUTDOWN/VALIDATE to ACTIVE mode	From receiving WAKE ping (RX ping going low-to-high) to device in ACTIVE mode (ready to do μ ART /SPI communication) (CVDD= 0.22 μ F capacitor DVDD = 0.22 μ F capacitor)		2	3.5	ms
$t_{SU(SLP2ACTCT)}$	Startup from SLEEP to ACTIVE mode (with Sleep2active ping)	From receiving SLP2ACT ping (RX ping going low-to-high) to device in ACTIVE mode (ready to do UART / SPIcommunication)			260	μ s
$t_{SU(WAKE_SLP)}$	Startup from SLEEP to ACTIVE mode (with WAKE ping)	From receiving WAKE ping (RX ping going low-to-high) to device in ACTIVE mode (ready to do UART / SPIcommunication)			600	μ s
t_{RST}	Reset time from ACTIVE mode to ACTIVE mode	From receiving WAKE ping (RX ping going low-to-high) or CONTROL1[SOFT_RESET]=1 to device in ACTIVE mode (ready to do UART /SPI communication)			600	μ s
t_{SLP}	From ACTIVE to SLEEP mode	From receiving SLEEP entry condition to enter in SLEEP mode			100	μ s
t_{SHTDN}	From ACTIVE/SLEEP/VALIDATE to SHUTDOWN mode	From receiving SHUTDOWN entry condition to enter in SHUTDOWN mode (CVDD<1.2V)			5	ms
t_{VALID_ENTRY}	From SHUTDOWN to VALIDATE	From fault tone toggling on COM port to DVDD hit above 1.75V			10	ms
$t_{VALID_TIMEOUT}$	time to validate fault tone before transition to SHUTDOWN state	Start from DVDD out of reset		150		ms
INH Driver TIMING						
t_{INH_DLY}	After device enters VALIDATE, from first couplet of fault tone to INH pulled high			720		μ s
PING SIGNAL TIMING						
t_{HLD_WAKE}	From user perspective, WAKE ping low time on MOSI/RX pin	VBAT > VPOR, RX pin (low-pulse width) VIO = 3.3 or 5V	2.5		3	ms
t_{HLD_SD}	From user perspective, SHUTDOWN ping low time on MOSI/RX pin	VBAT > VPOR, RX pin (low-pulse width) VIO = 3.3 or 5V	12.5			ms
t_{StA}	From user perspective, SLEEPtoACTIVE ping low time on MOSI/RX pin	VBAT > VPOR	250		300	μ s
Daisy-chain Communication Bus TIMING						
t_{PW_DC}	COMM data Pulse width of data (half bit time) for communication		230	250	270	ns
$t_{COMTONE}$	Time between pulses within a comm tone (HFO based).	Transmit. From the beginning of a pulse until the beginning of the next pulse.	10.67	11	11.33	μ s
$t_{COMTONE_PLS}$	Comm tone pulse width(HFO based)	Transmit	0.97	1	1.03	μ s
t_{FLTONE}	Time between pulses within a fault tone (LFO based).	Only transmit HB tone, not FAULT tone. From the beginning of a pulse until the beginning of the next pulse.	10.3	11.5	12.7	μ s
t_{FLTONE_PLS}	Fault tone or HB tone pulse width (analog delay based)			1		μ s
η_{HBDET}	HEARTBEAT: Number of pulses to detect as a valid tone (dig counter)	Detect		16		pulses
$\eta_{FTONEDET}$	FAULT TONE: Number of pulses to detect as a valid tone (dig counter)	Detect		64		pulses

over operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{HB_PERIOD}	HEARTBEAT: Period between HEARTBEAT Burst (from the beginning of a HEARTBEAT to the beginning of the next HEARTBEAT)		360	400	440	ms
$t_{HB_TIMEOUT}$	HEARTBEAT: Timeout to considered as not receiving HEARTBEAT		0.9	1	1.1	s
t_{HB_FAST}	HEARTBEAT: If HEARTBEAT is received within this time, it is considered receiving HEARTBEAT too fast			200		ms
t_{FTONE_PERIOD}	Defined by BQ7961X, FAULT TONE: Period between FAULT TONE Burst	From the beginning of a FAULT TONE to the beginning of the next FAULT TONE		50		ms
$t_{FT_LATENCY}$	Fault Tone latency in Base Device	From the time device receives the tone to the time asserts NFAULT		24		μ s
I/O TIMING (TX, RX, NFAULT)						
t_{RISE_TX}	Rise Time	$C_{LOAD} = 100pF$, VIO=3.3V or 5V		15		ns
t_{FALL_TX}	Fall Time	$C_{LOAD} = 100pF$, VIO=3.3V or 5V		15		ns
$t_{FALL/RISE_RX}$	RX pin rise/fall time				100	ns
UART TIMING						
UARTERR_BAUD	UART TX/RX baud rate (either 250K or 1Mbps) error		-1.5		1.5	%
$t_{UART(CLR)}$	UART Comm Clear low time		15		20	bit period
$t_{UART(RX_HIGH)}$	UART high time after Comm Clear, before sending Clear or Reset		1			bit period
SPI TIMING						
SCLK	SPI clock freq		2		6	MHz
$n_{SPI(CLR)}$	SPI Comm Clear low time			8		bit
t_{SPI_R}	SPI clock rising edge	25% to 75%			10	ns
t_{SPI_F}	SPI clock falling edge	25% to 75%			10	ns
t_{SPI_CLKH}	SPI clock high time		70			ns
t_{SPI_CLKL}	SPI clock low time		70			ns
t_8	Max SPI_RDY service interval. This time doesn't apply if total response bytes (payload + overhead) is less than 256 bytes	Read SCLK = 6MHz, with 30% SPI BUS idle time		1		ms
t_9	From nCS (25%) to SCLK rising (75%)		500			ns
t_{10}	From SCLK falling (25%) to nCS (75%)		500			ns
t_{11}	From nCS rising(75%) to nCS falling(25%)	Don't drop nCS while SPI_RDY is low	1			μ s
t_{12}	From nCS falling (25%) to stable MISO(L:20% H:80%)	Timing is defined at device pins, exclude propagation delay of PCB traces (from device perspective)			42	ns
t_{13}	From SCLK falling (25%) to stable MISO(L:20% H:80%)	Timing is defined at device pins, exclude propagation delay of PCB traces (from device perspective)			42	ns
t_{14}	From nCS rising (75%) to MISO drive to '1' (80%)	Timing is defined at device pins, exclude propagation delay of PCB traces (from device perspective)			42	ns
t_{SU}	Setup time, refer to 75% of SCLK rising		20			ns
t_H	Hold time, refer to 75% of SCLK rising		20			ns
SNIFF DETECTOR						

over operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
n_{VALIDATE}	Number of pulses needed (digital counter) to transition to validate mode	Sniffer is enabled, and device is in SHUTDOWN mode		64		pulses
$t_{\text{SNIFFIDLE}}$	Timer length. Once timer expired, it resets the 64 counter	Sniffer is enabled, and device is in SHUTDOWN mode	20		52	μs
OSCILLATOR						
f_{HFO}	HFO frequency		31.52	32	32.48	MHz
t_{HFOWDG}	HFO watchdog time	Reset digital if HFO is stuck or period is > than the watchdog timer			35	μs
f_{LFO}	LFO frequency		235.8	262	288.2	kHz
t_{LFOWDG}	LFO watchdog time	Reset digital if LFO is stuck or period is > than the watchdog timer			35	μs

6.7 Typical Characteristics

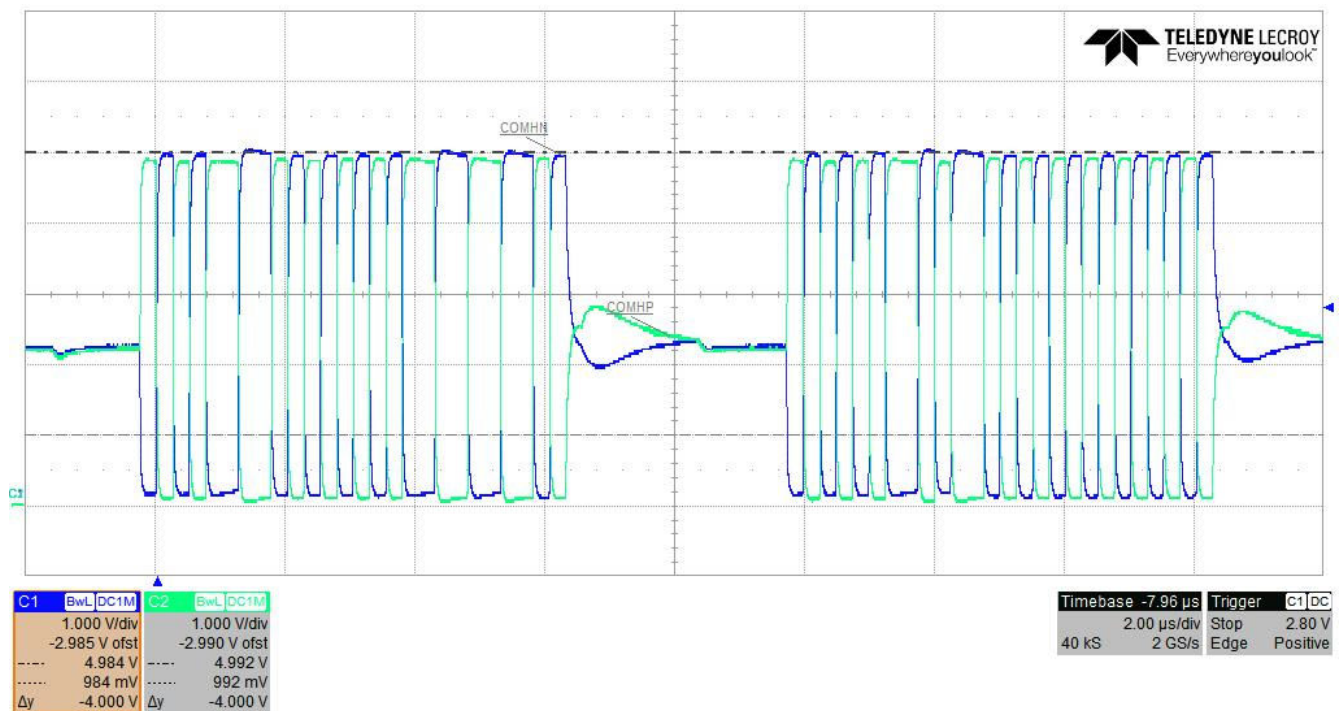


Figure 6-1. Typical COMHN and COMHP characteristic

7 Detailed Description

7.1 Overview

The BQ79600-Q1 is a bridge IC designed to interface between microcontroller (MCU) and TI battery monitoring ICs (BQ7961X-Q1 family). It translates between daisy chain interface and SPI/UART interface. It is fully compatible with BQ7961X-Q1 family. When working with BQ7961X-Q1 family, BQ79600-Q1 supports automatic host wakeup through INH pin when unmask fault is detected in high voltage battery pack. Details can be found in [Section 7.3.5](#) and [Section 7.3.4](#). For convenience of traceability, each device is marked by DIE IDs and PARTID which could be found in [Section 7.5](#) Register Summary Table.

7.2 Functional Block Diagram

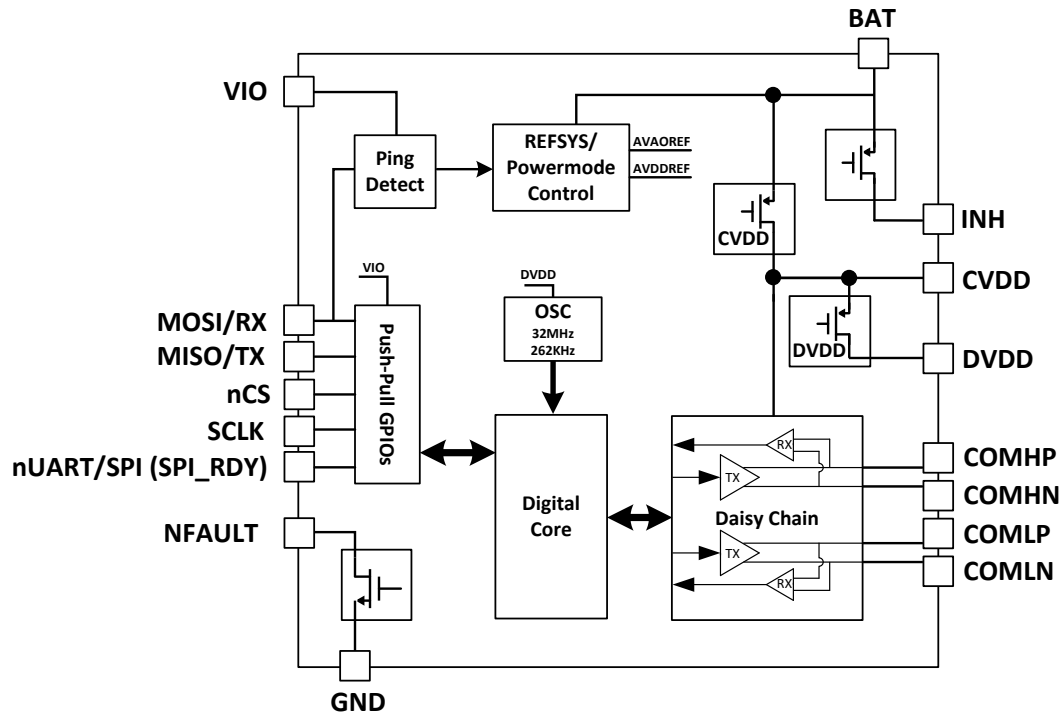


Figure 7-1. Functional Block Diagram

7.3 Feature Description

7.3.1 Functional Modes and Power Supply

7.3.1.1 Power Mode

The device has four power modes plus an Complete Off state. The functions supported under each power modes are summarized in [Table 7-1](#) and the power state diagram is shown in [Figure 7-2](#).

- **COMPLETE OFF:** The voltage at the BAT pin is less than $V_{BAT\ min}$, and all circuits are powered off.
- **SHUTDOWN:** The lowest power mode. Without VIO, device can only transition to VALIDATE. (If Sniffer used)
- **SLEEP:** A low power mode. Transition to ACTIVE is much faster compared to SHUTDOWN.
- **ACTIVE:** Full power mode. Device can communicate between MCU and daisy chain.
- **VALIDATE:** This state is to validate if there is real fault tone from stack devices. If fault tone is validated, drive INH pin towards VBAT (INH pin is latched until cleared by user). Device goes back to SHUTDOWN if $t_{VALID_TIMEOUT}$ or sleep timer expires. ($t_{VALID_TIMEOUT}$ timer is reset if fault tone is detected, detecting Heartbeat tone doesn't reset timer.) This state is bypassed if [Section 7.3.5](#) is disabled (by default). Once entered this state, a status bit [VALIDATE_DET] is set in next ACTIVE such that host knows what happened. Without VIO, device can only transition to SHUTDOWN. NFAULT pin is low in this mode.

Table 7-1. Functions Summary

FUNCTIONAL BLOCK ⁽¹⁾	SHUTDOWN	VALIDATE ⁽³⁾	SLEEP	ACTIVE
Data Communication RX/TX				If VIO_UV_R = 1
Time out		Sleep Timeout	Sleep Timeout	Comm Timeout
Tone RX (HB/Fault)		√	Depends on [DIR_SEL] and [TONE_RX_EN]	
Tone TX (WAKE/ SLP2ACT/ SHUTDOWN/ HWRST) ⁽⁵⁾				√
Tone TX (HB)			Depends on [DIR_SEL] and [HB_TX_EN]	
COM embedded fault				[FCOMM_EN] = 1
Sniff detector on COM*	If host enables this feature ⁽²⁾			
Wake/Shutdown Ping	If VIO_UV_R = 1	If VIO_UV_R = 1	If VIO_UV_R = 1	If VIO_UV_R = 1
SLP2ACT Ping			√	
NFAULT Driver		√	√	√
LFO		√	√	√
HFO				√
INH Driver	Holds State	√	INH_DIS[1:0] != 2'b11	INH_DIS[1:0] != 2'b11 ⁽⁴⁾
CVDD/DVDD		√	√	√
Thermal Shutdown		√	√	√

- (1) Once device in SLEEP/ACTIVE, losing VIO doesn't directly cause change of state, it causes loss of data communication to MCU.
(2) If host writes [SNIFDET_EN] = 1 & [SNIFDET_DIS] = 0 in ACTIVE mode, even device shuts down, enable signal is still valid. Sniff detector is enabled or disabled by a latch powered by always on power supply.
(3) This mode is bypassed if sniff detector is not enabled, see register DEV_CONF1.
(4) INH can only be triggered by [INH_SET_GO] bit in ACTIVE.
(5) Device does not recognize WAKE/ SLP2ACT/ SHUTDOWN/ HWRST tone sent by stack devices.

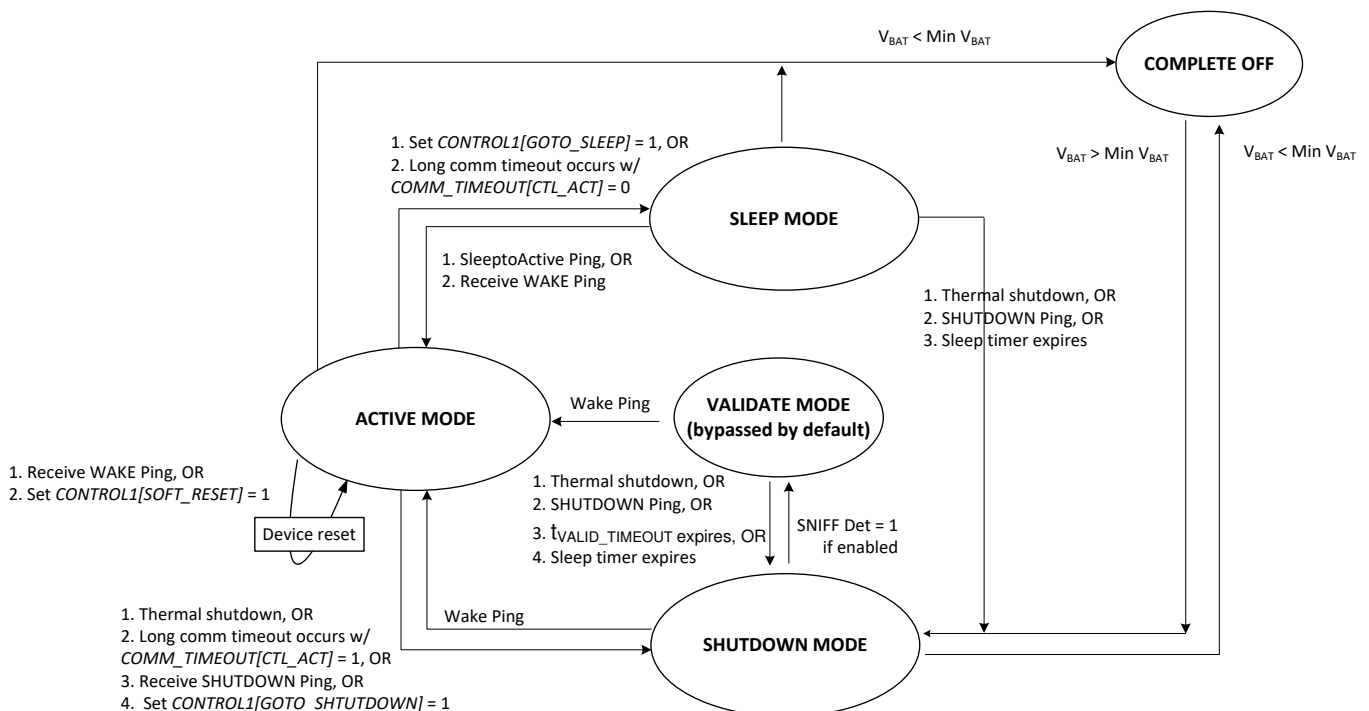


Figure 7-2. Power State Diagram

7.3.1.2 Pings

A “ping” is a specific signal pattern to change power mode of BQ79600-Q1. There are total of 3 different pings:

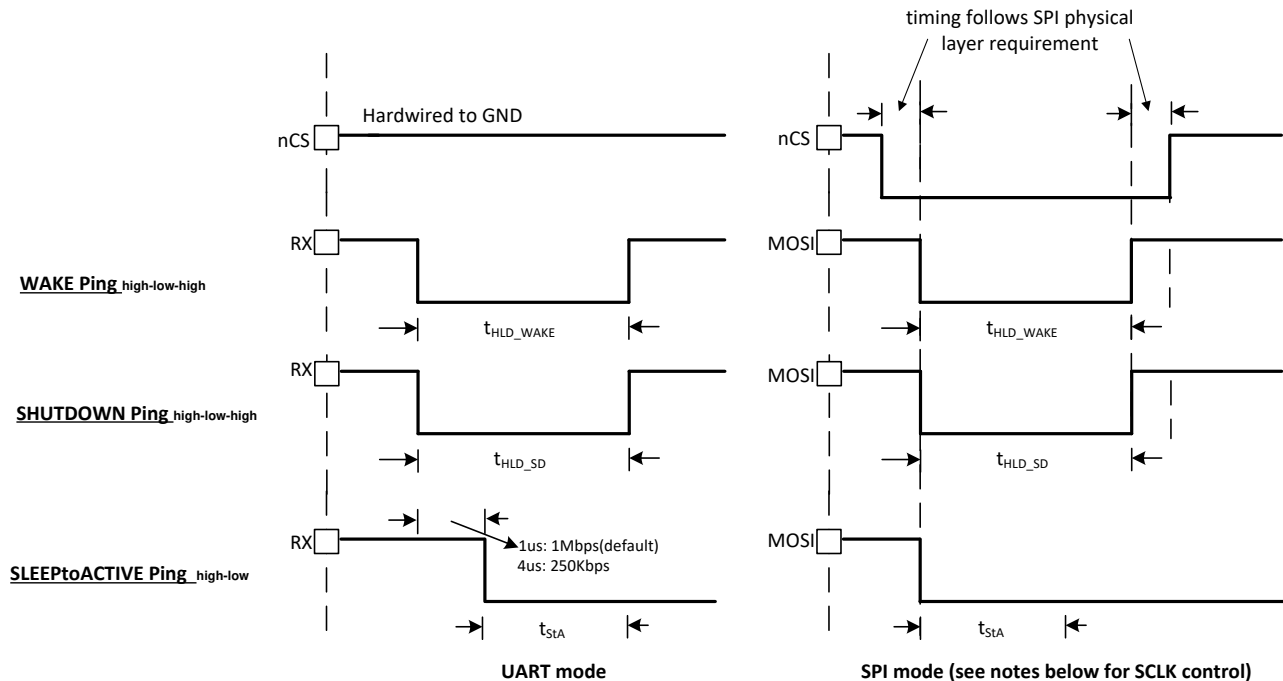


Figure 7-3. Communication Pings

- In SPI mode, ping can be sent without SCLK toggling or host can treat sending a ping like sending a series of logic '0' (drive nCS and SCLK properly, refer to SPI physical layer requirement, [Section 7.3.2.1.2.2](#)).
- Device does not transmit any tones to stack devices due to the receiving of pings.
- After sending in SHUTDOWN ping, host has to wait t_{SHDN} before sending another ping.
- If nCS = '1', all of the pings above are ignored by the device.

Note

If device is shut down through SHUTDOWN ping (COMH RX and COML RX are disabled at next wake up), host needs to send 1st WAKE ping, wait $t_{SU(WAKE_SHUT)}$, and then send 2nd WAKE ping.

Table 7-2. Device Behavior when SLP2ACT Ping is Sent

	UART	SPI WITH SCLK TOGGLING	SPI WITHOUT SCLK TOGGLING
ACTIVE	[COMMCLR_DET]=1, [STOP_DET]=1	[COMCLR_ERR]=1, [COMMCLR_DET]=0	[FMT_ERR]=1, [COMCLR_ERR]=0, [COMMCLR_DET]=0
SLEEP	[COMMCLR_DET]=1, [STOP_DET]=0, transitions to active mode	[COMCLR_ERR]=0, [COMMCLR_DET]=0, transitions to active mode	[FMT_ERR]=0, [COMCLR_ERR]=0, [COMMCLR_DET]=0, transitions to active mode

7.3.1.3 SPI/UART Selection

The SPI or UART interface is selected through hardware: connect pin nUART/SPI to VIO through resistor for SPI or connect to GND for UART. Device determines UART or SPI mode every time it transitions from SHUTDOWN to ACTIVE mode. Before transitioning to ACTIVE, the mode selected is locked in. VIO has to be above $V_{VIO_UV_R}$. nUART/SPI pin is used as output indication of SPI_RDY once in ACTIVE mode. See [Section 7.3.2.1.2.2.1](#) for more information.

7.3.1.4 Digital Reset

Digital reset is when digital core of the device in reset mode. It is not a power mode. Once device comes out of digital reset, [DRST] bit is set to '1', registers that are not included in NVM are set to RESET VALUE, registers included in NVM would be NVM program value. There are several conditions in which the device will go through a digital reset:

- A WAKE ping is received.
- The CONTROL1 [SOFT_RESET] = 1 command is sent in ACTIVE mode.
- Power supply faults. DVDD UV or CVDD UV is detected.
- A HFO or LFO watchdog fault will reset the digital.

7.3.1.5 Power Mode in BMS System

It is recommended to follow the power state combinations below to save system level power.

Table 7-3. Power Mode Combination Summary

STACK MONITORING IC	BQ79600-Q1	COMMENT
ACTIVE	ACTIVE	Recommended
SLEEP	SLEEP	Recommended
SLEEP	SHUTDOWN	Recommended
SHUTDOWN	SHUTDOWN	Recommended
Other combinations		Not recommended

7.3.1.6 Power Supply

This section provides an overview of each supplies for both user cases: without using Reverse Wakeup and with using Reverse Wakeup. See the [Section 7.3.6](#) for diagnostic control and fault detection on the power supplies block.

Table 7-4. Power Supply Summary

NAME	W/O REVERSE WAKEUP USER CASE	W/ REVERSE WAKEUP USER CASE
VIO	This supply is powered by regulated 3.3V or 5V from SBC (PMIC), it powers UART/SPI interface pins.	
BAT	This supply is powered by regulated 5V from SBC (PMIC).	This supply is powered by unregulated 12V battery.
AVAOREF	This supply is generated from VBAT. It is always on if VBAT exists. It powers REFSYS and Power mode control block.	
AVDDREF	This supply is derived from AVAOREF. AVDDREF and AVAOREF are connected by a switch, the switch is open in SHUTDOWN mode.	
DVDD	This supply is generated by the internal DVDD LDO. It is the supply for the digital circuits. It takes the input voltage from CVDD and generates a nominal 1.8V. It will not be used to power any external circuit.	
CVDD	This supply is powered by regulated 5V from SBC(PMIC). It is the supply for daisy chain interface.	This supply is generated by the internal CVDD LDO. It is the supply for the daisy chain interface (or vertical interface, VIF). It takes the input voltage from VBAT and generates a nominal 5V. It will not be used to power any external circuit.

7.3.1.7 Shutdown

Power Mode Transition Example

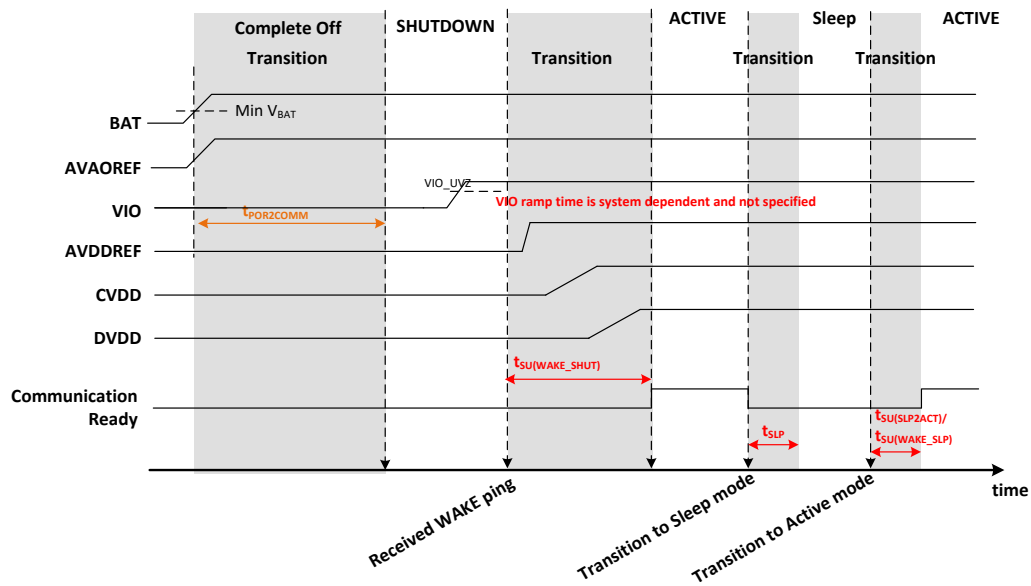


Figure 7-4. Shutdown -> Active -> Sleep -> Active

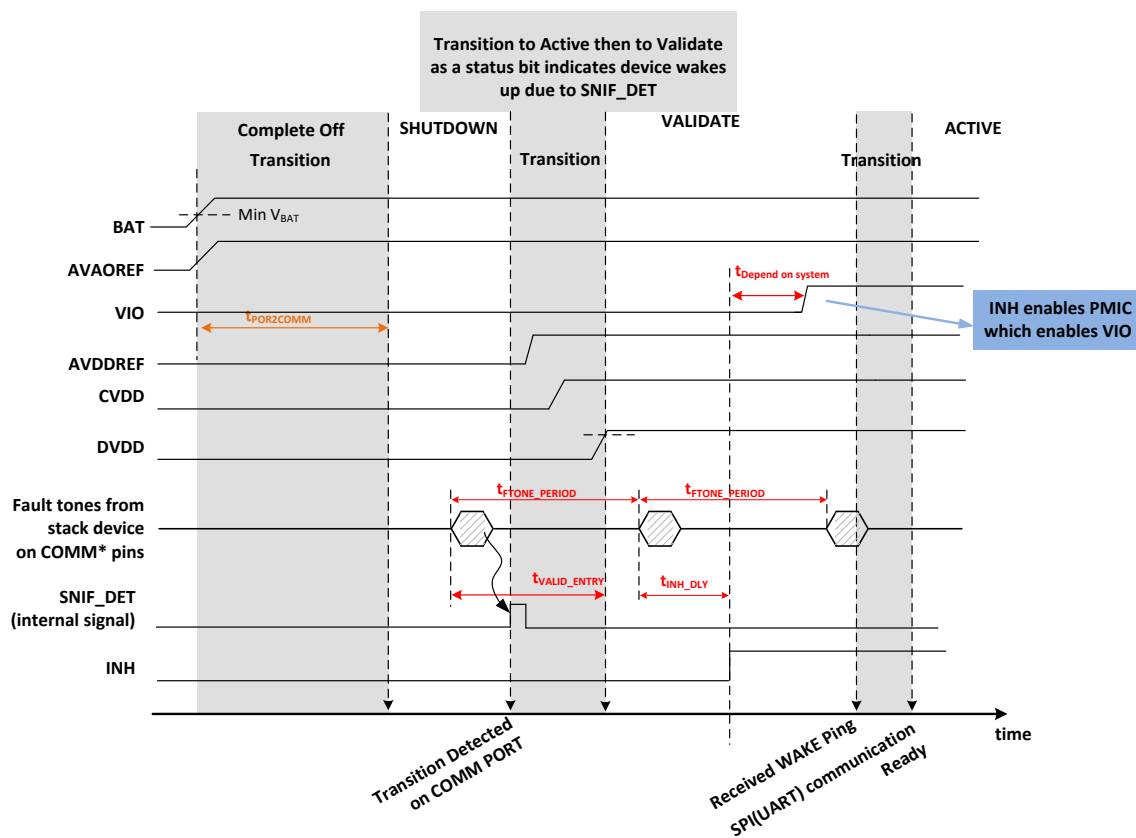


Figure 7-5. Shutdown -> Validate -> Active

7.3.2 Communication

This device is used as a bridge (base) device in daisy chain configuration, as showed in figure below. It communicates with MCU through UART or SPI and communicates with stack devices through daisy chain (proprietary interface). MCU always initiates communication with a Command frame. In the system, BQ devices would never send data back to MCU before MCU requests. And MCU needs to wait all expected response frames before sending next command frame. Thus, communication scenarios can be put into 2 categories:

- MCU sends Write Command frames to BQ devices. Write Command frames don't incur Response frame.
- MCU sends one Read Command frame to BQ devices and waits till all Response frames are received.

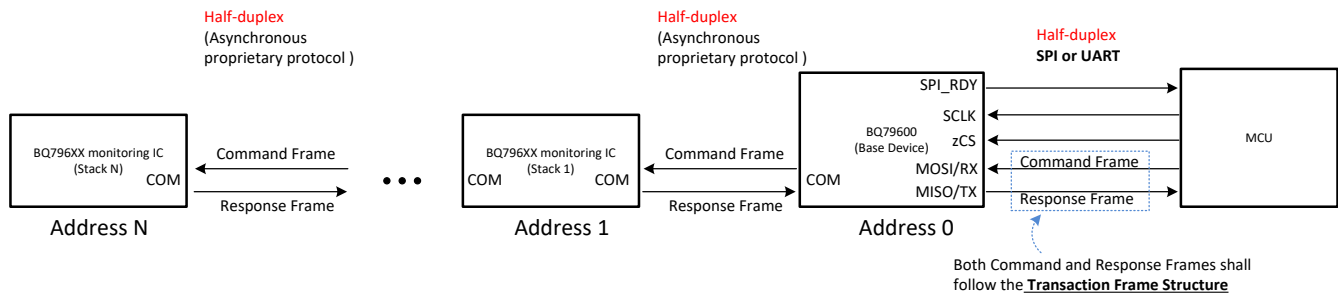


Figure 7-6. System Communication Diagram

Rest of the section talks about how data and tone are communicated among host, bridge, and stack devices: [Section 7.3.2.1](#) ([Section 7.3.2.1.1](#) and [Section 7.3.2.1.2](#) protocol), [Section 7.3.2.2](#). It also talks about [Section 7.3.2.3](#) and [Section 7.3.2.4](#).

7.3.2.1 Data Communication Protocol

7.3.2.1.1 Frame Layer

The communication frame is defined in figure below. It is made up of 5 types of information: initialization character (INIT), device address characters, register address character, data character(s) and CRC characters. Each character is transmitted at UART/ SPI/ Daisy Chain physical level, whose format is defined in following [Section 7.3.2.1.2](#) section. There are 3 types of transaction frames: **Read Command Frames**, **Write Command Frames** and **Response Frames**. They follow the structure in the figure below.

INIT [7:0] DEV ADR [7:0] REG ADR [15:0] DATA byte(s) CRC [15:0]						
Frames		INIT	DEV ADR	REG ADR	DATA	CRC
Read Command Frame	Single Device Read	Always 0x80	Y	Y	# of byte requested (max value 127, meaning 128)	Y
	Stack Read	Always 0xA0	N	Y		Y
	Broadcast Read	Always 0xC0	N	Y		Y
Write Command Frame	Single Device Write	Y	Y	Y	Actual payload (1-8 bytes)	Y
	Stack Write	Y	N	Y		Y
	Broadcast Write	Y	N	Y		Y
	Broadcast Write Reverse ⁽⁵⁾	Y ⁽⁵⁾	N	Y ⁽⁵⁾	Y ⁽⁵⁾	Y
Response Frame		Y	Y	Y	Actual payload (1-128 bytes)	Y

Figure 7-7. Command/Response Frame Structure

Notes:

- When BQ79600-Q1 is used as bridge device, to read BQ devices information, host **SHALL NOT** use Broadcast Read command but only Single Device Read or Stack Read. The reason is BQ79600-Q1 register address does not overlap with stack devices, it would only return 0x00 to Broadcast Read command.
- For Stack Read command, the response is broken into individual response frames from each device addressed. Each device (address N) in the stack waits until the device above it (address N+1) responds before device N sends its own data back.
- After a read command frame is transmitted, the host must wait for all expected responses to return (or timeout: $t_{WAIT_READ_MAX}$) before initiating a new command frame.
- A response frame is not mandatory. A response frame is only received after a read command frame.

- Broadcast Write Reverse command frame should only be used to config [DIR_SEL] bit, not for any other purposes. INIT byte is 0xE0, Reg address byte is 0x309 (BQ7961X-Q1), data byte is 0x80.
- Bytes received on COMH/COML are NOT propagated up to the stack; while bytes received on the SPI/UART are propagated to COMH or COML depending on [DIR_SEL].
- Even if there is a byte error, data is still forwarded from VIF to SPI (buffer)/UART; if there is a byte error, data doesn't forward from SPI/UART to VIF.

	Command Frame			Response Frame	
	Bit	Bit Name	Description	Bit Name	Description
INIT	7	FRAME_TYPE	1 = Command Frame	FRAME_TYPE	0 = Response Frame
	6	REQ_TYPE	000 = Single Device Read 001 = Single Device Write 010 = Stack Read 011 = Stack Write 100 = Broadcast Read 101 = Broadcast Write 110 = Broadcast Write Reverse 111 = RSVD ⁽¹⁾	RESPONSE_BYTE	Number of the data bytes 0x00 = 1 byte 0x01 = 2 bytes : 0x7F = 128 bytes
	5				
	4				
	3				
	3	RSVD	RSVD. This bit is ignored		
	2	DATA_SIZE	Number of DATA bytes, excluding device address, register address or CRC 000 = 1 byte 001 = 2 bytes : 111 = 8 bytes		
	1				
0					

	Bit	Bit Name	Description
DEV_ADR	7	RSVD	Should always write "0"
	6	RSVD	Should always write "0"
	5:0	Device Address	Set the device address range from 0x00 to 0x3F

	Bit	Bit Name	Description
REG_ADR	7:0	Register Address (MSB)	Target or beginning of the register address
	7:0	Register Address (LSB)	Target or beginning of the register address

	Bit	Bit Name	Description
DATA	7:0	Data Byte[0]	Data Byte[0]
	...		
	7:0	Data Byte [n]	Data Byte[n]

		Command/Response Frame	
	Bit	Bit Name	Description
CRC	7:0	CRC (MSB)	CRC-16-IBM polynomial (x ¹⁶ + x ¹⁵ + x ² + 1 or 11000000000000101) with 0xFFFF initialization
	7:0	CRC (LSB)	CRC-16-IBM polynomial (x ¹⁶ + x ¹⁵ + x ² + 1 or 11000000000000101) with 0xFFFF initialization

Figure 7-8. Frame Byte Definition

Notes:

- INIT character: (1) No function to this selection, but this selection sets the [RC_IERR] error flag.
- Device Address character: Bit 6 and 7 are reserved; 0x4F to 0xFF is decoded as 0x3F by device.
- Register Address characters: Register addresses are two bytes in length. They indicate the targeted register address on a single byte read/write, or the beginning of the register address on a multi-byte read/write. If an invalid register address is set on a write command, the command will be ignored. If an invalid register address is set on a read command, a 0x00 will be returned as response.
- Data characters can be:
 - Single data byte, it represents number of registers requested in Read Command Frame. The BQ79600-Q1 supports up to 128 byte reads. The valid data byte for read command frame is 0b00000000 - 0b11111111. The MSB of the data byte is ignored for read command frames. For example, 0b10011001 is read as 0b00011001 and returns data from 26 registers.
 - Actual payload in Write Command Frame (max 8 byte) or Response Frame (max 128 byte).
- CRC characters:
 - The CRC value is checked as the first step (assume no physical layer error, no [RC_IERR], no [RC_SOF], no [RC_BYTE_ERR]) after receiving the communication frame. If the CRC is incorrect, the entire frame is discarded and not processed. Any additional frame errors are not checked.
 - The frame with CRC error is still transferred up/down the stack. Every device processing this frame will also detect a CRC error. Hence, it is possible to have multiple devices indicating CRC fault on the same communication frame. If a CRC error occurs in the response frame from address N+1, device N does NOT append its own message and an invalid CRC fault is generated. For example, if device 15 finds response frame from device 16 has invalid CRC, device 15 doesn't send its own response frame.

- The device uses a CRC (cyclic redundancy check) to protect data integrity during transmission. The device uses the CRC-16-IBM polynomial ($x^{16} + x^{15} + x^2 + 1$) with 0xFFFF initialization.

7.3.2.1.1.1 Calculating Frame CRC Value

The CRC calculation by the transmitter is in bit-stream order across the entire transmission frame (except for the CRC). When determining bit-stream order for implementing the CRC algorithm, it is important to note that protocol bytes transmit serially, least-significant bit first. Figure 7-9 illustrates the bit-stream order concept.

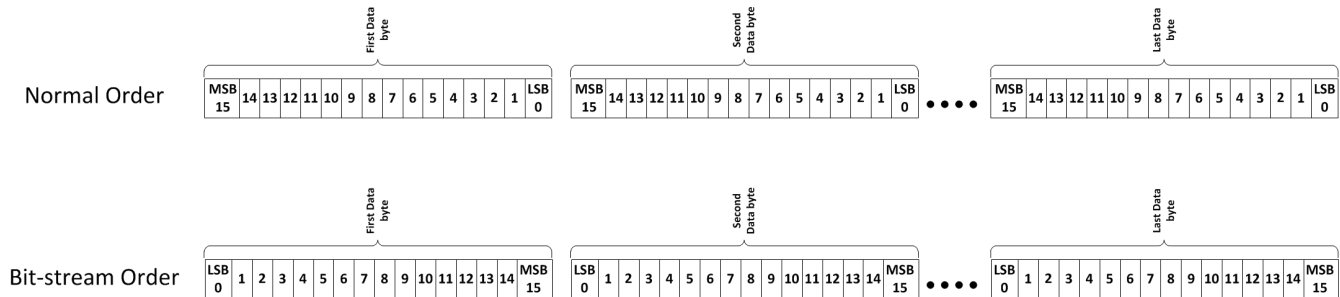


Figure 7-9. Bit-Stream Order Explanation

The CRC (0x0000) is appended to the end of the bit-stream. This bit-stream is then initialized by XOR'ing with 0xFFFF to catch any leading 0 errors. This new bit-stream is then divided by the polynomial (0xC002) until only the 2-byte CRC remains. During this process, the most significant 17 bits of the bit stream are XOR'd with the polynomial. The leading zeroes of the result are removed and that result is XOR'd with the polynomial once again. The process is repeated until only the 2-byte CRC remains. For example:

Example 1: CRC Calculation Using Polynomial Division

```
Command Frame = 0x80 00 02 0F 0B (0b1000 0000 0000 0000 0000 0010 0000 1111 0000 1011)
Command Frame in bit stream order = 0x01 00 40 F0 D0 (0b0000 0001 0000 0000 0100 0000 1111 0000 1101 0000)
After Initialization (XOR with 0xFFFF) = 0b1111 1110 1111 1111 0100 0000 1111 0000 1101 0000
1111 1110 1111 1111 0100 0000 1111 0000 1101 0000 0000 0000 0000 0000 #append 0x0000 for CRC
1100 0000 0000 0010 1 #XOR with polynomial
0011 1110 1111 1101 1100 0000 1111 0000 1101 0000 0000 0000 0000 0000
11 1110 1111 1101 1100 0000 1111 0000 1101 0000 0000 0000 0000 0000 #delete leading zeros from
previous result
11 0000 0000 0000 101 #XOR with polynomial
00 1110 1111 1101 0110 0000 1111 0000 1101 0000
.....
.....
.....
1100 0110 0000 0001 0000 0000
1100 0000 0000 0010 1 #XOR with polynomial
0000 0110 0000 0011 1000 0000
110 0000 0011 1000 0000
110 0000 0000 0001 01 #XOR with polynomial
000 0000 0011 1001 0100
0000 0011 1001 0100 #CRC result in bit stream order
1100 0000 0010 1001 #final CRC result in normal order
CRC final 0xC029
```

7.3.2.1.1.2 Verifying Frame CRC

There are several methods for checking the CRC of a frame. One method is to simply calculate the CRC for the transmitted command except the last two bytes (CRC bytes) using the method described in the previous section, and then compare that result with the transmitted CRC bytes. A more simple option is to run the entire transmission through the CRC algorithm. If the CRC is correct, the result is 0000. In this case, the initial zero padding of the bit-stream with 16 zeroes is not necessary. Using the previous result and running through the algorithm produces the following results:

Example 1: CRC Verification Using Polynomial Division:

```
Command Frame = 0x80 00 02 0F 0B (0b1000 0000 0000 0000 0000 0010 0000 1111 0000 1011)
CRC to Check = 0xC029
Command Frame w/ CRC in bit stream order = 0x80 00 02 0F 0B C0 29 (0b1000 0000 0000 0000 0000 0010
```

```

0000 1111 0000 1011 0000 0011 1001 0100)
After Initialization (XOR with 0xFFFF) = 0b0 1111 1110 1111 1111 0100 0000 1111 0000 1101 0000 0000
0011 1001 0100
1111 1110 1111 1111 0100 0000 1111 0000 1101 0000 0000 0011 1001 010 #delete leading zeros from
previous result
1100 0000 0000 0010 1 #XOR with polynomial
0011 1110 1111 1101 1100 0000 1111 0000 1101 0000 0000 0011 1001 0100
11 1110 1111 1101 1100 0000 1111 0000 1101 0000 0000 0011 1001 0100 #delete leading zeros from
previous result
11 0000 0000 0000 101 #XOR with polynomial
00 1110 1111 1101 0110 0000 1111 0000 1101 0000 0000 0011 1001 0100
.....
.....
.....
1100 0110 0000 0010 1001 0100
1100 0000 0000 0010 1 #XOR with polynomial
0000 0110 0000 0000 0001 0100
1 1000 0000 0000 0101 00
1 1000 0000 0000 0101 #XOR with polynomial
0 0000 0000 0000 0000 00
0x0000 #verfiy that CRC checks out valid

```

Note

The result of '0b0000 0000 0000 0000' for the CRC indicates a successful check.

7.3.2.1.2 Physical Layer

7.3.2.1.2.1 UART

Communication between host and BQ79600-Q1 can be configured to UART mode, refer to [Section 7.3.1.3](#). The UART interface baud rate is default to 1Mbps at power up or digital reset. The UART interface follows the standard serial protocol of 8-N-1 (see [Figure 7-10](#)), where it sends information as a START bit, followed by eight data bits, and then one STOP bit. The STOP bit indicates the end of the byte. The protocol also supports two STOP bits. When the device is configured as 2 stop bits ([TWO_STOP_EN] = 1, stack devices should also be set as two stop bits), the UART response frame from the device to MCU will always return with 2 stop bits.

The UART sends data on the TX pin and receives data on the RX pin. When idle, the TX and RX are high. TX is always pulled to VIO internally while in ACTIVE or SLEEP mode, whether enabled or disabled.

The UART interface is strictly a half-duplex interface. While transmitting, any attempted communication on RX is ignored. The only exception is [Section 7.3.2.1.2.1.2](#).

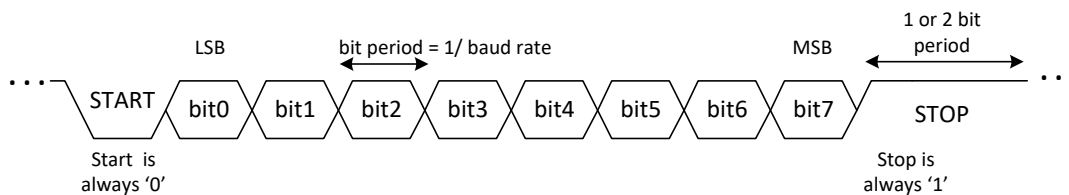


Figure 7-10. UART Character Definition

Note: User can change baud rate using register bit [UART_VIF_BAUD] for debug purpose.

7.3.2.1.2.1.1 TX HOLD OFF

UART transmitter is configurable to wait a specified number of bit periods after the last bit reception of Single Device read command frame from host before starting transmitting Response Frame using the TX_HOLD_OFF register, as showed in [Figure 7-11](#). This provides time for the host to switch the bus direction at the end of its transmission.

Note

Host does not need to configure register TX_HOLD_OFF in BQ79600-Q1 if Stack Read command is used. Host shall not use Broadcast Read Command.

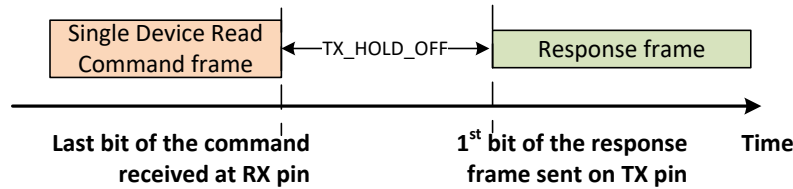


Figure 7-11. TX HOLD OFF Timing Diagram

7.3.2.1.2.1.2 UART COMM CLEAR

Note

Comm Clear concept only applies to bridge device not stack device.

Comm Clear command is used to clear the receiver and instruct it to look for a new start of frame. (Resync up with host) The next byte following the Comm Clear is considered a "start of frame" byte. The digital receiver continuously monitors the RX line for Comm Clear condition which is RX pin is held low for tUART (CLR) bit periods, showed in Figure 7-12.

When Comm Clear is detected, FAULT_COMM1 [COMMCLR_DET] and FAULT_COMM1 [STOP_DET] are set. [STOP_DET] flag is set because the Comm Clear timing violates the typical byte timing and the STOP bit is seen as '0'. The only exception to this is when a COMM CLEAR is sent while BQ79600-Q1 is in sleep. If this is the case, there is no STOP error flag.

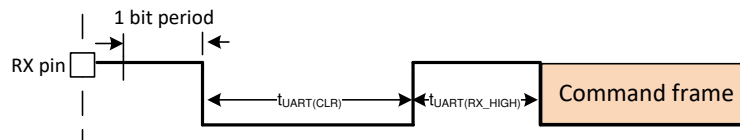


Figure 7-12. Comm Clear Timing

7.3.2.1.2.2 SPI

Note

To facilitate the communication between daisy chain (asynchronous protocol) and SPI (synchronous protocol), BQ79600-Q1 requires the use of Section 7.3.2.1.2.2.1.

Communication between host and BQ79600-Q1 can be configured to SPI mode, refer to Section 7.3.1.3. The host is always a SPI master while BQ79600-Q1 is always a slave. At physical layer, SPI is a five pin interface including 4 common pins (nCS, SCLK, MOSI, MISO) plus SPI_RDY. At SPI interface, each bit is captured on low to high clock transitions and propagated on high to low clock transition and byte includes 8 bits as shown in Figure 7-13. Please note, MISO is driven high in idle mode. If MCU talks to multiple slaves, please add a tri-state buffer between BQ79600 MISO and MCU.

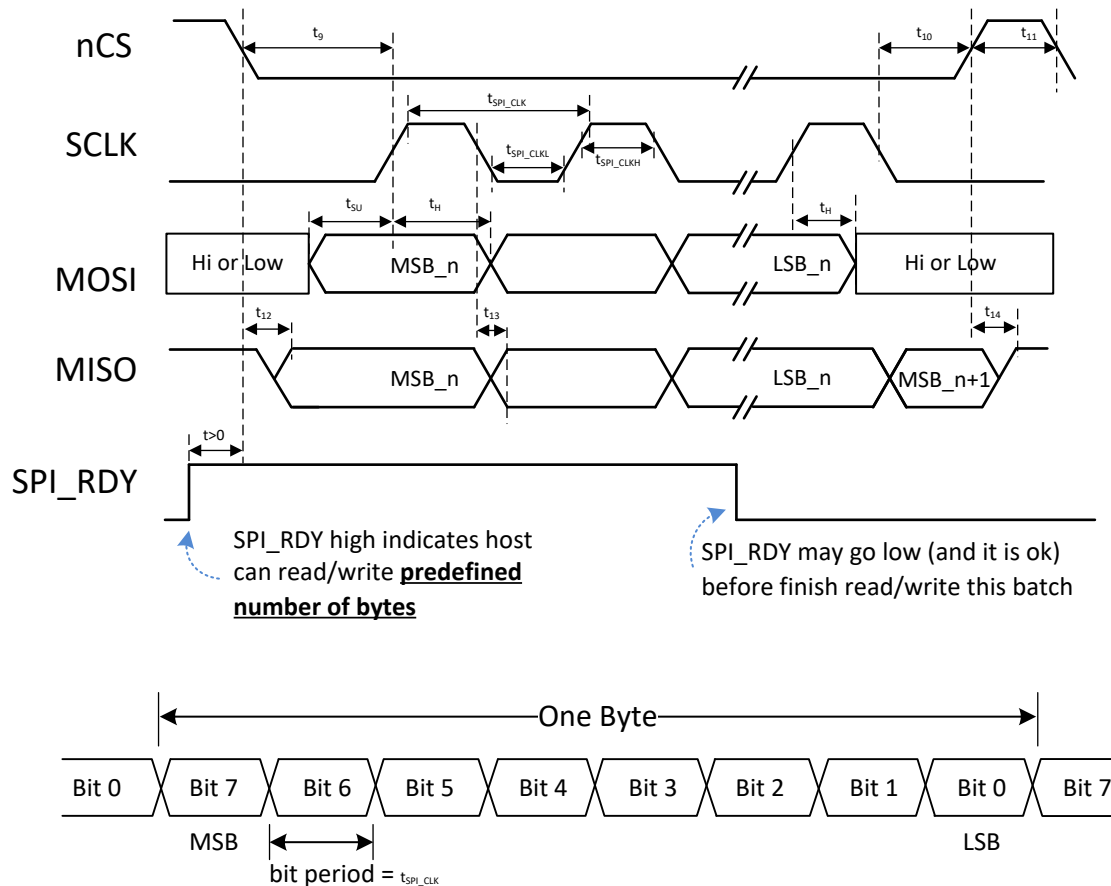


Figure 7-13. SPI Timing and Byte Definition

Note

Although SPI interface is full duplex at physical layer, at frame layer, it is effectively half duplex because daisy chain only supports half duplex. It means, at the given time, only a command frame OR a response frame is transmitted between MCU and the device.

- When command is being sent from MCU to the device, BQ79600 TX FIFO (2 buffers) should be empty, 0xFF is sent to MCU (except in FIFO diagnostic mode)
- When response is being sent from device to MCU, MCU shall clock in 0xFF

- Host shall provide SPI clock in the range between 2MHz and 6MHz. This range is set by the predefined size of FIFO. Even SPI can run at 6MHz, it does not increase the throughput of total system as daisy chain speed still limit the throughput.
- To avoid collisions on the daisy chain interface, the MCU must wait until all expected response frames are received (or waiting timer expires) before sending another command frame to bq79600. Refer to flow chart in [Figure 7-16](#).
- Master shall always drive MOSI to '1' when not sending command frame into device.
- **Host Read mode:** from host perspective, read mode is between first byte of a valid read command and last expected number of byte received.
- **Device Read mode:** start from first byte of a valid read command, device exists read mode when TX FIFO times out and FIFO is empty. Device read mode is subset of host read mode. (concept used for understanding of communication fault registers [0x2301](#), [0x2302](#))
- Other than Comm Clear, device rejects any data from MOSI before exiting Device Read mode.
- SPI module rejects any data from daisy chain (stack device) or from itself after TX FIFO timeout [List item](#). until it enters Device Read mode again.

- For command frame, device uses falling edge of nCS as indication of start of frame, and rising edge as end of frame. MCU needs to toggle and keep nCS low for the entire frame (up to 14 characters), toggle nCS back high at the end of this frame. It is legitimate to freeze SCLK while nCS is low. Pulsing nCS in the middle of a command frame is not supported.
- For response frame, nCS is not required to remain low throughout, although it can just like sending command frame. Host can toggle nCS high, stop SPI reading in the middle of the frame (at byte boundary).

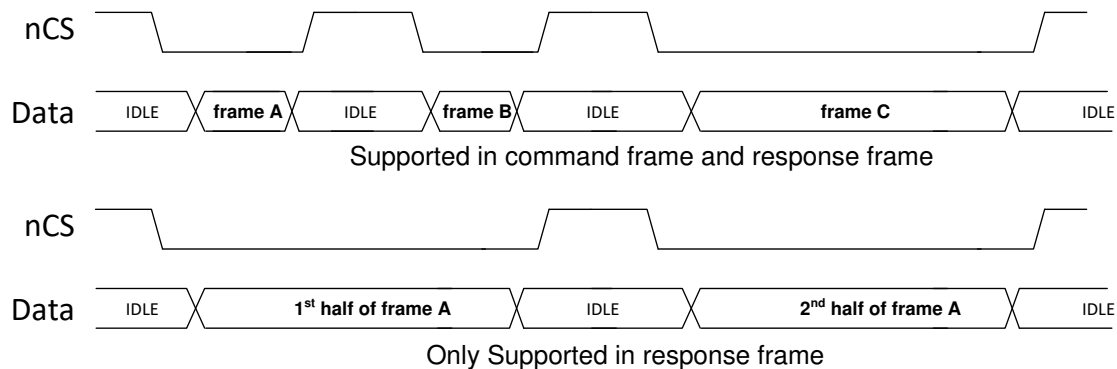


Figure 7-14. nCS Behavior when Sending Frames

7.3.2.1.2.2.1 SPI_RDY and SPI FIFOs

SPI_RDY is a output signal indicating to host that data is ready to communicate. SPI FIFOs [Figure 7-15](#) are buffers in the device for temporary storage of incoming/outgoing data. They are required for the following reasons:

- Daisy chain baud rate is approximately 1Mbps by default, once host requests large amount of data, e.g. 400 bytes, device daisy chain receiver would try to send it back to host, but since device does not own the SCLK, it cannot control when the data would be read out. Thus when host is not reading, device need to store the incoming data in the TX FIFO. And even when host is reading, FIFOs are still needed to handle the baud rate difference between SPI and daisy chain.
- SPI_RDY is required because there is limited depth of TX FIFOs. If host requests more than 256 bytes, and host does not service (read data out) device in time, data overflow would happen. SPI_RDY indicates to host that certain amount of data is ready to be read or written, e.g. If host request 129 bytes, first time SPI_RDY flags 128 bytes is ready, second time SPI_RDY flags 1byte is ready. For details refer to [Table 7-5](#)

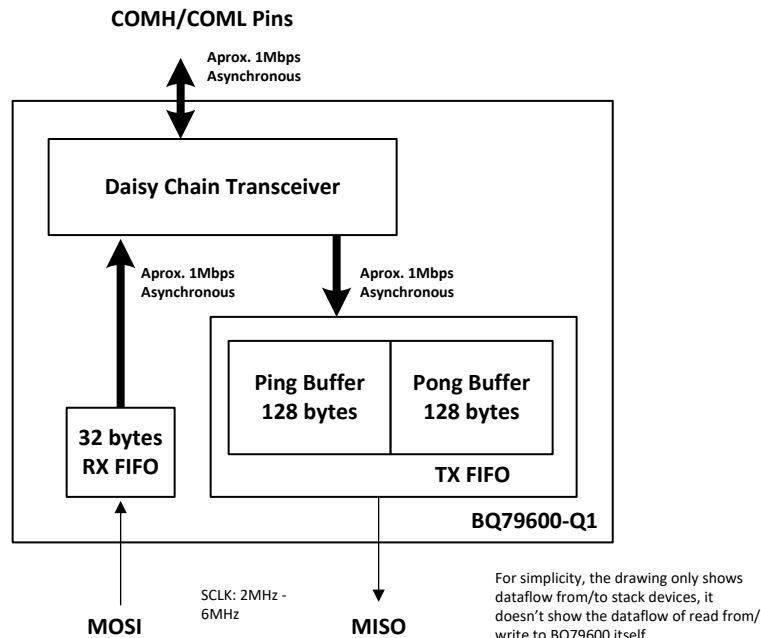


Figure 7-15. SPI FIFO Simplified Diagram

TX FIFO consists of two 128 bytes buffers (working together as Ping-Pong buffer).

1. When Ping buffer is filled up, Pong buffer should be empty, to store incoming data.
2. While Pong buffer is being filled, Ping buffer is being read. Each byte in buffer is reset to 0xFF once being read. Ping buffer shall be empty (read out) before Pong buffer is full.
3. After Pong buffer is filled up, Ping buffer catches up.
4. Device goes through this loop (step 1 to 3) till all response data are received.
5. Host has to read TX FIFO fast enough such that Ping (Pong) buffer is read out and ready to store data from daisy chain before Pong (Ping) buffer is full.

Table 7-5. SPI_RDY Behavior Summary

	CASE #	HI->LOW WHEN	LO->HI WHEN
		a	b
Host Writes	1	In 2μs when RX FIFO has >= 16 bytes.	After event a1, in 2μs when RX FIFO has < 8 bytes.
	2	In 5μs after device receives 1st byte of read command frame.	In 1us after ping (pong) buffer is full.
Host Reads	3	TX buffer being read becomes empty (before transmitting out last bit of last byte in the buffer) Note: once going low, SPI_RDY remains low for 2μs no matter what.	TX FIFO time out happened Note: TX FIFO timeout could happen while SPY_RDY is high (while host is reading TX FIFO), in this case, after event a3, SPI_RDY goes low for approximately 2μs, then come back high.

Notes:

- SPI_RDY sets flag only, doesn't gate data flow into or out of device.
- Once devices enters Device Read mode, device rejects any data from host other than COMM CLEAR. a1, b1 doesn't apply anymore.
- TX FIFO Timeout: after SPI module receives one byte of data from either daisy chain or BQ79600-Q1 local, a timer starts; this timer expires if there is no data received for 30μs.

7.3.2.1.2.2.2 Flow to Read/Write BQ79600-Q1

User shall follow flow chart [Figure 7-16](#) to do read from device and [Figure 7-17](#) to do write to device activities.

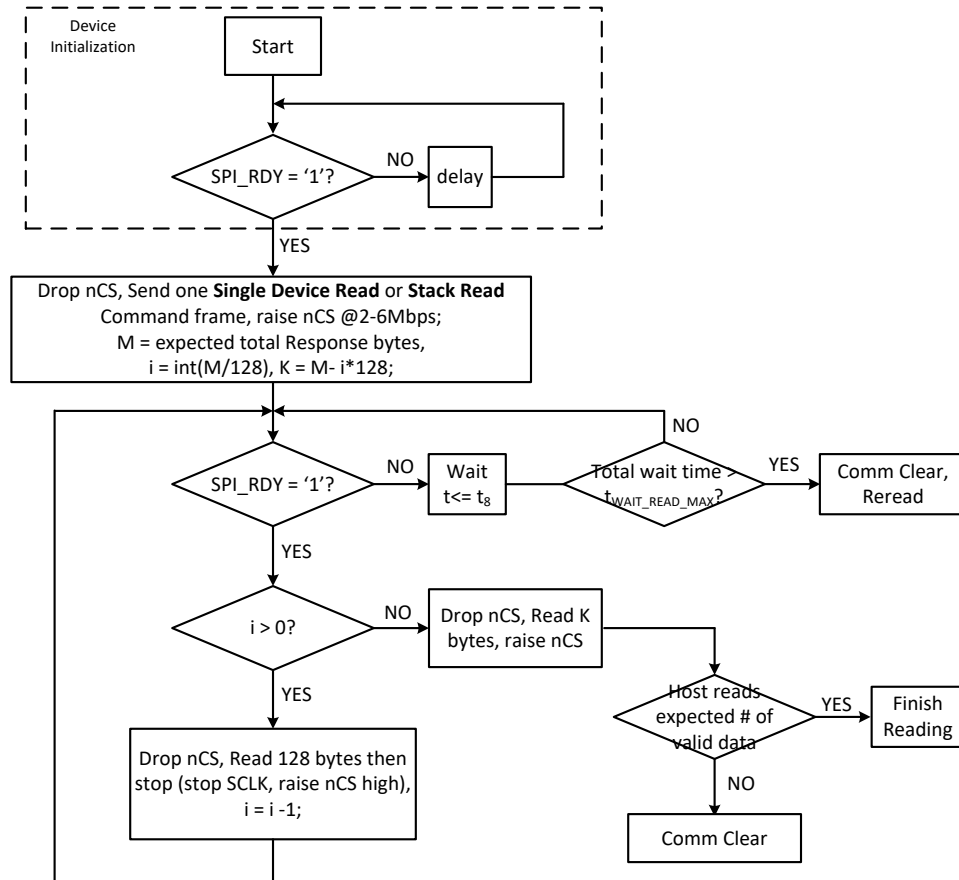


Figure 7-16. Flow Chart to Read from Device

Note

MCU shall check SPI_RDY pin at least every t_8 (max service interval). $t_8 = 1\text{ms}$ at SCLK = 6MHz / $890\mu\text{s}$ at SCLK = 4MHz / $550\mu\text{s}$ at SCLK = 2MHz assuming host starts to read TX FIFO right after detecting SPI_RDY = '1' and SPI bus has 30% idle time in the process of reading 128 bytes.

- For response frame, nCS has to be toggled high after reading the last byte of data in the current buffer.
- $t_{\text{WAIT_READ_MAX}}$
 - stack read/single device read from stack devices, with n stack devices, request m bytes in total (payload + overheads, from all stack devices), wait time: $(n-1)*3\mu\text{s} + m*10\mu\text{s} + 100\mu\text{s}$.
 - single device read from BQ79600-Q1, request m bytes, wait time: $100\mu\text{s} + m*10\mu\text{s}$.

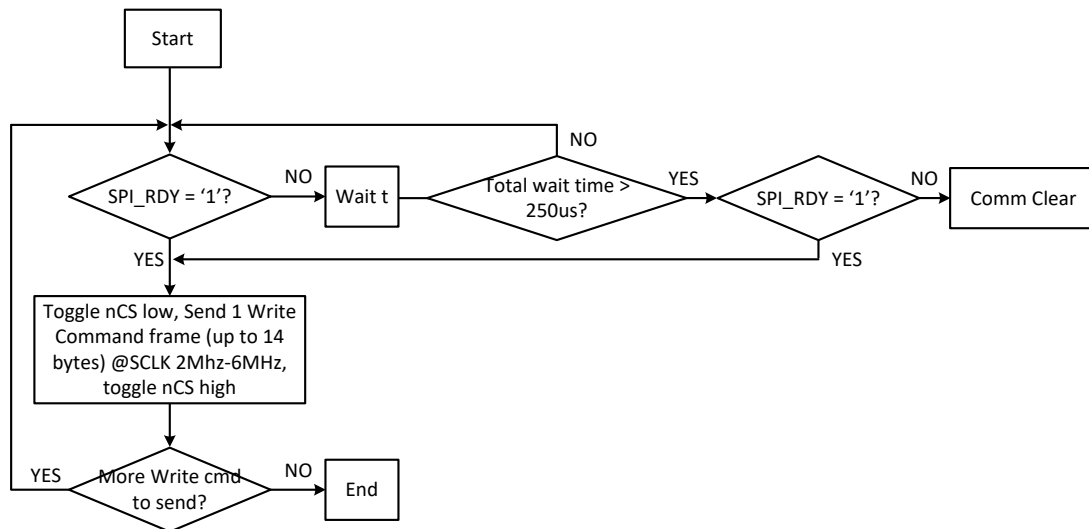


Figure 7-17. Flow Chart to Send Write Command Frame to Device

Notes:

- Since write command frame doesn't incur response frames, host shall discard data from MISO pin.
- If host sends partial data, the device would keep waiting the rest of command data till communication time out happens. Refer to [Section 7.3.2.4](#) for details.

7.3.2.1.2.2.3 SPI COMM CLEAR

Note

SPI Comm Clear, applied only to bridge device, is used to clear the FIFOs and reset SPI module.

It stops BQ79600-Q1 sending response; it cannot stop stack devices sending response data back to BQ79600-Q1. If host still couldn't communicate to device, host can ultimately use SHUTDOWN ping followed by WAKE ping to reboot device. Device only responds to Comm Clear in ACTIVE. Use the SPI Comm Clear command when:

- SPI_RDY being low exceeds expected time, either $t_{\text{SPI_RD_WAIT_MAX}}$ in READ mode in [Figure 7-16](#) or when 220 μ s while sending write command frame into device. Don't send Comm Clear before this wait time has elapsed.
- Data read back by host has CRC error.
- Host cannot communicate to the device.

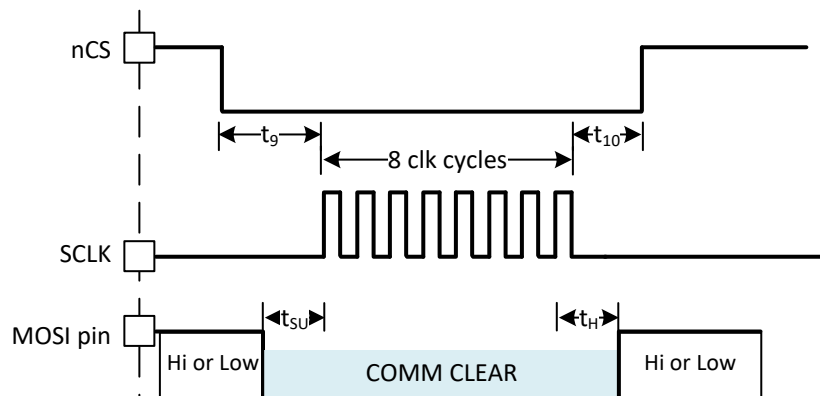


Figure 7-18. SPI Comm Clear

SPI Comm Clear is strictly defined as nCS toggling low, 8 bits of '0', nCS toggling high like shown in [Figure 7-18](#). nCS must remain low while sending the Comm Clear. If additional data is detected on MOSI pin before nCS going high, [COMCLR_ERR] bit is set, device ignores malformed Comm Clear. A correctly formed Comm Clear will only trigger [COMMCLR_DET]. If Comm Clear is used, it can trigger DEBUG_SPI_FRAME [TR_SOF] (while transmitting local data), [RC_SOF](receiving partial data) and [TS_WAIT] (while transmitting daisy chain data).

7.3.2.1.2.3 Daisy Chain

Daisy chain is the interface (COMH/COML) communicating to stack devices. It is bi-directional and half duplex, and, therefore, has a transmitter (TX) and receiver (RX) on both COMH and COML interfaces, [Figure 7-1](#). Signal going in and out of daisy chain port is taken care by the device. To use the device, host does not need to know daisy chain physical layer protocol (bit definition, byte definition and byte transferring). Host just needs to control SPI or UART port properly. Still, for user's information, daisy chain physical layer protocol is described below.

Daisy chain bit is transmitted between COM*P and COM*N in fully differential fashion, see [Figure 7-19](#).

Daisy chain byte uses an asynchronous 13-bit byte-transfer protocol. The definition of each bit in the byte is defined in [Table 7-6](#). Byte to byte transmission is captured in [Figure 7-20](#).

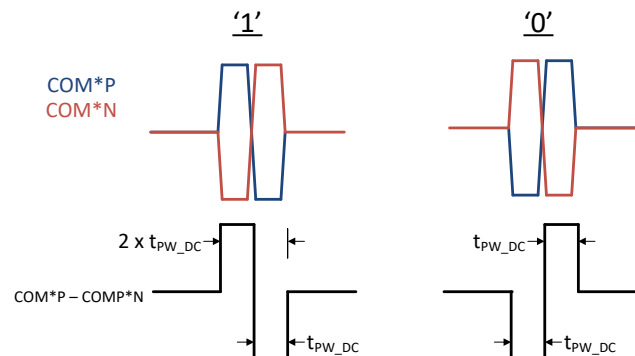


Figure 7-19. Daisy Chain Bit Definition

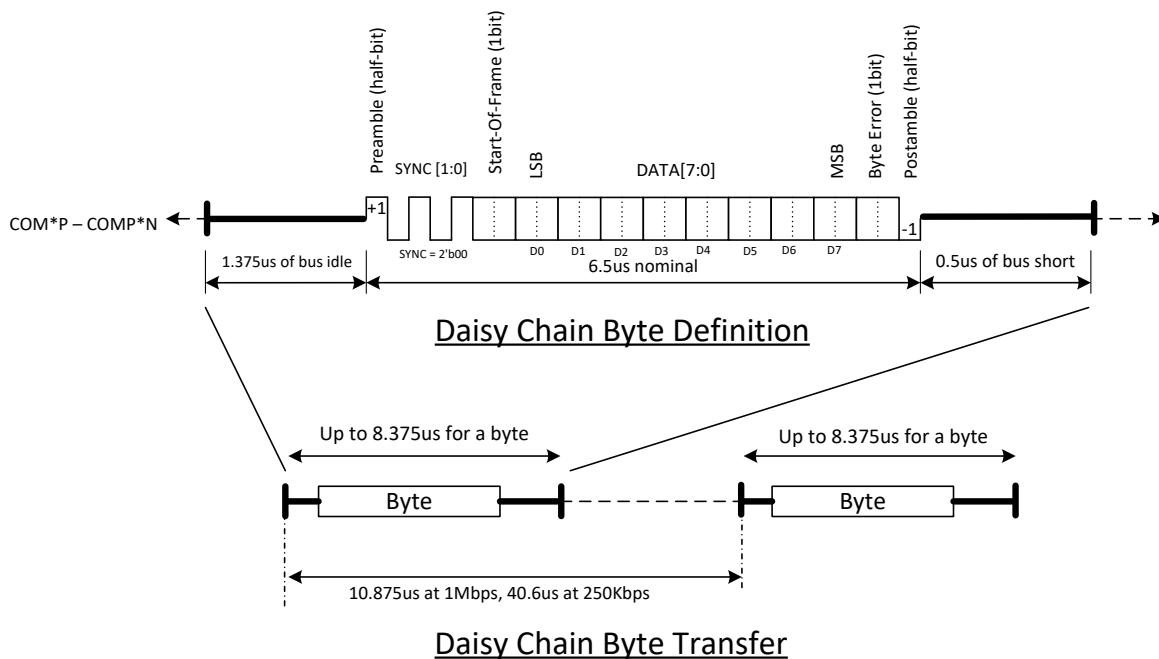


Figure 7-20. Daisy Chain Byte/ Byte Transfer Definition

Table 7-6. Daisy Chain Byte Definition

BIT NAME	DESCRIPTION
Preamble (half-bit)	Indicates a start of byte transaction, signaling the receiver to start sampling. This half-bit and the following 2 SYNC bits are used for extra timing information.
SYNC[1:0]	Always 0b00. The SYNC bits are used for the digital to assess the timing and noise level on the byte, improving the detection of a '1' and '0' in a noisy environment.
Start-Of-Frame (1-bit)	The Start-Of-Frame (SOF) bit defines the byte as the INIT byte (initialization byte) in the frame, refer to Figure 7-8 . Stack device needs this information in order to process the communication.
Data[7:0]	The actual 8-bit payload.
Byte Error BERR (1-bit)	For BQ79600-Q1, BERR is always '0' in command frames sent to stack device. While in received response frames, if it is '1', it indicates last device DEBUG_COM*_PHY[PERR] = 1.
Postamble (half-bit)	Indicates the end of byte transaction.

7.3.2.2 Tone Communication Protocol

Other than data, certain information is transmitted using tone: signals to change power state of stack device (SLP2ACT tone, WAKE tone, SHUTDOWN tone, HWRST tone), signals related to faults (FAULT tone and HEARTBEAT tone). The definition of each tone is defined in [Figure 7-22](#) and [Figure 7-21](#).

Device can transmit and receive tones in summary below:

Table 7-7. Available Tones to BQ79600-Q1

DIRECTION	NAME
Receive	FAULT tone and HEARTBEAT tone
Transmit	SLP2ACT tone, WAKE tone, SHUTDOWN tone, HWRST tone, HEARTBEAT tone

Note:

- Device does not transmit Fault Tone as it uses [Section 7.3.3.2.1](#) to signal fault if enabled.
- SLP2ACT/WAKE/SHUTDOWN/HWRST tone transmitting is on demand when corresponding bit in register CONTROL1 and CONTROL2 is set.
- When bridge device in SHUTDOWN, wakeup bridge device doesn't change power mode of stack devices.
- Receiving threshold value is defined as n_{HBDET} , $n_{FTONEDET}$ in section 6.6.
- Transmitting number is predefined by device in [Table 7-8](#).

Table 7-8. Transmitting Tones Summary Table

	SLP2ACT	WAKE	SHUTDOWN	HWRST	HEARTBEAT (SLEEP only)
N of Couplets Transmitted in Single Burst	30	90	270	810	30
Burst Period	NA	NA	NA	NA	400ms

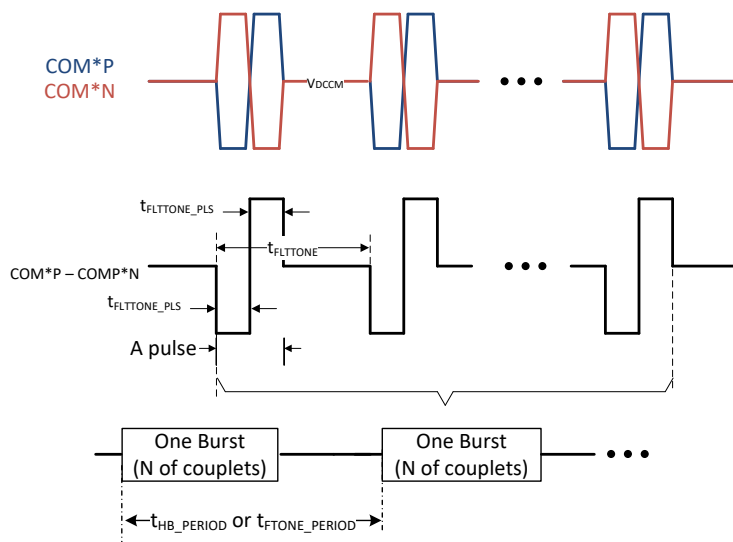


Figure 7-21. FAULT Tone and HEARTBEAT Tone Definition

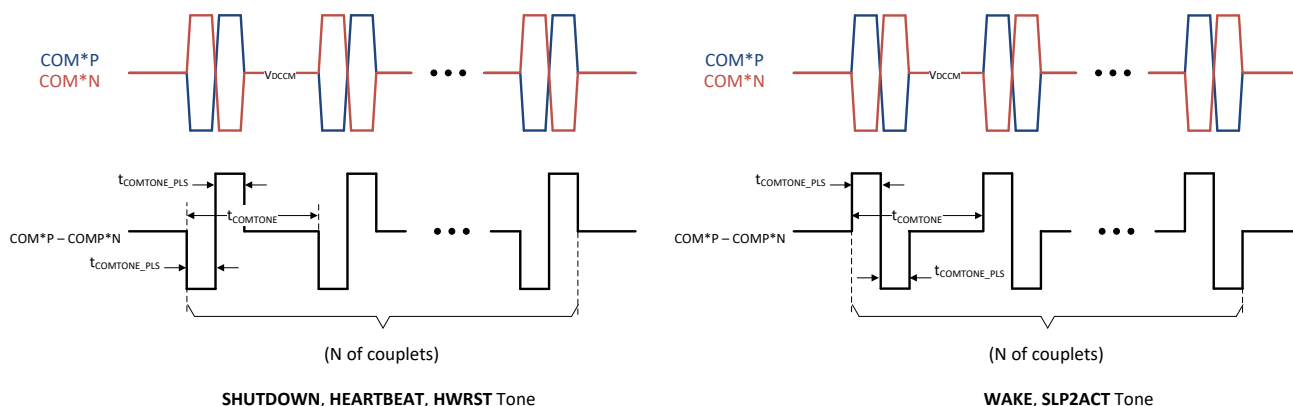


Figure 7-22. SHUTDOWN, HEARTBEAT, HWRST, SLP2ACT Tone Definition

7.3.2.3 Device Auto Addressing / Ring Communication

Note

The host starts communication at least 100µs after changing the *[DIR_SEL]* setting to ensure the device finishes the COMH/COML reconfiguration.

7.3.2.3.1 Auto-Addressing

To properly communicate to every device in daisy chain, host has to assign a unique device address to every device. This process is called Auto-addressing. This step is required every time devices come out of SHUTDOWN or digital reset. [Table 7-9](#) describes a procedure to bring up a system of 1 bridge device and 3 stack devices from SHUTDOWN to a state ready to do read/write communication.

- All device addresses must be sequential

Table 7-9. Auto-Addressing with Figure 7-22(a), assume all devices are in SHUTDOWN

STEP	WORK WITH BQ7961X-Q1
1	send WAKE ping on RX (wake up BQ79600-Q1)
2	single device write to BQ79600-Q1 CONTROL1 [SEND_WAKE] = 1 (wake up stack devices)
3	dummy stack write data 0x00 to register 0x343 to 0x34A (sync up internal DLL). These are 8 separate write commands.
4	brdcast write 0x01 to address 0x309 (enable auto addressing)

Table 7-9. Auto-Addressing with Figure 7-22(a), assume all devices are in SHUTDOWN (continued)

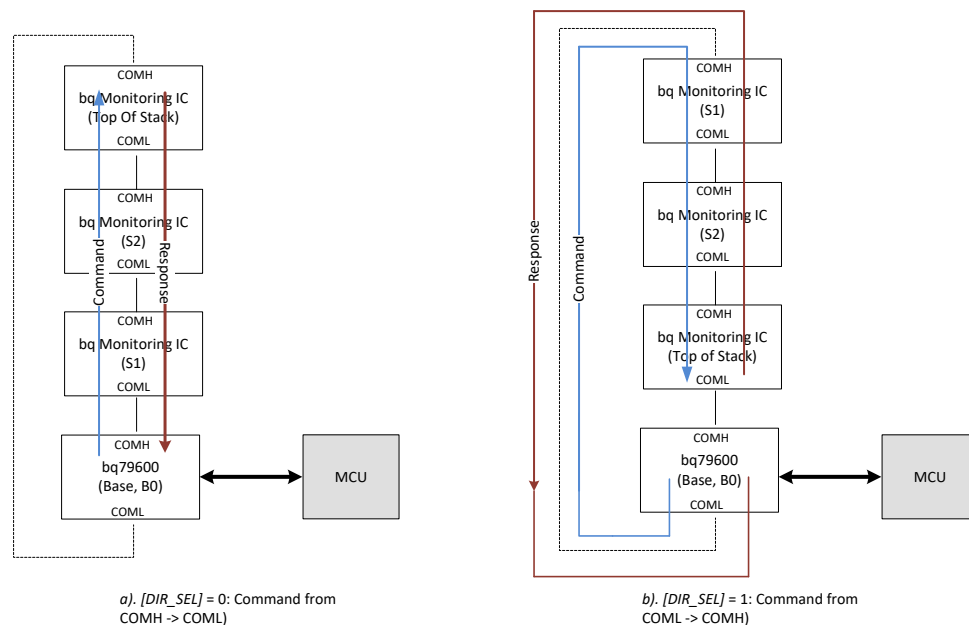
STEP	WORK WITH BQ7961X-Q1
5	brdcast write consecutively to 0x306 = 0,1,2,3 (address 1-3 assigned to BQ7961X-Q1, 0 assigned to BQ79600-Q1)
6	brdcast write 0x02 to address 0x308 (set BQ7961X-Q1 as stack device)
7	single device write to device 3: data 0x03 to address 0x308 (set 3rd BQ7961X-Q1 as top of stack, BQ79600-Q1 is default to base)
8	dummy stack read registers 0x343 to 0x34A (sync up internal DLL). These are 8 separate read commands.
9	stack read address 0x306 (read back to verify address are correct for stack devices)
10	single device read to BQ79600-Q1, verify 0x2001 = 0x14
11	finish initialization

7.3.2.3.2 Ring Communication (optional)

A ring communication (optional) allows the system to establish communication from either direction. This allows the system to continue communicating to all stack devices even if one piece of daisy chain cable is broken.

Table 7-10 describes a procedure auto address Figure 7-23(b): to bring up a system of 1 bridge device and 3 stack devices from SHUTDOWN to a state ready to do read/write communication in **reverse direction**.

To change communication direction from Figure 7-23(a) to Figure 7-23(b), follow the steps 2, 4-14. (Assuming all devices in (a) are already in ACTIVE and auto addressed as described in Table 7-9)

**Figure 7-23. Example to Change Communication Direction in Daisy Chain****Table 7-10. Auto-Addressing Figure 7-22(b), assume all devices are in SHUTDOWN**

STEP	WORK WITH BQ7961X-Q1
1	send WAKE ping on RX (wake up BQ79600-Q1)
2	single device write to BQ79600-Q1 control 1 [DIR_SEL] = 1 (change BQ79600-Q1 direction)
3	single device write to BQ79600-Q1 CONTROL1 [SEND_WAKE] = 1 (wake up stack devices)
4	dummy stack write data 0x00 to registers 0x343 to 0x34A (sync up internal DLL). These are 8 separate write commands.
5	brdcast write reverse 0x80 to address 0x309 (change stack devices direction DIR_SEL =1)
6	brdcast Write 0x02 to address 0x308 ⁽¹⁾
7	brdcast Write 0x81 to address 0x309 (enable BQ7961X-Q1 auto addressing)
8	brdcast Write consecutively to address 0x307 = 0,1,2,3 (address 1-3 assigned to BQ7961X-Q1, 0 assigned to BQ79600-Q1)
9	brdcast write 0x02 to address 0x308 (set BQ7961X-Q1 as stack device)

Table 7-10. Auto-Addressing Figure 7-22(b), assume all devices are in SHUTDOWN (continued)

STEP	WORK WITH BQ7961X-Q1
10	single device write to device 3: data 0x03 to address 0x308 (set 3rd BQ7961X-Q1 as top of stack, BQ79600-Q1 is default to base)
11	dummy stack read registers 0x343 to 0x34A (sync up internal DLL). These are 8 separate read commands.
12	stack read address 0x307 (read back to verify address are correct for stack device)
13	single device read to BQ79600-Q1, verify 0x2001 = 0x14
14	finish initialization

- (1) Clear the previous TOP_STACK flag after communication direction is changed because top of stack device cannot be reached if one cable is broken

7.3.2.4 Communication Timeout

In ACTIVE, there are two programmable communication timeout timers, **comm timeout short** (once expires, flag fault) and **comm timeout long** (once expired, transition to SLEEP or SHUTDOWN). They monitor the absence of a valid frame from either UART/SPI or daisy chain communications. A valid frame is defined as any frame (response or command) that does NOT contain any errors that prevent the frame from being processed.

In SHUTDOWN, the timers are disabled and reset. In SLEEP, the last timer values are held frozen. The timer is reset every time a valid response or command frame is received.

How to set the timer, timer expiration action are described in COMM_TIMEOUT. In order to avoid entering SHUTDOWN mode before a communications timeout fault, ensure the COMM_TIMEOUT [CTS_TIME] is shorter than the COMM_TIMEOUT [CTL_TIME].

7.3.2.5 Communication Debug Mode

The device provides a communication debug mode to ease the initial development phase. Enter/exit debug is controlled by setting of register DEBUG_CTRL_UNLOCK. Once device is in debug mode, user is able to control the UART/daisy chain baud rate and on/off of COMH/COML RX/TX. Please refer to register DEBUG_COMM_CTRL. User can always read DEBUG_COMM_STAT register for comm status disregard the setting/mode of device.

In addition to that, device provides communication low level faults (physical and frame layer) to facilitate debug. Refer to registers from address [0x2301](#) – [0x2307](#).

7.3.3 Fault Handling

7.3.3.1 Fault Status Hierarchy/Reset/Mask

7.3.3.1.1 Fault Status Hierarchy

BQ79600-Q1 reports faults in hierarchy, as shown in [Figure 7-24](#):

- Level 1 is FAULT_SUMMARY register in which each bit represents an OR function of all the bits in its own hierarchy captured in level 2.
- Level 2 bit is the OR function of level 3 bits in its own hierarchy.
- Level 3 contains debug register bits meant to inform host frame and physical layer fault. Level 3 fault is useful in firmware development.
- Any bit triggered in lower level would trigger higher level bit in its hierarchy, e.g. if [TXFIFO_OV] is set, [SPI_PHY], [FAULT_COMM] would also be set.

Note

Host system can periodically poll the FAULT_SUMMARY to check the fault status and only read the lower level fault registers if needed.

7.3.3.1.2 Fault Reset and Mask

Once fault is detected, the fault status bit is latched until cleared using the reset bit.

When a specific fault reset bit is set, the same color coded bits in level 1 to level 3 are cleared if the fault condition is gone. If the fault condition persists and the reset bit is written, the fault status bit is not reset. For example, if [TXFIFO_OV], [DVDD_OV] bits are set, [SPI_PHY], [FAULT_COMM] and [FAULT_PWR] are set, if fault conditions are eliminated and write '1' to [RST_UART_SPI] and [RST_PWR], 5 faults bits would be '0'.

When a specific fault mask bit is set, the same color coded bits would be masked, meaning the fault bits will still be set, but the faults will not be reflected in level 1, FAULT_SUMMARY register. For example, if [MSK_UART_SPI] = 1, any bits being set marked green in level 2 and 3 won't set [FAULT_COMM] bit.

When fault is masked, it will also prevent the device from asserting the NFAULT pin when the masked faults occur. See [Section 7.3.3.2](#) for details.

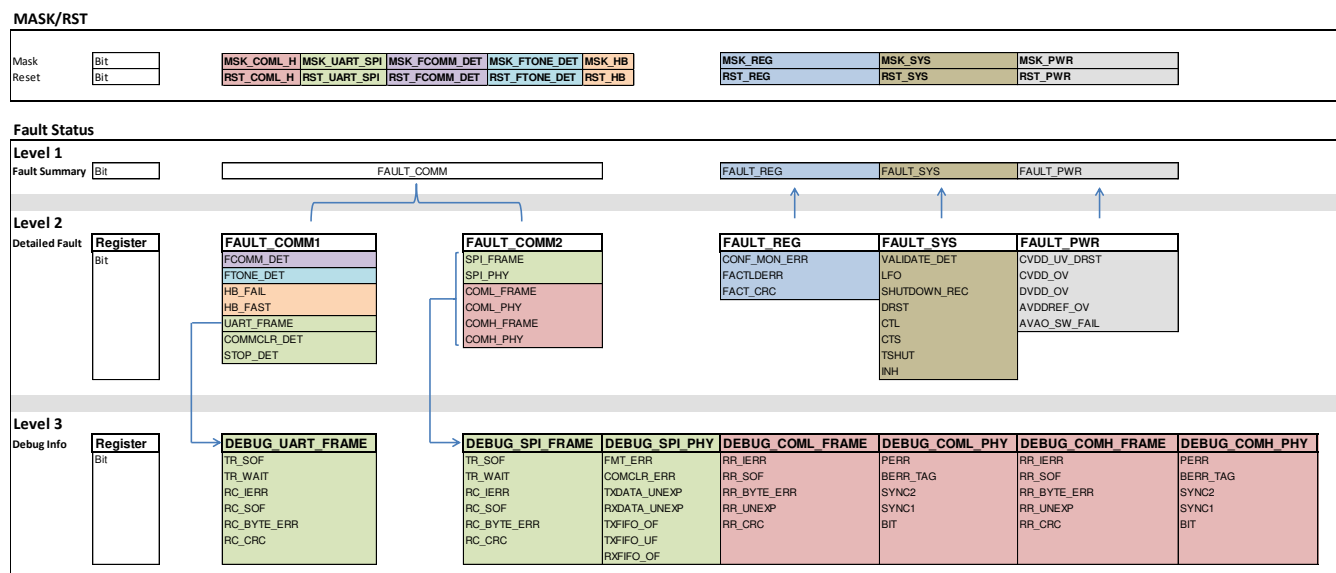


Figure 7-24. Fault Status Hierarchy, Mask and Reset

7.3.3.2 Fault Interface

Host can acquire the fault status with the following two methods:

- Host ignores NFAULT pin, constantly polls the FAULT_SUMMARY register of each device. If FAULT_SUMMARY is non-zero, read the low level fault status registers to obtain more information.
- Host monitors NFAULT pin status. Enable fault status to pass down the daisy chain to bridge device. Enable bridge device's NFAULT pin to be asserted when the FAULT_SUMMARY is non-zero. When NFAULT is triggered, host polls fault information to diagnose further.

7.3.3.2.1 NFAULT

Device integrates an NMOS open-drain output (NFAULT) to signal the MCU that a fault has occurred in the system (either fault from BQ79600-Q1 or from monitoring IC). The NFAULT driver is enabled when [NFAULT_EN] = 1. When BQ79600-Q1 detects an unmasked fault, NFAULT asserts low. When NFAULT is disabled, the device will set the corresponding flag in FAULT_SUMMARY register but will not assert NFAULT.

If the fault information of stack devices are not transmitted to bridge device through Section 7.3.3.2.2, NFAULT output only indicates faults in BQ79600-Q1.

7.3.3.2.2 Daisy Chain (COMH and COML)

When using BQ79600-Q1 NFAULT pin to signal the host under a fault detection, the stack devices have to transfer their fault status information to the base device. The information is transmitted through COMH/L interface through the same communication cables:

- In ACTIVE, BQ79600-Q1 detects embedded fault info in response frame from stack device.
- In SLEEP, stack device sends Heartbeat and Fault tone to BQ79600-Q1.
- In SHUTDOWN, use Sniff Detector of BQ79600-Q1 monitors stack device Fault tone.

7.3.3.2.2.1 Fault Transmitting when BQ79600-Q1 in ACTIVE

In ACTIVE mode, stack devices can embed their fault status in their response frames (refer to Figure 7-7) that are sent to BQ79600-Q1. The BQ79600-Q1 can detect their embedded fault info and sets [FTONE_DET] bit once criteria in Figure 7-25 is met. Please refer to BQ7961X-Q1 on how to use embed fault feature.

To pass on the fault status of the stack devices, the host sends a stack read which will result with response frame pass through every device in the daisy chain, giving each device an opportunity to embed their fault status to response frame.

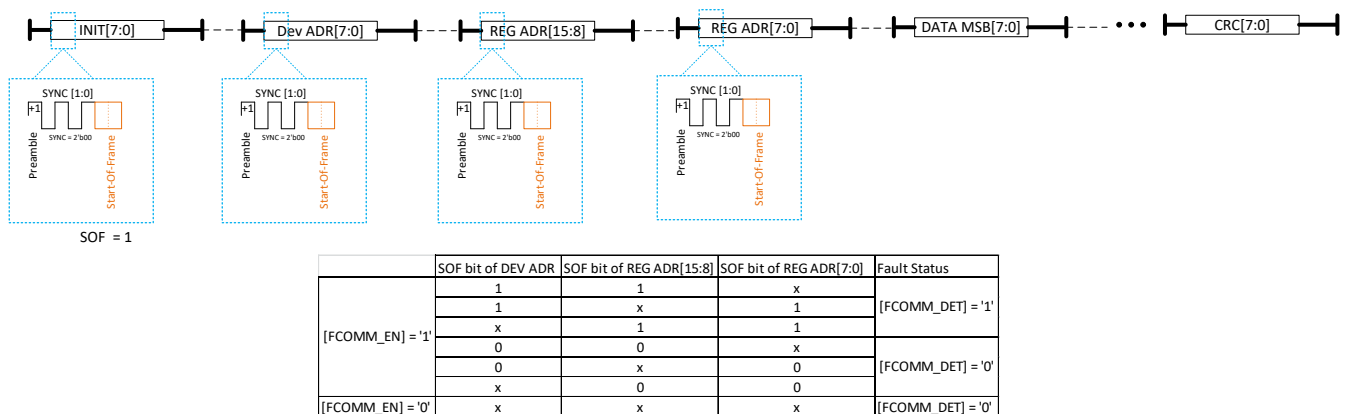


Figure 7-25. Embed Fault Detection in Response Frame

7.3.3.2.2.2 Fault Transmitting when BQ79600-Q1 in SLEEP

Because data communication is not available in SLEEP mode, the device provides following options to transmitting fault information:

- Transmit the Heartbeat tone (enabled by [HB_TX_EN], used to check integrity of cable between bridge and first stack device). Device does not transmitted fault tone as it has NFAULT.

- Detect Heartbeat and Fault tone, enabled, by [TONE_RX_EN].

These tones are transmitted in the same direction as a communication command frame, which is based on the CONTROL1[DIR_SEL] setting. For the tone signal to return back to BQ79600-Q1 (so NFAULT can be triggered if needed), a Ring architecture must be used to support transmitting fault status in SLEEP mode.

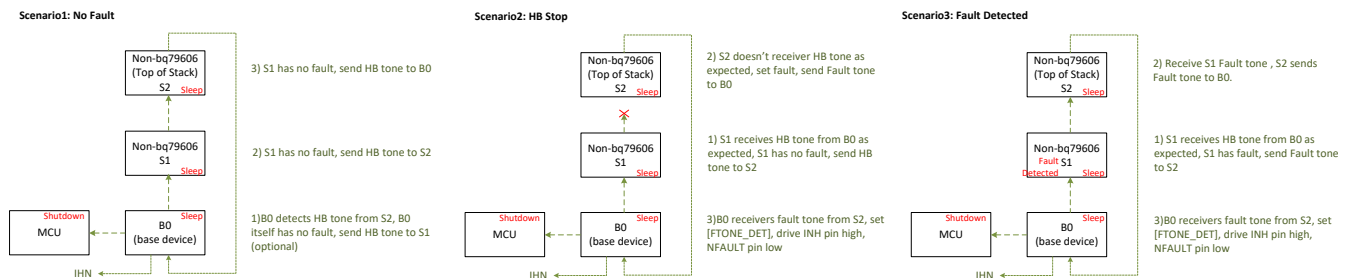


Figure 7-26. Heartbeat and Fault Tone Examples

Both the Heartbeat and Fault Tone are a type of tone similar to the communication. One main difference is a communication tone only transmits with a single burst of couplets, while Heartbeat and Fault Tones are sent with a burst of couplets periodically. See [Figure 7-21](#) for details.

7.3.3.2.2.3 Fault Transmitting (Automatic Host Wakeup/Reverse Wakeup) when BQ79600-Q1 in SHUTDOWN

Note

This feature (Auto Host Wakeup/Reverse Wakeup) is only available if [Section 7.3.5](#) is enabled.

The purpose of this user case is to keep BQ79600-Q1 in lowest power mode while still being able to detect fault information from stack devices. In this case, fault information transmission is similar to that of SLEEP: top of stack device sends HB or Fault tone to BQ79600-Q1. The difference lies in the detection of those tones in BQ79600-Q1. In SHUTDOWN, TONE RX is off, only low power [Section 7.3.5](#) is available. Once sniffer detects FAULT tone, it puts device into VALIDATE mode in which full power TONE RX is available, device would validate if true Fault tone exists or not. If yes, it triggers INH. See [Figure 7-27](#) for different case.

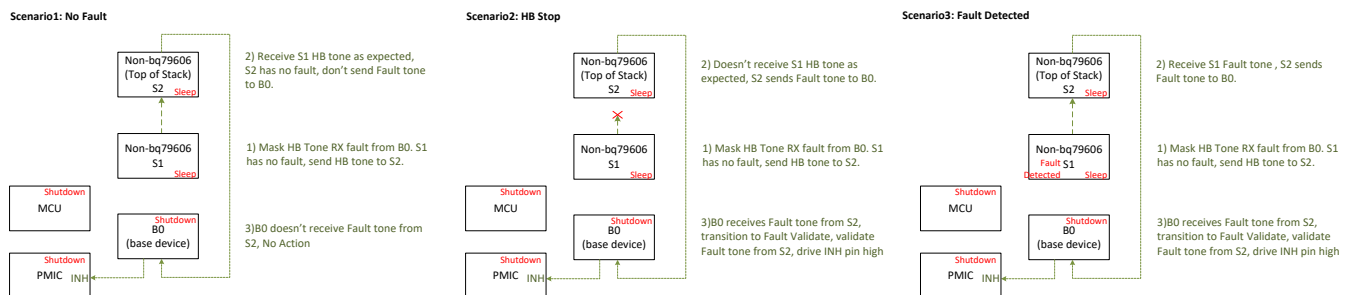


Figure 7-27. Reverse Wakeup User Case

7.3.4 INH/ Reverse Wakeup

Note

INH pin is used if Reverse Wakeup feature is used. If this feature is not used, connect this pin to BAT pin, refer to schematic in [Figure 8-1](#).

Reverse wakeup feature is a mechanism where BQ79600-Q1 can wakeup the host, through INH pin, on faulty status from either BQ79600-Q1 or stack devices like BQ7961X-Q1. MCU and its supply (PMIC/SBC) are in SHUTDOWN for power saving on low voltage battery side.

The INH pin is a high voltage output pin that provides voltage from the BAT minus V_{DROP_INH} to enable an external high voltage regulators (SBC, PMIC). These regulators are usually used to support the microprocessor and VIO pin. When INH PMOS pullup is not activated, INH pin goes to a high Z state, it relies on external circuit to define the pin voltage (in application circuit, 100kohm resistor to GND is used.)

INH PMOS pullup can be triggered:

- In SLEEP mode or VALIDATE mode if following faults are detected regardless of setting of register FAULT_MSK: [FTONE_DET], [HB_FAIL], [HB_FAST], [AVAO_SW_FAIL], [FACT_CRC], [CONF_MON_ERR].
- In ACTIVE, INH can only be triggered by setting [INH_SET_GO] = 1.

Once INH triggered, it remains latched in all modes as long as VBAT is not removed.

INH function described above can be disabled by configuring INH_DIS[1:0] = 2'b11.

Every time INH PMOS is activated, fault bit [INH] is set. To clear the fault, set INH_DIS[1:0] = 2'b11 (disarm INH driver), then write [RST_SYS] = 1. After this, to use INH feature, set INH_DIS[1:0] = 2'b00.

As part of safety diagnostic (SM202 in Safety Manual), host can trigger INH in ACTIVE and check if pin voltage is set properly: If INH pin voltage is higher than V_{INH_DET} , [INH_STAT] = 1.

Note

INH pin should be considered a "high voltage logic" terminal, thus should be used to drive the EN terminal of the system's power management device. It should be not used as a switch for power management supply itself. This terminal is not reverse battery protected and thus should not be connected outside of the system module.

7.3.5 Sniff Detector

Note

Sniff detector is only used if reverse wakeup feature is used.

Sniff detector, powered by AVAOREF, is used to detect fault tone from stack device on COMH* or COML*. This detector would trigger if it detects/counts no less than nVALIDATE of continuous couplets (either “-” or “+”) with amplitude larger than V_{VAL_THR} . If any couplet timing interval is larger than $t_{SNIFFIDLE}$, detector/counter is reset.

This sniff detector rejects HB tone since nVALIDATE is more than 30, both HB/Fault tone are “-” tone; detector doesn’t expect “+” tone.

Note

The usage assumption of this detector is when system is in idle mode, BQ79600-Q1 in SHUTDOWN.

Sniff detector is only effective in SHUTDOWN. Once detector is triggered, device transitions from SHUTDOWN to VALIDATE. The sniff detector is by default disabled when first transition from COMPLETE OFF to SHUTDOWN. To enable the feature, host has to keep [SNIFDET_EN] = ‘1’ & [SNIFDET_DIS] = ‘0’ before transitioning to SHUTDOWN. After enabling the detector, if device doesn’t transition to COMPLETE OFF, the only way to disable the detector is to keep [SNIFDET_DIS] = ‘1’. (Disable bit has higher priority so don’t care about the setting of [SNIFDET_EN]) before transitioning to SHUTDOWN.)

7.3.6 Device Diagnostic

The product is developed as a safety element out of context (SEooC), with a target safety goal of ASIL-D for communication. The following sub-sections describe the diagnostic control and fault status that can be used as part of the safety mechanisms.

The Safety Manual and FMEDA for BQ79600-Q1 are available separately from Texas Instruments. Contact TI Sales Associate or Applications Engineer for further information.

7.3.6.1 Power Supplies Check

7.3.6.1.1 Power Supply Diagnostic Check

The internal power supply circuits have overvoltage, undervoltage, and/or current limit checks. The table below summarizes the diagnostics that apply for each power supply and the corresponding action when failure is detected.

Table 7-11. Power Supply Diagnostic Summary

SUPPLY	OV CHECK	UV CHECK	CURRENT LIMIT
DVDD	If fails, set [DVDD_OV]	If fails, digital reset	Limit current to EC table current limit specification
CVDD	If fails, set [CVDD_OV]	If fails, digital reset, set [CVDD_UV_DRST]	Limit current to EC table current limit specification
AVAOREF	If fails, set [AVDDREF_OV] (AVAOREF supplies AVDDREF)	If fails, device shuts down	
AVDDREF	If fails, set [AVDDREF_OV]	If fails, set [AVAO_SW_FAIL] or digital reset	
VIO		If fails, device losses data communication and Ping function	

7.3.6.1.2 Power Supply BIST

The device implemented a power supply BIST (Built-In Self-Test) function to test CVDD, DVDD, AVDDREF OV detection comparator integrity. It is a command base function initiated by host. Steps below explains how it works, and further details can be found in Safety Manual (SM017).

1. Host shall read the register FAULT_PWR to verify [CVDD_OV], [DVDD_OV], [AVDDREF_OV] are low.
2. Host shall write [PWR_DIAG_GO] = 1.
3. After 1.7ms, host shall read if [PWR_DIAG_RDY] = 1, else shall, keep waiting, reread.
4. If yes, host shall read FAULT_PWR register, [CVDD_OV], [DVDD_OV], [AVDDREF_OV] to verify the bits are asserted.
5. Host shall reset faults above.

7.3.6.2 Thermal Shutdown

Thermal shutdown (TSHUT) event occurs when the Thermal Shutdown sensor value exceeds the thermal shutdown temperature threshold. The sensor operates without interaction and is separated from the ADC measured die sensor. The thermal shutdown function has a register-status indicator flag (*FAULT_SYS[TSHUT]*) that is saved during the shutdown event and can be read after the device is waken back up. When a TSHUT fault occurs, the device immediately enters the SHUTDOWN mode. Any pending transactions on UART or daisy chain are discarded. There is no fault signaling when a thermal shutdown event occurs, as the device immediately shuts down.

To awaken the device, host shall ensure the ambient temperature is below T_{SD_FALL} and sends a WAKE ping to the base device. Host shall not attempt to wake the device if the ambient temperature is still above T_{SD_FALL} .

Upon waking up, the *FAULT_SYS[TSHUT]* bit is set. The *FAULT_SYS[SHUTDOWN_REC]* = 1 indicating the prior shutdown was caused by abnormal event. See [Section 7.5.17](#) for more details. If the system faults are unmasked, *FAULT_MSK1[MSK_SYS]* = 0, the thermal shutdown will be reflected as FAULT and will be indicated in the *FAULT_SUMMARY* register and the assertion of the NFAULT pin.

7.3.6.3 Oscillators Watchdog

The oscillators are monitored by watchdog circuits. There are two oscillators in the device, the HFO and the LFO. If these oscillators are not functioning, the device does not operate. If the HFO or LFO does not transition within the expected time, the watchdog circuits trigger Digital Reset.

When such unexpected reset occurs, it is recommended for the host to send a SHUTDOWN ping/tone to the problem device and follow up a WAKE ping to reset the daisy chain. If the oscillators are truly damaged, the device will not restart and must be replaced.

In addition to the watchdog, the LFO frequency is monitored to ensure it stays within acceptable limits. If the LFO frequency falls outside of the expected range, the *FAULT_SYS_FAULT[LFO]* bit is set.

7.3.6.4 Register Bit Flip Monitor

This bit flip checker monitors 2 configuration registers: DEV_CONF1, FAULT_MSK. It is always running when device is out of SHUTDOWN. Whenever user changes those 2 register settings or any of the register bit flips, fault bit [CONF_MON_ERR] is set.

Once user changes the setting, user shall write [CONF_MON_GO]=1 (resample 2 register values), write [RST_REG] =1 to clear the [CONF_MON_ERR] fault, after this point, if any bit flips among those 2 registers, [CONF_MON_ERR] is set. After device resets (receive WAKE ping or [SOFT_RESET] = 1), [CONF_MON_ERR] = 0.

This device does not have customer register CRC check and the register bit flip monitor provides the protection for the above mentioned customer registers.

7.3.6.5 SPI FIFO Diagnostic

The FIFO diagnostic mode provides a method for host to exercise the RX/TX FIFOs. Please refer to safety manual for detailed FIFO diagnostic safety mechanism implementation (SM132).

7.4 Device Functional Modes

See [Section 7.3.1](#)

7.5 Register Maps

	Addr Hex	R/W Type	Reset Value	Data							
				Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
DIR0_ADDR	306	R/W	"0000 0000"	RSVD	RSVD	ADDRESS[5]	ADDRESS[4]	ADDRESS[3]	ADDRESS[2]	ADDRESS[1]	ADDRESS[0]
DIR1_ADDR	307	R/W	"0000 0000"	RSVD	RSVD	ADDRESS[5]	ADDRESS[4]	ADDRESS[3]	ADDRESS[2]	ADDRESS[1]	ADDRESS[0]
CONTROL1	309	R/W-AC	"0000 0000"	DIR_SEL	SEND_SHUTDOWN	SEND_WAKE	SEND_SLPTOACT	GOTO_SHUTDOWN	GOTO_SLEEP	SOFT_RESET	ADDR_WR
CONTROL2	30A	R/W-AC	"0000 0000"	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	SEND_HW_RESET	RSVD
DIAG_CTRL	2000	R/W-AC	"0000 0000"	RSVD	RSVD	CONF_MON_GO	PWR_DIAG_GO	SPL_FIFO_DIAG_GO	FLIP_FACT_CRC	FLIP_TR_CRC	INH_SET_GO
DEV_CONF1	2001	R/W	"0001 0100"	SNIFDET_EN	SNIFDET_DIS	TONE_RX_EN	FCOMM_EN	TWO_STOP_EN	NFAULT_EN	RESERVED	HB_TX_EN
DEV_CONF2	2002	R/W	"0000 0000"	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	INH_DIS[1]	INH_DIS[0]
TX_HOLD_OFF	2003	R/W	"0000 0000"	DLY[7]	DLY[6]	DLY[5]	DLY[4]	DLY[3]	DLY[2]	DLY[1]	DLY[0]
SLP_TIMEOUT	2004	R/W	"0000 0011"	RSVD	RSVD	RSVD	RSVD	RSVD	SLP_TIME[2]	SLP_TIME[1]	SLP_TIME[0]
COMM_TIMEOUT	2005	R/W	"0011 0100"	RSVD	CTS_TIME[2]	CTS_TIME[1]	CTS_TIME[0]	CTL_ACT	CTL_TIME[2]	CTL_TIME[1]	CTL_TIME[0]
SPL_FIFO_UNLOCK	2010	R/W	"0000 0000"	RSVD	RSVD	RSVD	RSVD	CODE[3]	CODE[2]	CODE[1]	CODE[0]
FAULT_MSK	2020	R/W	"0000 0000"	MSK_COML_H	MSK_UART_SPI	MSK_FCOMM_DET	MSK_FTONE_DET	MSK_HB	MSK_REG	MSK_SYS	MSK_PWR
FAULT_RST	2030	R/W-AC	"0000 0000"	RST_COML_H	RST_UART_SPI	RST_FCOMM_DET	RST_FTONE_DET	RST_HB	RST_REG	RST_SYS	RST_PWR
FAULT_SUMMARY	2100	R	"0000 0000"	RSVD	RSVD	RSVD	RSVD	FAULT_COMM	FAULT_REG	FAULT_SYS	FAULT_PWR
FAULT_REG	2101	R	"0000 0000"	RSVD	RSVD	RSVD	RSVD	RSVD	CONF_MON_ERR	FACTLDERR	FACT_CRC
FAULT_SYS	2102	R	"0000 0000"	VALIDATE_DET	LFO	SHUTDOWN_REC	DRST	CTL	CTS	TSHUT	INH
FAULT_PWR	2103	R	"0000 0000"	RSVD	RSVD	RSVD	CVDD_UV_DRST	CVDD_OV	DVDD_OV	AVDDREF_OV	AVAO_SW_FAIL
FAULT_COMM1	2104	R	"0000 0000"	RSVD	FCOMM_DET	FTONE_DET	HB_FAIL	HB_FAST	UART_FRAME	COMMCLR_DET	STOP_DET
FAULT_COMM2	2105	R	"0000 0000"	RSVD	SPI_FRAME	SPI_PHY	COML_FRAME	COML_PHY	COML_FRAME	COML_PHY	COMH_PHY
DEV_DIAG_STAT	2110	R	"0000 0000"	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	PWR_DIAG_RDY	INH_STAT
PARTID	2120	R	"0000 0000"	REV[7]	REV[6]	REV[5]	REV[4]	REV[3]	REV[2]	REV[1]	REV[0]
DIE_ID1	2121	R	"0000 0000"	ID[7]	ID[6]	ID[5]	ID[4]	ID[3]	ID[2]	ID[1]	ID[0]
DIE_ID2	2122	R	"0000 0000"	ID[7]	ID[6]	ID[5]	ID[4]	ID[3]	ID[2]	ID[1]	ID[0]
DIE_ID3	2123	R	"0000 0000"	ID[7]	ID[6]	ID[5]	ID[4]	ID[3]	ID[2]	ID[1]	ID[0]
DIE_ID4	2124	R	"0000 0000"	ID[7]	ID[6]	ID[5]	ID[4]	ID[3]	ID[2]	ID[1]	ID[0]
DIE_ID5	2125	R	"0000 0000"	ID[7]	ID[6]	ID[5]	ID[4]	ID[3]	ID[2]	ID[1]	ID[0]
DIE_ID6	2126	R	"0000 0000"	ID[7]	ID[6]	ID[5]	ID[4]	ID[3]	ID[2]	ID[1]	ID[0]
DIE_ID7	2127	R	"0000 0000"	ID[7]	ID[6]	ID[5]	ID[4]	ID[3]	ID[2]	ID[1]	ID[0]
DIE_ID8	2128	R	"0000 0000"	ID[7]	ID[6]	ID[5]	ID[4]	ID[3]	ID[2]	ID[1]	ID[0]
DIE_ID9	2129	R	"0000 0000"	ID[7]	ID[6]	ID[5]	ID[4]	ID[3]	ID[2]	ID[1]	ID[0]
DEBUG_CTRL_UNLOCK	2200	R/W	"0000 0000"	CODE[7]	CODE[6]	CODE[5]	CODE[4]	CODE[3]	CODE[2]	CODE[1]	CODE[0]
DEBUG_COMM_CTRL	2201	R/W	"0111 1000"	RSVD	COML_TX_EN	COML_RX_EN	COMH_TX_EN	COMH_RX_EN	UART_VIF_BAUD	USER_UART_EN	USER_DAISSY_EN
DEBUG_COMM_STAT	2300	R	"0001 0011"	RSVD	RSVD	RSVD	HW_DAISSY_DRV	COML_TX_ON	COML_RX_ON	COMH_TX_ON	COMH_RX_ON
DEBUG_SPI_PHY	2301	R	"0000 0000"	RSVD	FMT_ERR	COMCLR_ERR	TXDATA_UNEXP	RXDATA_UNEXP	TXFIFO_OF	TXFIFO_UF	RXFIFO_OF
DEBUG_SPI_FRAME	2302	R	"0000 0000"	RSVD	RSVD	TR_SOF	TR_WAIT	RC_IERR	RC_SOF	RC_BYTE_ERR	RC_CRC
DEBUG_UART_FRAME	2303	R	"0000 0000"	RSVD	RSVD	TR_SOF	TR_WAIT	RC_IERR	RC_SOF	RC_BYTE_ERR	RC_CRC
DEBUG_COMH_PHY	2304	R	"0000 0000"	RSVD	RSVD	RSVD	PERR	BERR_TAG	SYNC2	SYNC1	BIT
DEBUG_COMH_FRAME	2305	R	"0000 0000"	RSVD	RSVD	RSVD	RR_IERR	RR_SOF	RR_BYTE_ERR	RR_UNEXP	RR_CRC
DEBUG_COML_PHY	2306	R	"0000 0000"	RSVD	RSVD	RSVD	PERR	BERR_TAG	SYNC2	SYNC1	BIT
DEBUG_COML_FRAME	2307	R	"0000 0000"	RSVD	RSVD	RSVD	RR_IERR	RR_SOF	RR_BYTE_ERR	RR_UNEXP	RR_CRC

Figure 7-28. Register Summary

7.5.1 Register Summary Table

ADDRESS	REGISTER	DESCRIPTION	R/W MODE	RESET VALUE	NVM VALUES	INCLUDED IN NVM?
0x306	DIR0_ADDR	Device Address North Direction	R/W	0x00	n/a	no
0x307	DIR1_ADDR	Device Address South Direction	R/W	0x00	n/a	no
0x309	CONTROL1	Control1	R/W	0x00	n/a	no
0x30A	CONTROL2	Control2	R/W	0x00	n/a	no
0x2000	DIAG_CTRL	Diagnostic Control	R/W	0x00	n/a	no
0x2001	DEV_CONF1	Device Configure1	R/W	0x14	n/a	no
0x2002	DEV_CONF2	Device Configure2	R/W	0x00	n/a	no
0x2003	TX_HOLD_OFF	Transmitter Hold off Control	R/W	0x00	n/a	no
0x2004	SLP_TIMEOUT	Sleep Timer	R/W	0x03	n/a	no
0x2005	COMM_TIMEOUT	Communication Timeout Control	R/W	0x34	n/a	no
0x2010	SPI_FIFO_UNLOCK	FIFO Diagnostic Unlock	R/W	0x00	n/a	no
0x2020	FAULT_MSK	Fault Mask	R/W	0x00	n/a	no
0x2030	FAULT_RST	Fault Reset	R/W	0x00	n/a	no
0x2100	FAULT_SUMMARY	Fault Summary	R	0x00	n/a	no
0x2101	FAULT_REG	Register Fault	R	0x00	n/a	no
0x2102	FAULT_SYS	System Fault	R	0x00	n/a	no
0x2103	FAULT_PWR	Power Fault	R	0x00	n/a	no
0x2104	FAULT_COMM1	Communication Fault1	R	0x00	n/a	no
0x2105	FAULT_COMM2	Communication Fault2	R	0x00	n/a	no
0x2110	DEV_DIAG_STAT	Diagnostic Status	R	0x00	n/a	no
0x2120	PARTID	Part ID	R	0x00	various	yes
0x2121	DIE_ID1	Die ID1	R	0x00	various	yes
0x2122	DIE_ID2	Die ID2	R	0x00	various	yes
0x2123	DIE_ID3	Die ID3	R	0x00	various	yes
0x2124	DIE_ID4	Die ID4	R	0x00	various	yes
0x2125	DIE_ID5	Die ID5	R	0x00	various	yes
0x2126	DIE_ID6	Die ID6	R	0x00	various	yes
0x2127	DIE_ID7	Die ID7	R	0x00	various	yes
0x2128	DIE_ID8	Die ID8	R	0x00	various	yes
0x2129	DIE_ID9	Die ID9	R	0x00	various	yes
0x2200	DEBUG_CTRL_UNLOCK	Debug Control Unlock	R/W	0x00	n/a	no
0x2201	DEBUG_COMM_CTRL	Debug Communication Control	R/W	0x78	n/a	no
0x2300	DEBUG_COMM_STAT	Debug Communication Status	R	0x13	n/a	no
0x2301	DEBUG_SPI_PHY	SPI Physical Layer Error	R	0x00	n/a	no
0x2302	DEBUG_SPI_FRAME	SPI Frame Layer Error	R	0x00	n/a	no
0x2303	DEBUG_UART_FRAME	UART Frame Layer Error	R	0x00	n/a	no
0x2304	DEBUG_COMH_PHY	COMH Physical Layer Error	R	0x00	n/a	no
0x2305	DEBUG_COMH_FRAME	COMH Frame Layer Error	R	0x00	n/a	no
0x2306	DEBUG_COML_PHY	COML Physical Layer Error	R	0x00	n/a	no
0x2307	DEBUG_COML_FRAME	COML Frame Layer Error	R	0x00	n/a	no

7.5.2 Register: DIR0_ADDR

Address: 0x306							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD		ADDRESS[5:0]					
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
ADDRESS [5:0]	Follow steps in section "Device Addressing" to config this register. Always shows the current Device Address used by the device when [DIR_SEL] = 0. Default to 0x00 when digital core out of digital reset. MCU can re-address the device by writing a different device address to this register, and the device will take on the new address immediately.						

7.5.3 Register: DIR1_ADDR

Address: 0x307							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD		ADDRESS[5:0]					
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
ADDRESS [5:0]	Follow steps in section "Device Addressing" to config this register. Always shows the current Device Address used by the device when [DIR_SEL] = 1. Default to 0x00 when digital core out of digital reset. MCU can re-address the device by writing a different device address to this register, and the device will take on the new address immediately.						

7.5.4 Register: CONTROL1

Address: 0x0309							
B7	B6	B5	B4	B3	B2	B1	B0
DIR_SEL	SEND_SHUTDO WN	SEND_WAKE	SEND_SLPTOAC T	GOTO_SHUTDO WN	GOTO_SLEEP	SOFT_RESET	ADDR_WR
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
DIR_SEL	Select daisy chain comm direction. Not self-clear bit. 0: In a device, command frame travels from MCU to COMH 1: In a device, command frame travels from MCU to COML						
SEND_SHUTDO WN	Sends SHUTDOWN tone to next device up the stack. The device receiving this bit is unaffected. Self-clear bit. 0: Ready 1: Send SHUTDWON tone up the stack						
SEND_WAKE	Send WAKE tone up the stack. Self-clear bit. 0: Ready 1: Send WAKE tone up the stack, then reset its own						
SEND_SLPTOAC T	Send SLEEPtoWAKE tone up the stack. Self-clear bit. 0: Ready 1: Send SLEEPtoWAKE tone up the stack						
GOTO_SHUTDO WN	Transition device to SHUTDOWN mode. Self-clear bit. 0: Ready 1: Enter SHUTDOWN mode						
GOTO_SLEEP	Transition device to SLEEP mode. Self-clear bit. 0: Ready 1: Enter SLEEP mode						
SOFT_RESET	Reset the digital to OTP default. Self-clear bit. 0: Ready 1: Reset device						
ADDR_WR	Enable device to start auto-addressing. See auto-addressing chapter for detail. Self cleared. 0: Not performing auto-address. Device forwards communication transaction as normal 1: Device is being auto-addressed; the 1st communication transaction it received will not be forwarded.						

7.5.5 Register: CONTROL2

Address: 0x30A							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD						SEND_HW_RESET	SPARE
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
SEND_HW_RESET	Send HW_RESET tone up the stack. Self-clear bit. 0: Ready 1: Send HW_RESET tone to next stack device up						

7.5.6 Register: DIAG_CTRL

Address: 0x2000							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD		CONF_MON_GO	PWR_DIAG_GO	SPI_FIFO_DIAG_GO	FLIP_FACT_CRC	FLIP_TR_CRC	INH_SET_GO
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
CONF_MON_GO	Resample the 2 registers (DEV_CONF1, FAULT_MSK) setting that bit flip monitor checks against. self-clear bit. For usage, refer to paragraph: Register Bit Flip Monitor 0: No action 1: Resample 2 registers setting						
PWR_DIAG_GO	Indicates a power supply BIST diagnostic is initiated, self-clear bit. 0: No action 1: Initiate power BIST diagnostic						
SPI_FIFO_DIAG_GO	Write the unlock code 0x0A to SPI_FIFO_UNLOCK and followed by writing [SPI_FIFO_DIAG_GO] = 1 to do FIFO diagnostic. For detailed steps of FIFO diagnostic, refer to safety manual. This bit is self-cleared. 0: No action 1: Initiate SPI FIFO diagnostic						
FLIP_FACT_CRC	An enable bit to flip the factory CRC expected value. This is for factory CRC diagnostic. 0: No action 1: flip the CRC expected value. This will cause a factory CRC fault, FAULT_REG[FACT_CRC]						
FLIP_TR_CRC	Sends a purposely incorrect communication (during transmitting response) CRC by inverting all of the calculated CRC bits 0: Send CRC as calculated 1: Send inverted CRC						
INH_SET_GO	This bits intentionally activates INH PMOS pull up, sets [INH] and [INH_STAT] to 1. (self-cleared) 0: mission mode 1: Trigger INH PMOS pull up						

7.5.7 Register: DEV_CONF1

Address: 0x2001							
B7	B6	B5	B4	B3	B2	B1	B0
SNIFDET_EN	SNIFDET_DIS	TONE_RX_EN	FCOMM_EN	TWO_STOP_EN	NFAULT_EN	RESERVED	HB_TX_EN
0	0	0	1	0	1	0	0
RW	RW	RW	RW	RW	RW	RW	RW
SNIFDET_EN	Enable the Sniff detector on COM* port, this bit is latched into AVAOREF domain, once latched in, this bit is still effective in SHUTDOWN 0: no effect 1: SNIF DET is enabled in SHUTDOWN						
SNIFDET_DIS	Enable the Sniff detector on COM* port, this bit is latched into AVAOREF domain, once latched in, this bit is still effective in SHUTDOWN 0: no effect 1: SNIF DET is disabled in SHUTDOWN, if both SNIFDET_EN and SNIFDET_DIS are '1', SNIFDET_DIS takes priority.						
TONE_RX_EN	Enable the Tone receiver, depends on [DIR_SEL], one of the COML/COMH tone recover is enabled. COMH/COML tone receivers are enabled in VALIDATE disregarding setting of this bit. 0: Disable 1: Enable						
FCOMM_EN	Enable the fault state detection through communication in ACTIVE mode 0: Disable 1: Enable						
TWO_STOP_EN	Enables two stop bits for the UART in case of severe oscillator error in both the host and device 0: One STOP bit 1: Two STOP bit						
NFAULT_EN	Enables the NFAULT function 0: NFAULT driver is disabled 1: NFAULT pulls low to indicate an unmasked fault is detected						
RESERVED	Reserved. Default value is 0. Please don't alter.						
HB_TX_EN	Enable HEARTBEAT transmitter when device is in SLEEP mode 0: Disable 1: Enable						

7.5.8 Register: DEV_CONF2

Address: 0x2002							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD						INH_DIS[1:0]	
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
INH_DIS[1:0]	Disable INH driver (PMOS pull up). In all modes, this bit overwrites [INH_SET_GO] and fault tone detection event 00: INH function is enabled 01: INH function is enabled 10: INH function is enabled 11: INH function is DISABLED						

7.5.9 Register: TX_HOLD_OFF

Address: 0x2003							
B7	B6	B5	B4	B3	B2	B1	B0
DLY[7:0]							
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
DLY[7:0]	Set the number of bit period delay from 0 to 255, after receiving the STOP bit of a command frame and before transmitting the 1st bit of response frame						

7.5.10 Register: SLP_TIMEOUT

Address: 0x2004							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD				SLP_TIME[2:0]			
0	0	0	0	0	0	1	1
RW	RW	RW	RW	RW	RW	RW	RW
SLP_TIME[2:0]	This timer starts counting when device enters SLEEP or VALIDATE. When the timer expires, the device enters SHUTDOWN. The timer reset if device wakes up to ACTIVE. 000: no timeout. Device remains in SLEEP 001: 5s 010: 10s 011: 1min 100: 10mins 101: 30mins 110: 1hour 111: 2hours						

7.5.11 Register: COMM_TIMEOUT

Address: 0x2005							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD	CTS_TIME[2:0]			CTL_ACT	CTL_TIME[2:0]		
0	0	1	1	0	1	0	0
RW	RW	RW	RW	RW	RW	RW	RW
CTS_TIME[2:0]	Set the short communication timeout. When this timer expires, the device set the FAULT_SYS [CTS] bit. This can be used as an alert to the system to prevent a long communication timeout. 000: disable short communication timeout 001: 100ms 010: 2s 011: 10s (default) 100: 1min 101: 10mins 110: 30min 111: 1hour						
CTL_ACT	Configure the device action when long communication timeout timer expires 0: set FAULT_SYS[CTL] and send device to SLEEP mode (default at reset) 1: Send the device to SHUTDOWN. FAULT_SYS[CTL] bit will not be set						
CTL_TIME[2:0]	Set the long communication timeout. When this timer expires, the device takes the action configured by the [CTL_ACT] bit. 000: disable long communication timeout 001: 100ms 010: 2s 011: 10s 100: 1min (default) 101: 10mins 110: 30min 111: 1hour						

7.5.12 Register: SPI_FIFO_UNLOCK

Address: 0x2010							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD				CODE[3:0]			
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
CODE[3:0]	In UART mode - write has no impact and read always returns 0 In SPI mode - Write the unlock code 0x0A to SPI_FIFO_UNLOCK (MSB 4 bits are don't care, e.g. 0x2A would also unlock) and followed by writing [SPI_FIFO_DIAG_GO] = 1 to do FIFO diagnostic. For detailed steps of FIFO diagnostic, refer to safety manual. After these bits are written, read cmd doesn't affect them while any write cmd clears them.						

7.5.13 Register: FAULT_MSK

Address: 0x2020							
B7	B6	B5	B4	B3	B2	B1	B0
MSK_COML_H	MSK_UART_SPI	MSK_FCOMM_DET	MSK_FTONE_DET	MSK_HB	MSK_REG	MSK_SYS	MSK_PWR
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
MSK_COML_H	If this bit is set, [FAULT_COMM] bit in FAULT_SUMMARY register is not set (NFAULT won't toggle) due to bits: [COML_FRAME], [COMH_FRAME], [COML_PHY], [COMH_PHY] and registers: DEBUG_COMH_PHY, DEBUG_COMH_FRAME, DEBUG_COML_PHY, DEBUG_COML_FRAME 0: Assert NFAULT and set [FAULT_COMM] if fault above is detected 1: No NFAULT action, [FAULT_COMM] not set due to faults above						
MSK_UART_SPI	If this bit is set, [FAULT_COMM] bit in FAULT_SUMMARY register is not set (NFAULT won't toggle) due to [STOP_DET], [COMMCLR_DET], [UART_FRAME], [SPI_FRAME], [SPI_PHY] bits and DEBUG_UART_FRAME, DEBUG_SPI_PHY, DEBUG_SPI_FRAME registers 0: Assert NFAULT and set [FAULT_COMM] if fault above is detected 1: No NFAULT action, [FAULT_COMM] not set due to faults above						
MSK_FCOMM_DET	If this bit is set, [FAULT_COMM] bit in FAULT_SUMMARY register is not set (NFAULT won't toggle) due to [FCOMM_DET] 0: Assert NFAULT and set [FAULT_COMM] if fault above is detected 1: No NFAULT action, [FAULT_COMM] not set due to faults above						
MSK_FTONE_DET	If this bit is set, [FAULT_COMM] bit in FAULT_SUMMARY register is not set (NFAULT won't toggle) due to [FTONE_DET] 0: Assert NFAULT and set [FAULT_COMM] if fault above is detected 1: No NFAULT action, [FAULT_COMM] not set due to faults above						
MSK_HB	If this bit is set, [FAULT_COMM] bit in FAULT_SUMMARY register is not set (NFAULT won't toggle) due to [HB_FAIL] and [HB_FAST] 0: Assert NFAULT and set [FAULT_COMM] if fault above is detected 1: No NFAULT action, [FAULT_COMM] not set due to faults above						
MSK_REG	If this bit is set, [FAULT_REG] bit in FAULT_SUMMARY register is not set and NFAULT won't toggle due to OTP fault. It doesn't affect FAULT_REG register. 0: Mask disabled 1: Mask enabled to prevent fault signaling						
MSK_SYS	If this bit is set, [FAULT_SYS] bit in FAULT_SUMMARY register is not set and NFAULT won't toggle due to SYS fault. It doesn't affect FAULT_SYS register. 0: Assert NFAULT if any bit from FAULT_SYS is set to '1' 1: No NFAULT action regardless of FAULT_SYS status						
MSK_PWR	If this bit is set, [FAULT_PWR] bit in FAULT_SUMMARY register is not set and NFAULT won't toggle due to PWR fault. It doesn't affect FAULT_PWR register. 0: Mask disabled 1: Mask enabled to prevent fault signaling						

7.5.14 Register: FAULT_RST

Address: 0x2030							
B7	B6	B5	B4	B3	B2	B1	B0
RST_COML_H	RST_UART_SPI	RST_FCOMM_DET	RST_FTONE_DET	RST_HB	RST_REG	RST_SYS	RST_PWR
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
RST_COML_H	Reset bits: COML_FRAME, COMH_FRAME, COML_PHY, COMH_PHY and registers: DEBUG_COMH_PHY, DEBUG_COMH_FRAME, DEBUG_COML_PHY, DEBUG_COML_FRAME. self-cleared bit. 0: Do not reset 1: Reset to 0x00						
RST_UART_SPI	Reset STOP_DET, COMMCLR_DET, UART_FRAME, SPI_FRAME, SPI_PHY bits and DEBUG_UART_FRAME, DEBUG_SPI_PHY, DEBUG_SPI_FRAME registers. self-cleared bit. 0: Do not reset 1: Reset to 0x00						
RST_FCOMM_DET	Reset FCOMM_DET bit, self-cleared bit. 0: Do not reset 1: Reset to '0'						
RST_FTONE_DET	Reset FTONE_DET bit, self-cleared bit. 0: Do not reset 1: Reset to '0'						
RST_HB	Reset HB_FAIL and HB_FAST bit, self-cleared bit. 0: Do not reset 1: Reset to '0'						
RST_REG	Resets FAULT_SUMMARY [FAULT_REG] to '0', self-cleared bit. 0: Do not reset 1: Reset to '0'						
RST_SYS	This bit is self-clear to 0 after writing to 1. 0: Do not reset 1: Reset register FAULT_SYS and [FAULT_SYS] to 0x00 and '0'						
RST_PWR	Resets FAULT_SUMMARY [FAULT_PWR] to '0' and register FAULT_PWR, self-cleared bit. 0: Do not reset 1: Reset						

7.5.15 Register: FAULT_SUMMARY

Address: 0x2100							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD				FAULT_COMM	FAULT_REG	FAULT_SYS	FAULT_PWR
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
FAULT_COMM	Indicate communication related fault is detected (any bits in register FAULT_COMM1, FAULT_COMM2 is set) 0: No fault 1: Fault						
FAULT_REG	Indicate registers related fault is detected 0: No fault 1: Fault						
FAULT_SYS	Indicates system fault is detected (any bits in register FAULT_SYS). 0: No fault 1: Fault						
FAULT_PWR	Indicates a power supply fault is detected (any bits in register FAULT_PWR) 0: No fault 1: Fault						

7.5.16 Register: FAULT_REG

Address: 0x2101							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD					CONF_MON_ERR	FACTLDERR	FACT_CRC
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
CONF_MON_ERR	Indicates monitored 2 registers (DEV_CONF1, FAULT_MSK) have at least one bit flip. For usage, refer to paragraph: Register Bit Flip Monitor. 0: No fault 1: Fault						
FACTLDERR	Indicates the factory NVM registers could not be loaded from OTP. 0: No fault 1: Fault						
FACT_CRC	Indicates a CRC error has occurred in the factory register space. 0: No fault 1: Fault						

7.5.17 Register: FAULT_SYS

Address: 0x2102							
B7	B6	B5	B4	B3	B2	B1	B0
VALIDATE_DET	LFO	SHUTDOWN_REC	DRST	CTL	CTS	TSHUT	INH
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
VALIDATE_DET	Indicates device transitioned to VALIDATE mode 0: no transitioned to VALIDATE MODE 1: transitioned to VALIDATE MODE						
LFO	Indicated LFO frequency is off by +/-25% 0: no fault 1: fault						
SHUTDOWN_REC	Indicates the device was shut down using SHUTDOWN ping. If this bit is set, the COML and COMH RX are both disabled at wake up as a way to isolate itself from the stack. Send another wake ping or [SOFT_RESET] to re-enable COMH/L RX. 0: The previous SHUTDOWN was normal 1: The previous SHUTDOWN was caused by ping, which is not a usual SHUTDOWN method, indicating potential communication issue with the device						
DRST	Indicates a digital reset has occurred. 0: no digital reset 1: digital reset has occurred						
CTL	Indicates a long communication timeout occurred. Device action is configured by [CTL_ACT]. This bit is not observable if the action is set to device shutdown. 0: No fault 1: long communication timeout occurs. Observable if long timeout action is set to SLEEP						
CTS	Indicates a short communication timeout occurred. No action from the device. This can be served as an alert to system before reaching long communication timeout. 0: No fault 1: short communication timeout occurs						
TSHUT	Indicates the previous shutdown was due to thermal shutdown 0: die temp is < thermal shutdown threshold 1: the previous shutdown was to due thermal shutdown						
INH	Indicates INH PMOS is enabled 0: INH driver is off 1: INH driver enabled, pulls pin up to BAT						

7.5.18 Register: FAULT_PWR

Address: 0x2103							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD			CVDD_UV_DRST	CVDD_OV	DVDD_OV	AVDDREF_OV	AVAO_SW_FAIL
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
CVDD_UV_DRST	Indicates CVDDUV fault caused DRST. Shutdown/power up event shall not trigger this fault. 0: no fault 1: fault						
CVDD_OV	Indicates an over voltage fault on the CVDD pin 0: no fault 1: fault						
DVDD_OV	Indicates an over voltage fault on the DVDD pin 0: no fault 1: fault						
AVDDREF_OV	Indicates an over voltage fault on the AVDD_REF internal supply 0: no fault 1: fault						
AVAO_SW_FAIL	Indicates a fault is detected on the AVAO_REF switch. 0: no fault 1: fault						

7.5.19 Register: FAULT_COMM1

Address: 0x2104							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD	FCOMM_DET	FTONE_DET	HB_FAIL	HB_FAST	UART_FRAME	COMMCLR_DET	STOP_DET
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
FCOMM_DET	Indicates the fault status bit in comm frame is set by stack devices. At least a fault in stack devices happened. Only apply to non-bq79606 family 0: no fault 1: fault						
FTONE_DET	Indicates the fault tone is detected, COM Port in sleep mode if work with non-bq79606 (Detection is monitoring the COML side if [DIR_SEL] = 0 and vice versa.) 0: no fault 1: fault						
HB_FAIL	Indicates HB tone is NOT detected within tHB_PERIOD, not supported with BQ79606A-Q1 (Detection is monitoring the COML side if [DIR_SEL] = 0 and vice versa.) 0: no fault, HB tone detected as expected 1: fault						
HB_FAST	Indicates HB tone is detected within tHB_FAST (too frequent), not supported with BQ79606A-Q1 (Detection is monitoring the COML side if [DIR_SEL] = 0 and vice versa.) This bit could also be set when [FTONE_DET]=1 depends on how soon the Fault Tone is detected from previous Heartbeat tone. 0: no fault 1: fault						
UART_FRAME	Indicates a UART FAULT detected when receiving command or transmitting response frames Further detail of the fault information is available in the DEBUG_UART_FRAME register 0: no fault 1: fault						
COMMCLR_DET	A UART/SPI communication clear signal is detected. This bit is set when Sleep2active ping is sent in UART mode. While in SPI mode, it is not set when sleep2active ping is sent. 0: no UART/SPI Comm Clear 1: UART/SPI Comm Clear detected						
STOP_DET	Indicates and unexpected STOP condition is received. Apply to UART mode. This bit is set when Sleep2active ping is sent in ACTIVE. 0: no fault 1: fault						

7.5.20 Register: FAULT_COMM2

Address: 0x2105							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD		SPI_FRAME	SPI_PHY	COML_FRAME	COML_PHY	COMH_FRAME	COMH_PHY
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
SPI_FRAME	Indicates a SPI frame level FAULT detected when receiving command or transmitting response frames. Further detail of the fault information is available in the DEBUG_SPI_FRAME register 0: no fault 1: fault						
SPI_PHY	Indicates a SPI bit level FAULT detected when receiving command or transmitting response frames. Further detail of the fault information is available in the DEBUG_SPI_PHY register 0: no fault 1: fault						
COML_FRAME	Indicate a COML byte level fault detected when receiving response frames. Further details of the fault information is available in DEBUG_COML_FRAME 0: no fault 1: fault						
COML_PHY	Indicate a COML bit level fault detected, which would cause at least a byte level (RC_RR) fault. Further details of the fault information are available in DEBUG_COML_PHY register. 0: no fault 1: fault						
COMH_FRAME	Indicate a COMH byte level fault detected when receiving response frames. Further details of the fault information is available in DEBUG_COMH_FRAME 0: no fault 1: fault						
COMH_PHY	Indicate a COMH bit level fault detected, which would cause at least a byte level (RC_RR) fault. Further details of the fault information are available in DEBUG_COMH_PHY register. 0: no fault 1: fault						

7.5.21 Register: DEV_DIAG_STAT

Address: 0x2110							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD						PWR_DIAG_RDY	INH_STAT
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
PWR_DIAG_RDY	Indicates a power supply BIST test is done. It is cleared when [PWR_DIAG_GO] bit is set. 0: BIST is not done 1: BIST is done						
INH_STAT	Indicates INH Pin see VINH_DET, this bit reflects real time value of pin INH 0: Voltage not detected 1: Voltage detected						

7.5.22 Register: PARTID

Address: 0x2120							
B7	B6	B5	B4	B3	B2	B1	B0
REV[7:0]							
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
REV[7:0]	Device revision 0x00 = A0, 0x01 = A1, 0x02 = A2, etc.						

7.5.23 Register: DIE_ID1

Address: 0x2121							
B7	B6	B5	B4	B3	B2	B1	B0
ID[7:0]							
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
ID[7:0]	Digital reset value 0x00, after factory NVM loaded successfully, it is value in the NVM.						

7.5.24 Register: DIE_ID2

Address: 0x2122							
B7	B6	B5	B4	B3	B2	B1	B0
ID[7:0]							
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
ID[7:0]	Digital reset value 0x00, after factory NVM loaded successfully, it is value in the NVM.						

7.5.25 Register: DIE_ID3

Address: 0x2123							
B7	B6	B5	B4	B3	B2	B1	B0
ID[7:0]							
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
ID[7:0]	Digital reset value 0x00, after factory NVM loaded successfully, it is value in the NVM.						

7.5.26 Register: DIE_ID4

Address: 0x2124							
B7	B6	B5	B4	B3	B2	B1	B0
ID[7:0]							
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
ID[7:0]	Digital reset value 0x00, after factory NVM loaded successfully, it is value in the NVM.						

7.5.27 Register: DIE_ID5

Address: 0x2125							
B7	B6	B5	B4	B3	B2	B1	B0
ID[7:0]							
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
ID[7:0]	Digital reset value 0x00, after factory NVM loaded successfully, it is value in the NVM.						

7.5.28 Register: DIE_ID6

Address: 0x2126							
B7	B6	B5	B4	B3	B2	B1	B0
ID[7:0]							
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
ID[7:0]	Digital reset value 0x00, after factory NVM loaded successfully, it is value in the NVM.						

7.5.29 Register: DIE_ID7

Address: 0x2127							
B7	B6	B5	B4	B3	B2	B1	B0
ID[7:0]							
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
ID[7:0]	Digital reset value 0x00, after factory NVM loaded successfully, it is value in the NVM.						

7.5.30 Register: DIE_ID8

Address: 0x2128							
B7	B6	B5	B4	B3	B2	B1	B0
ID[7:0]							
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
ID[7:0]	Digital reset value 0x00, after factory NVM loaded successfully, it is value in the NVM.						

7.5.31 Register: DIE_ID9

Address: 0x2129							
B7	B6	B5	B4	B3	B2	B1	B0
ID[7:0]							
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
ID[7:0]	Digital reset value 0x00, after factory NVM loaded successfully, it is value in the NVM.						

7.5.32 Register: DEBUG_CTRL_UNLOCK

Address: 0x2200							
B7	B6	B5	B4	B3	B2	B1	B0
CODE[7:0]							
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
CODE[7:0]	Write the unlock code (0xA5) to this register to activate the setting in the DEBUG_COMM_CTRL register. Any other value than the unlock code will deactivate any effect in the DEBUG_COMM_CTRL setting and return to the device's normal setting. This register doesn't affect DEBUG_COMM_STAT register.						

7.5.33 Register: DEBUG_COMM_CTRL

Address: 0x2201							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD	COML_TX_EN	COML_RX_EN	COMH_TX_EN	COMH_RX_EN	UART_VIF_BAUD	USER_UART_EN	USER_DAISSY_EN
0	1	1	1	1	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
COML_TX_EN	Enable COML transmitter 0: disable 1: enable						
COML_RX_EN	Enable COML receiver 0: disable 1: enable						
COMH_TX_EN	Enable COMH transmitter 0: disable 1: enable						
COMH_RX_EN	Enable COMH receiver 0: disable 1: enable						
UART_VIF_BAUD	This bit changes the baud rate of UART and Daisy Chain (Also called VIF) to 250kb/s. Useful on Daisy Chain debug. When system set all the daisy chain devices to the 250kb/s baud rate, it slows down the response byte through the Daisy Chain to increase the robustness of the Daisy Chain for debug purpose. 0: 1Mbps 1: 250kbps						
USER_UART_EN	This bit enables [UART_VIF_BAUD] setting 0: disable 1: enable						
USER_DAISSY_EN	This bit enables the setting of bits [6:3] in this register 0: the setting of bits [6:3] of current register has no effect 1: the device will configure the COML and COMH per current register setting						

7.5.34 Register: DEBUG_COMM_STAT

Address: 0x2300							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD			HW_DAISSY_DRV	COML_TX_ON	COML_RX_ON	COMH_TX_ON	COMH_RX_ON
0	0	0	1	0	0	1	1
R	R	R	R	R	R	R	R
HW_DAISSY_DRV	Indicates the COML and COMH are controlled by the device itself or by MCU control 0: the DEBUG_COMM_CTRL [USER_DAISSY_EN] = 1. COML and COMH are under manual control through DEBUG_COMM_CTRL register 1: COML and COMH are controlled by the device						
COML_TX_ON	Show the current COML transmitter status 0: Disabled 1: Enabled						
COML_RX_ON	Show the current COML receiver status 0: Disables 1: Enables						
COMH_TX_ON	Show the current COMH transmitter status 0: Disabled 1: Enabled						
COMH_RX_ON	Show the current COMH receiver status 0: Disabled 1: Enabled						

7.5.35 Register: DEBUG_SPI_PHY

Address: 0x2301							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD	FMT_ERR	COMCLR_ERR	TXDATA_UNEXP	RXDATA_UNEXP	TXFIFO_OF	TXFIFO_UF	RXFIFO_OF
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
FMT_ERR	When not in Host Read mode (after sending a write cmd frame), indicates malformed cmd is received: 1. first byte of data after nCS falling edge is 0xFF 2. devices receives not multiple of 8bit . 3. In ACTIVE, nCS pulses low without SCLK toggling (send SLP2ACT ping) Comm clear doesn't trigger this fault. This fault doesn't lock down SPI interface. 0: No fault 1: Fault						
COMCLR_ERR	More than 8 SCLK pulses are received during a comm clear. Once this fault is detected, device ignores the malformed comm clear. This fault doesn't affect SPI_RDY nor data storing into RX/TX FIFO. 0: No fault 1: Fault						
TXDATA_UNEXP	This fault occurs when: 1. device receives unexpected data from itself or daisy chain after a Comm clr OR 2. device receives unexpected data from daisy chain(or from bq79600) after a TX FIFO timeout. Once this fault is detected, device doesn't store any new data into RX/TX FIFO. Device sends out 0xFF on MISO after current byte finished transmission(FIFO data mask happens at byte boundary). SPI_RDY held low till Comm Clear detected. Comm Clear needed for recovery. 0: No fault 1: Fault						
RXDATA_UNEXP	Host sends data other than 0xFF during Device Read mode OR initiates SPI communication when SPI_RDY = 0 (e.g. While FIFO2 is being filled up, host continues reading FIFO2 right after FIFO1 is read out. SPI_RDY is low at this point) Once this fault is detected, device doesn't store any new data into RX/TX FIFO. Device would send out 0xFF on MISO after current byte finished transmission(FIFO data mask happens at byte boundary). SPI_RDY held low till Comm Clear detected. Comm Clear needed for recovery. 0: No fault 1: Fault						
TXFIFO_OF	If SPI module receives data (from itself or daisy chain) when current FIFO is full and 2nd FIFO is not empty (host hasn't complete reading the 2nd FIFO). Once this fault is detected, device doesn't store any new data into RX/TX FIFO. Device sends out 0xFF on MISO after current byte finished transmission(FIFO data mask happens at byte boundary). SPI_RDY held low till Comm Clear detected. Comm Clear needed for recovery. 0: No fault 1: Fault						
TXFIFO_UF	In Host Read mode, when both FIFOs are empty and the host continues to send clocks for more read data (send 0xFF through MOSI). SPI Slave sets this fault flag and continues to send 0xFF on MISO when host requests data. Once this fault is detected, device doesn't store any new data into RX/TX FIFO. Device sends out 0xFF on MISO after current byte finished transmission(FIFO data mask happens at byte boundary). SPI_RDY held low till Comm Clear detected. Comm Clear needed for recovery. (FIFO 1 is read out, FIFO2 is still buffering, if reading happens, it is RXDATA_UNEXP fault) Once FIFO is empty, device rejects any 0xFF until nCS going high. 0: No fault 1: Fault						
RXFIFO_OF	While device is receiving command frame, more data is sent than device can accommodate in RX FIFO. Once this fault is detected, device doesn't store any new data into RX/TX FIFO. Device sends out 0xFF on MISO after current byte finished transmission(FIFO data mask happens at byte boundary). SPI_RDY held low till Comm Clear detected. Comm Clear needed for recovery. Follow "Flow to send write cmd frame into bq79600" to avoid this fault. 0: No fault 1: Fault						

7.5.36 Register: DEBUG_SPI_FRAME

Address: 0x2302							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD		TR_SOF	TR_WAIT	RC_IERR	RC_SOF	RC_BYTE_ERR	RC_CRC
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
TR_SOF	Indicates that a SPI COMM CLEAR is received while there is data in output FIFO (SPI_RDY is 0 or SPI_RDY = 1 and receiving new data from VIF.) If SPI_RDY = 1 and not receiving new data from VIF while receiving COMM CLEAR, it doesn't assert this bit. 0: No fault 1: Fault						
TR_WAIT	Indicates that a SPI COMM CLEAR is received while there is no data in output FIFO regardless of SPI_RDY. 0: No fault 1: Fault						
RC_IERR	Detects initialization byte error in the received command frame. Example: • reserved command bits in the "INIT byte" is set, • expected to a cmd "INIT byte", but it is configured as response frame instead • extra data is sent other than one cmd frame during one nCS low All bytes that follow are discarded until a Comm Clear is received or a SOF due to a properly formed new command is received (nCS falling, etc). 0: No fault 1: Fault						
RC_SOF	Detects a start-of-frame (SOF) error. Mostly, this is SPI COMM CLEAR is received on the SPI before the current frame is finished, e.g. partial cmd frame is sent 0: No fault 1: Fault						
RC_BYTE_ERR	Device receives data from the VIF that needs to go out on SPI, while host send data into device through SPI. (TXDATA_UNEXP should also be set) CRC is not calculated and doesn't forward communication in daisy chain. All bytes that follow are ignored until a communication CLEAR is received. 0: No fault 1: Fault						
RC_CRC	Detects CRC error in the received command frame from SPI. The frame will be considered as discarded frame. Frame CRC is calculated only if no physical level error, IERR, BERR, SOF error detected. 0: No fault 1: Fault						

7.5.37 Register: DEBUG_UART_FRAME

Address: 0x2303							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD		TR_SOF	TR_WAIT	RC_IERR	RC_SOF	RC_BYTE_ERR	RC_CRC
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
TR_SOF	Indicates that a UART COMM CLEAR is received while the device is still transmitting data to host. 0: No fault 1: Fault						
TR_WAIT	The device is waiting for its turn to transfer a response out but the action is terminated because: 1. the device receives a UART COMM CLEAR signal, or 2. the device receives a new command This bit is valid when broadcast or stack read command has been issued. 0: No fault 1: Fault						
RC_IERR	Detects initialization byte error in the received command frame. Example: • a STOP error is detected, • reserved command bits in the "INIT byte" is set, • expected a "INIT byte", but it is configured as response frame instead (e.g. 1st byte received after COMM CLEAR signal should but a "INIT byte") All bytes that follow are ignored until a communication CLEAR is received. 0: No fault 1: Fault						
RC_SOF	Detects a start-of-frame (SOF) error. Mostly, this is UART COMM CLEAR is received on the UART before the current frame is finished. 0: No fault 1: Fault						
RC_BYTE_ERR	Other than INIT byte, if STOP error is detected in rest of byte in the frame or TX receives data while RX is using. CRC is not calculated and doesn't forward communication in daisy chain. All bytes that follow are ignored until a communication CLEAR is received. 0: No fault 1: Fault						
RC_CRC	Detects CRC error in the received command frame from UART. The frame will be considered as discarded frame. Frame CRC is calculated only if no physical level error, IERR, BERR, SOF error detected 0: No fault 1: Fault						

7.5.38 Register: DEBUG_COMH_PHY

Address: 0x2304							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD			PERR	BERR_TAG	SYNC2	SYNC1	BIT
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
PERR	Pertain to received response frame from stack device only. PERR is a lump bit of all physical errors. Any error bit that is set in this register will also set the [PERR] bit. However, abnormality that isn't classified in the register can also trigger the [PERR] bit (e.g. detecting missing of data or wrong data order.) 0: no comm error detected 1: detected abnormality of the received comm frame. BERR is asserted to the forwarded communication.						
BERR_TAG	Indicates BERR bit is set in at least one byte in received response frame from stack device 0: received response frame doesn't have BERR 1: received response frame has BERR						
SYNC2	Timing information extracted from the demodulation of the preamble half-bit and the two full bits of synchronization is outside of the expected window, this bit is set and the byte is not processed. 0: No fault 1: Fault						
SYNC1	If the demodulation of the preamble half-bit and the two full bits of synchronization data have errors and the timing is likely not correct, this bit is set and the byte is not processed. 0: No fault 1: Fault						
BIT	The device has detected a data bit, however, the detection samples is not enough to guarantee a strong '1' or '0'. 0: No fault 1: Fault						

7.5.39 Register: DEBUG_COMH_FRAME

Address: 0x2305							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD			RR_IERR	RR_SOF	RR_BYTE_ERR	RR_UNEXP	RR_CRC
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
RR_IERR	Detects initialization byte error in the received response frame. This may due to improper formatting of a byte such as a frame initialization byte is expected, but start-of-frame (SOF) bit is not set, or an invalid frame type is selected. All bytes from VIF that follows are ignored until a SOF bit is received (BQ79600 still forwards data from VIF to UART/SPI). 0: No fault 1: Fault						
RR_SOF	Valid when [DIR_SEL] = 0. Detects a start-of-frame (SOF) error on COMH. The SOF bit shall only be set in the initialization frame but the SOF bit is set in the current frame that is not expected. 0: No fault 1: Fault						
RR_BYTE_ERR	Valid when [DIR_SEL] = 0. Detected any byte errors, other than the error in the initialization byte, in the received response frame. If the byte err is on the last byte of the CRC, it would check CRC in addition to discarding the frame. This error can be triggered by one or more error bit set in the DEBUG_COMH_PHY register. All bytes from VIF that follows are ignored until a SOF bit is received (BQ79600 still forwards data from VIF to UART/SPI). 0: No fault 1: Fault						
RR_UNEXP	If [DIR_SEL] = 1, but device received response frame from COMH which is an invalid condition and device will set this error bit. 0: No fault 1: Fault						
RR_CRC	Indicates one or more COMH response frames being discarded due to CRC error. 0: No fault 1: Fault						

7.5.40 Register: DEBUG_COML_PHY

Address: 0x2306							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD			PERR	BERR_TAG	SYNC2	SYNC1	BIT
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
PERR	Pertain to received response frame from stack device only. PERR is a lump bit of all physical errors. Any error bit that is set in this register will also set the [PERR] bit. However, abnormality that isn't classified in the register can also trigger the [PERR] bit (e.g. detecting missing of data or wrong data order.) 0: no comm error detected 1: detected abnormality of the received comm frame. BERR is asserted to the forwarded communication.						
BERR_TAG	Indicates BERR bit is set in at least one byte in received response frame from stack device 0: received response frame doesn't have BERR 1: received response frame has BERR						
SYNC2	Timing information extracted from the demodulation of the preamble half-bit and the two full bits of synchronization is outside of the expected window, this bit is set and the byte is not processed. 0: No fault 1: Fault						
SYNC1	If the demodulation of the preamble half-bit and the two full bits of synchronization data have errors and the timing is likely not correct, this bit is set and the byte is not processed. 0: No fault 1: Fault						
BIT	The device has detected a data bit, however, the detection samples is not enough to guarantee a strong '1' or '0'. 0: No fault 1: Fault						

7.5.41 Register: DEBUG_COML_FRAME

Address: 0x2307							
B7	B6	B5	B4	B3	B2	B1	B0
RSVD			RR_IERR	RR_SOF	RR_BYTE_ERR	RR_UNEXP	RR_CRC
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R
RR_IERR	Detects initialization byte error in the received response frame. This may due to improper formatting of a byte such as a frame initialization byte is expected, but start-of-frame (SOF) bit is not set, or an invalid frame type is selected. All bytes from VIF that follows are ignored until a SOF bit is received (BQ79600 still forwards data from VIF to UART/SPI). 0: No fault 1: Fault						
RR_SOF	Valid when [DIR_SEL] = 1. Detects a start-of-frame (SOF) error on COML. The SOF bit shall only be set in the initialization frame but the SOF bit is set in the current frame that is not expected. 0: No fault 1: Fault						
RR_BYTE_ERR	Valid when [DIR_SEL] = 1. Detected any byte errors, other than the error in the initialization byte, in the received response frame. If the byte err is on the last byte of the CRC, it would check CRC in addition to discarding the frame. This error can be triggered by one or more error bit set in the DEBUG_COML_PHY register. All bytes from VIF that follows are ignored until a SOF bit is received (BQ79600 still forwards data from VIF to UART/SPI). 0: No fault 1: Fault						
RR_UNEXP	If [DIR_SEL] = 0, but device received response frame from COML which is an invalid condition and device will set this error bit. 0: No fault 1: Fault						
RR_CRC	Indicates one or more COML response frames being discarded due to CRC error. 0: No fault 1: Fault						

8 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

8.1 Application Information

This section is concerned with the external operation, what external components are required to add to the device to make it usable in a particular design, and how to calculate the values for those external components.

8.2 Typical Applications

8.2.1 Bridge With Reverse Wakeup in UART

The following application circuit is used when user chooses to use UART interface and any of those features: reverse wakeup, ring architecture, fault/heartbeat tone.

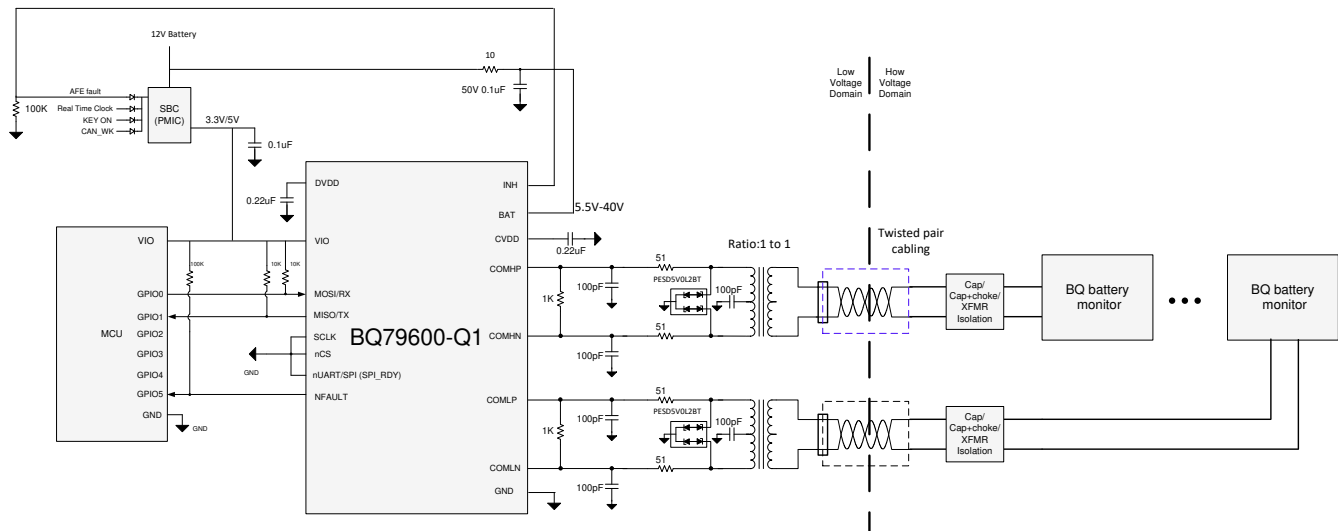


Figure 8-1. Typical Bridge with Reverse Wakeup in UART Applications Circuit

8.2.1.1 Design Requirements

Table 8-1 describes the design parameters.

Table 8-1. Recommended Design Requirement

PARAMETER	VALUE
UART speed	1Mbps

8.2.1.2 Detailed Design Procedure

8.2.1.2.1 MCU Interface (UART, NFAULT)

To select UART interface, connect nCS, nUART/SPI and SCLK pins to GND. UART interface includes RX/TX pins. They are pulled up through a 10-100kΩ resistor to VIO like figure above.

NFAULT pin, if not used, connect to GND. Otherwise, pull it up with 100kΩ to VIO.

8.2.1.2.2 Daisy Chain Interface

Given that galvanic isolation is expected between BQ79600-Q1 and stack BQ devices, transformer isolation is recommended. Section 8.2.1 shows the interface components values.

Contact TI for transformer recommendations.

8.2.1.2.3 INH Connection

INH pin is connected to the power management IC (PMIC) enable pin such that when reverse wakeup is triggered, INH would be pulled towards BAT and enable PMIC. INH pin should always be lower than BAT pin. The 100kΩ connected to INH in [Section 8.2.1](#) is to make sure INH pin potential is defined when INH driver is off.

8.2.1.3 Application Performance Plot



Figure 8-2. Command and Response Frame for Read from 2 Stack Devices

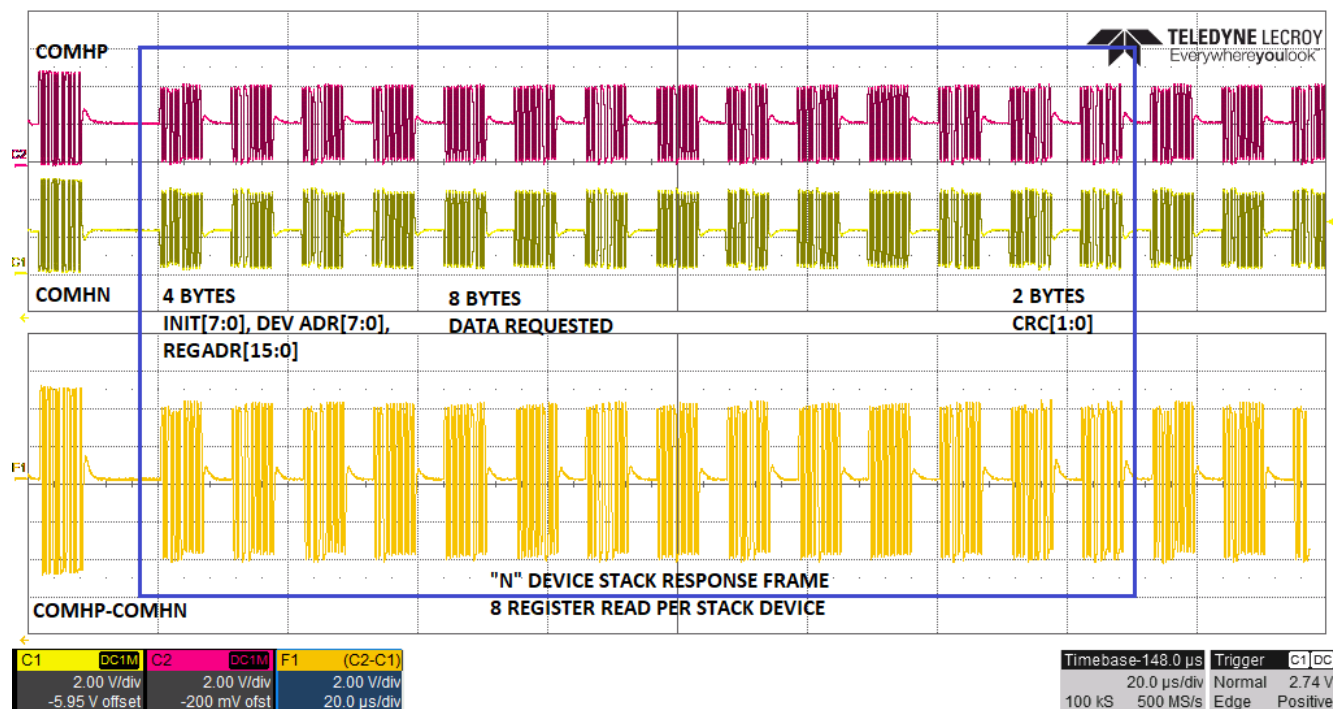


Figure 8-3. Zoomed View of Response Frame for 8 Registers Read from Stack Devices

8.2.2 Bridge Without Reverse Wakeup in SPI

The following application circuit is used when user chooses to use SPI interface and none of those features: reverse wakeup, ring architecture, fault/heartbeat tone.

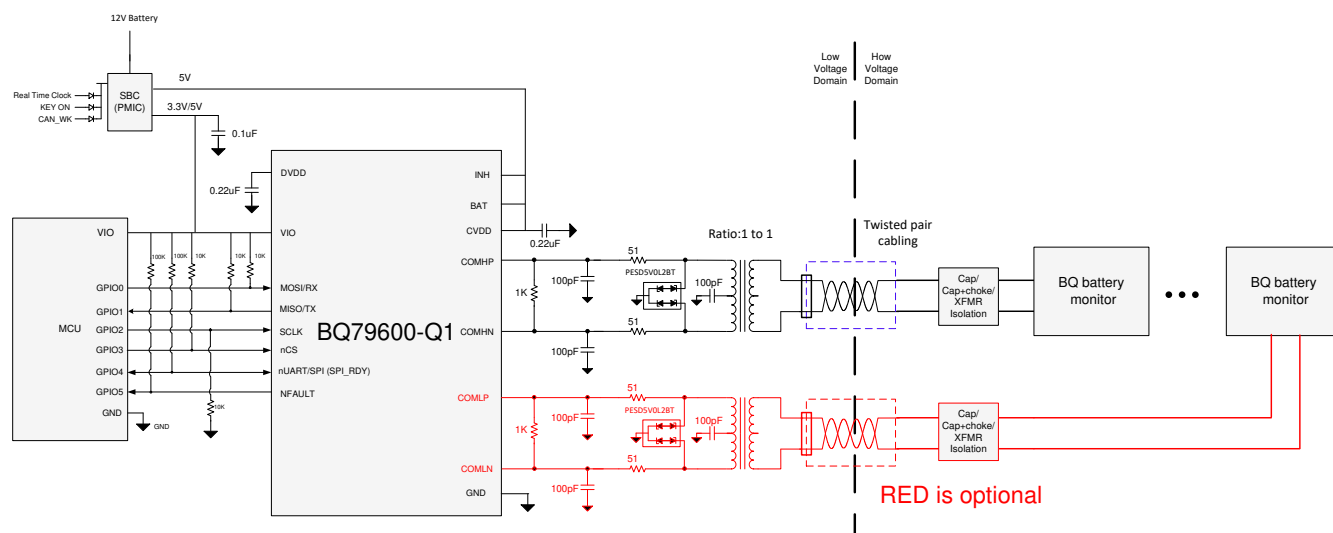


Figure 8-4. Typical Bridge without Reverse Wakeup in SPI Applications Circuit

8.2.2.1 Design Requirements

Table 8-2 describes the design parameters.

Table 8-2. Key Requirements

PARAMETER	VALUE
SPI speed	2 - 6Mbps

8.2.2.2 Detailed Design Procedure

8.2.2.2.1 MCU Interface (SPI, SPI_RDY, NFAULT)

To select SPI interface, connect MOSI, MISO, SCLK, nCS, nUART/SPI as figure above shows. Connect SPI_RDY to MCU GPIO port for SPI flow control use, [Section 7.3.2.1.2.2.1](#).

NFAULT pin, if not used, connect to GND. Otherwise, pull it up with 100K Ω to VIO.

8.2.2.2.2 Daisy Chain Interface

Refer to [Section 8.2.1.2.2](#).

8.2.2.3 Application Performance Plot

See [Application Performance Plot](#) for application performance curve.

9 Power Supply Recommendations

The BQ79600-Q1 can be powered by either directly from 12-V battery (nominal 9 - 16V) or regulated 5-V supply. The design consideration for both options are described in the table below.

Table 9-1. Supply Design Considerations

	12-V BATTERY (nominal 9 - 16 V)	REGULATED 5-V SUPPLY
VIO	3.125 - 5.25V, decouple 0.1-μF capacitor to gnd	3.125 - 5.25V, decouple 0.1-μF capacitor to gnd
BAT	Put RC between supply battery and BAT pin, R = 10Ω, C = 0.1-μF capacitor. Make sure BAT pin sees no less than 5.5V and no larger than 40V.	4.75 - 5.25V, decouple 0.22-μF capacitor to gnd
CVDD	Decouple 0.22-μF capacitor to gnd	shorted to BAT
DVDD	Decouple 0.22-μF capacitor to gnd	Decouple 0.22-μF capacitor to gnd

10 Layout

The layout for this device must be designed carefully. Any design outside these guidelines can affect the communication robustness and EMI performance. Care must be taken in the layout of signals to and from the device to avoid coupling noise onto sensitive inputs. The layout of ground and power connections, as well as communication signals, should also be made carefully.

10.1 Layout Guidelines

10.1.1 Ground Planes

It is very important to establish a clean grounding scheme to ensure best performance of the device. There is one ground pin (GND) on the device. It is a good practice to use top and bottom PCB layers for signal routing, and use middle layers as ground planes. Even on a PCB layer that is mainly for signal routing, it is good practice to have a small island of ground pour if possible to provide a low-impedance ground, rather than simply a via through the ground trace to an lower ground plane. Create a keep-out area (no other traces and no ground plane) around the daisy chain components in all PCB layers.

There is a strong recommendation to have a minimum of 4 layers in the PCB, with one fully dedicated layer as an unbroken VSS plane (except thermal reliefs). Avoid placing tracks on this layer to maintain the unbroken integrity of the plane structure.

10.1.2 Bypass Capacitors for Power Supplies

The bypass capacitors of the following pins must be placed as close to the device pins as possible to ensure proper performance.

- BAT, VIO, CVDD, DVDD

10.1.3 UART/SPI communication

The UART/SPI interface (MISO/TX, MOSI/RX, SCLK, nCS, nUART/SPI_RDY) between MCU and BQ79600-Q1 shall be kept as short and straight as possible for optimized EMC performance.

10.1.4 Daisy Chain Communication

It is important to have proper layout on the COMHP/N and COMLP/N circuits in order to have the best robust daisy chain communication.

- Keep differential traces as short as possible and as straight as possible. Minimize turns and avoid any looping on the traces.
- Keep the differential traces on the same layers. Run the trace in parallel with shielding and matching trace impedance.
- Place the isolation components close to the connectors.
- Create a keep-out area (no other traces and no ground plane) around the daisy chain components in all PCB layers.

10.2 Layout Example

This section presents the BQ79600-Q1 Evaluation Module (EVM) design as a layout example. Given the EVM doesn't have an MCU, the example of UART/SPI connection layout is not optimized.

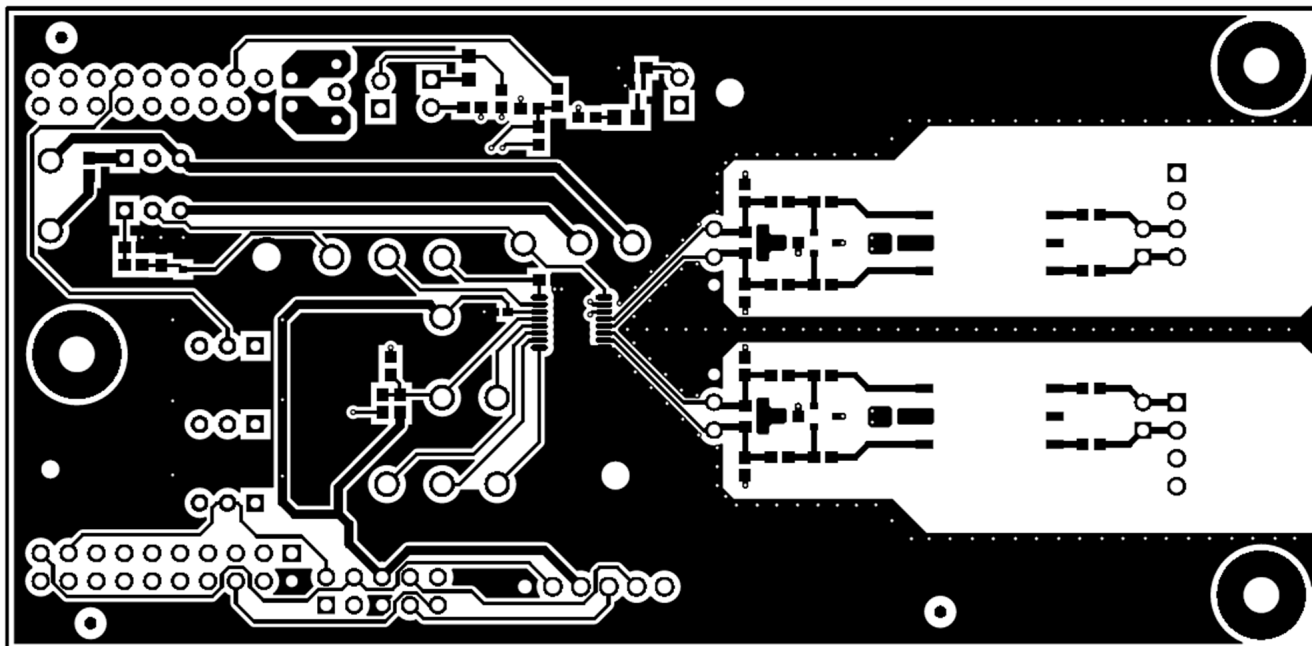


Figure 10-1. Top Layer Layout

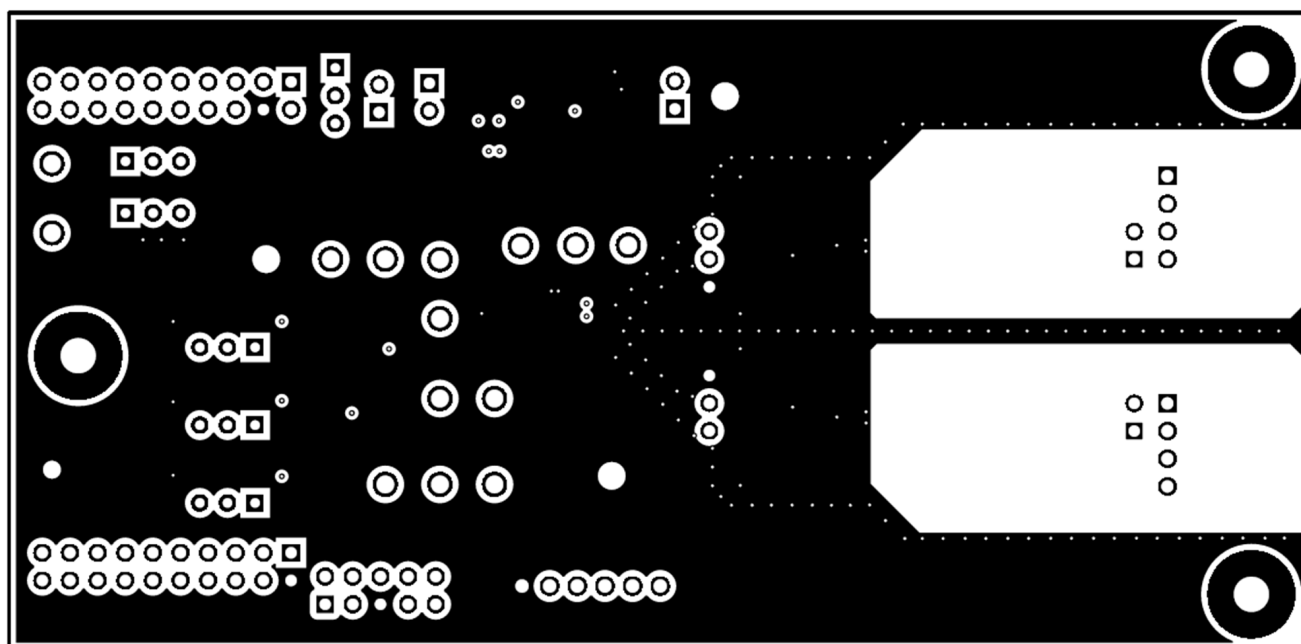


Figure 10-2. Signal 1 Layer Layout

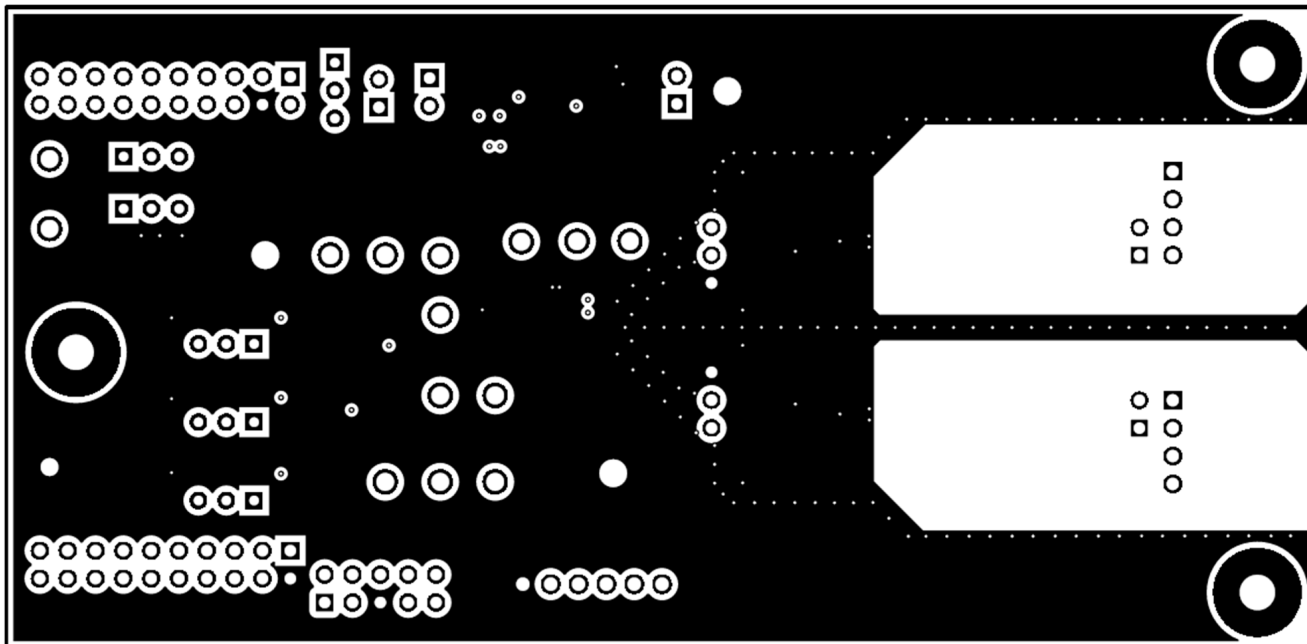


Figure 10-3. Signal 2 Layer Layout

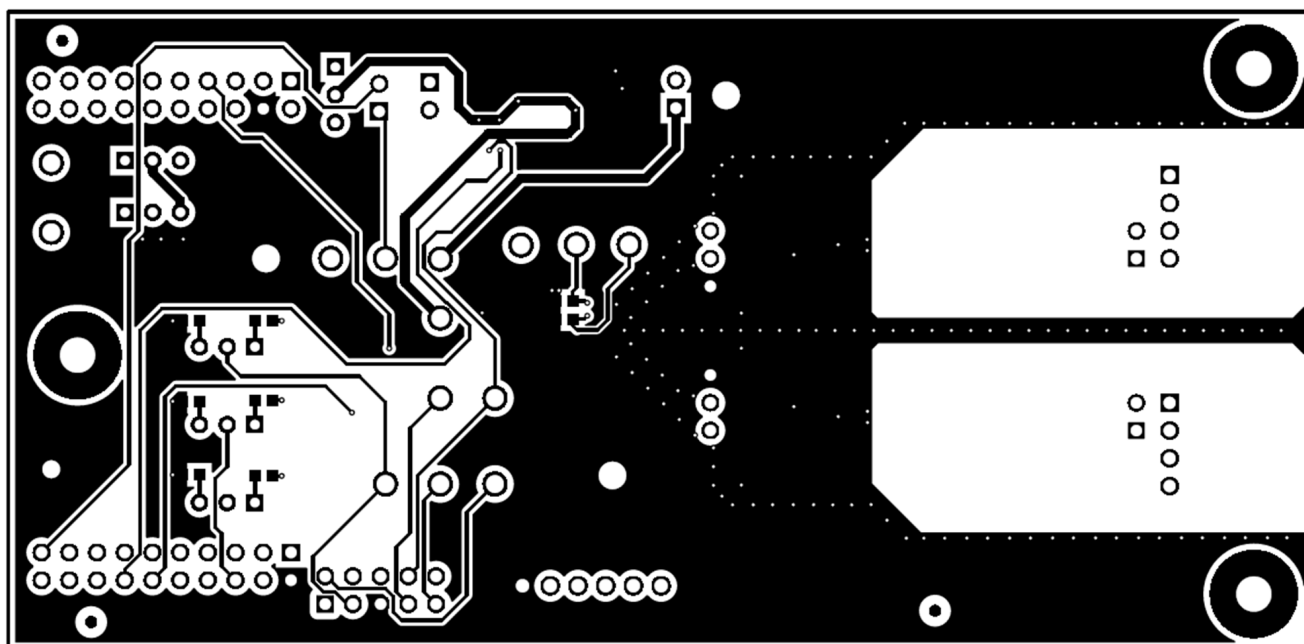


Figure 10-4. Bottom Layer Layout

11 Device and Documentation Support

11.1 Device Support

11.2 Third-Party Products Disclaimer

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11.3 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Subscribe to updates* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

11.4 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

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11.6 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

11.7 Glossary

[TI Glossary](#) This glossary lists and explains terms, acronyms, and definitions.

12 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
BQ79600PWRQ1	Active	Production	TSSOP (PW) 16	2000 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	BQ79600
BQ79600PWRQ1.A	Active	Production	TSSOP (PW) 16	2000 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	BQ79600

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

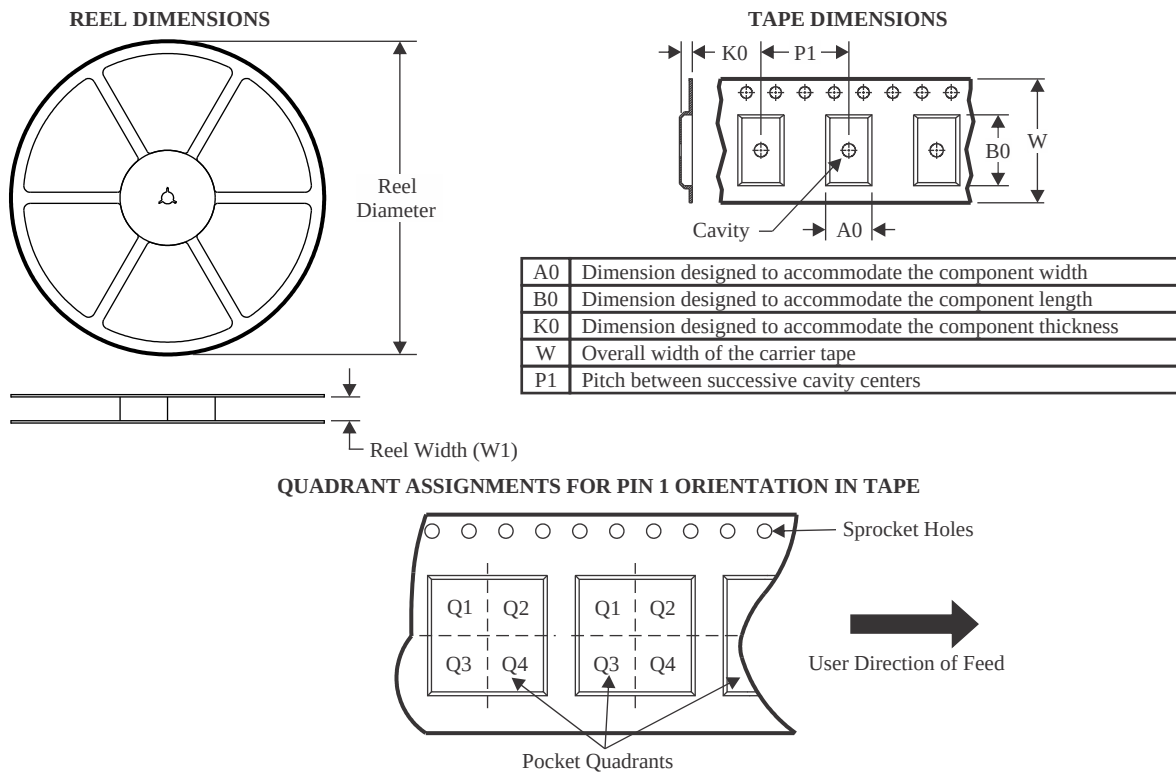
⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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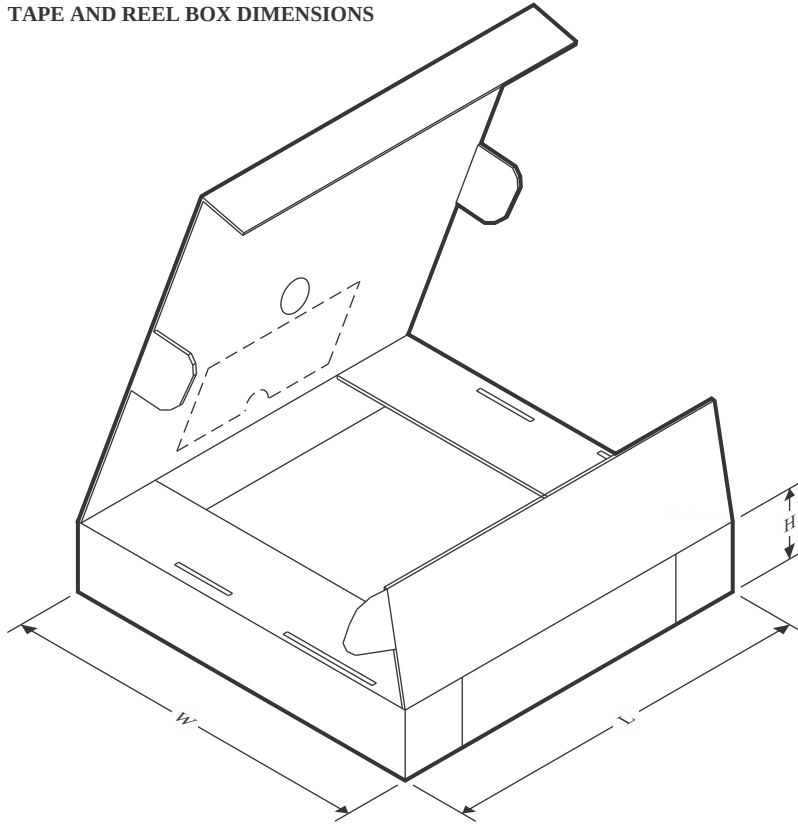
TAPE AND REEL INFORMATION



*All dimensions are nominal

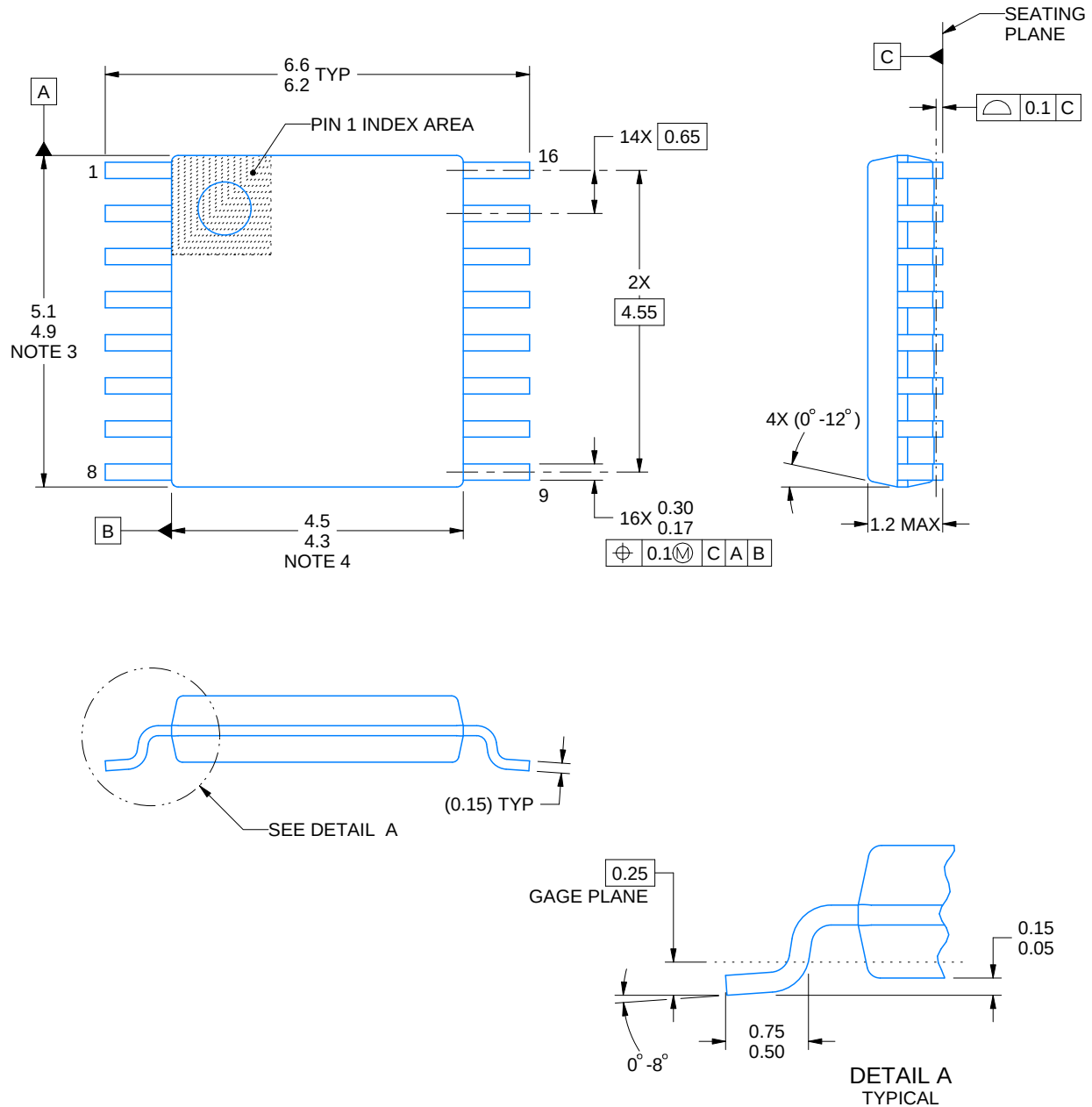
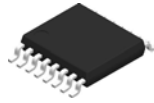
Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
BQ79600PWRQ1	TSSOP	PW	16	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
BQ79600PWRQ1	TSSOP	PW	16	2000	353.0	353.0	32.0



4220204/B 12/2023

NOTES:

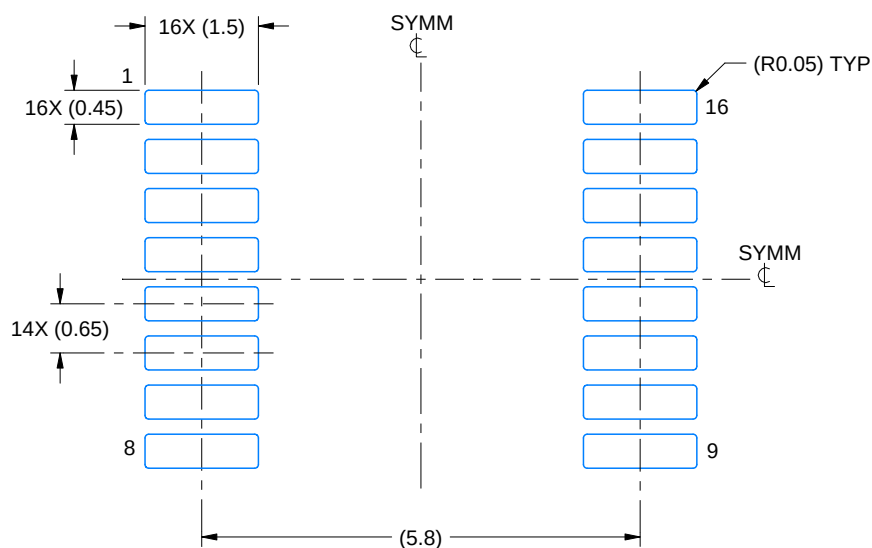
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.

EXAMPLE BOARD LAYOUT

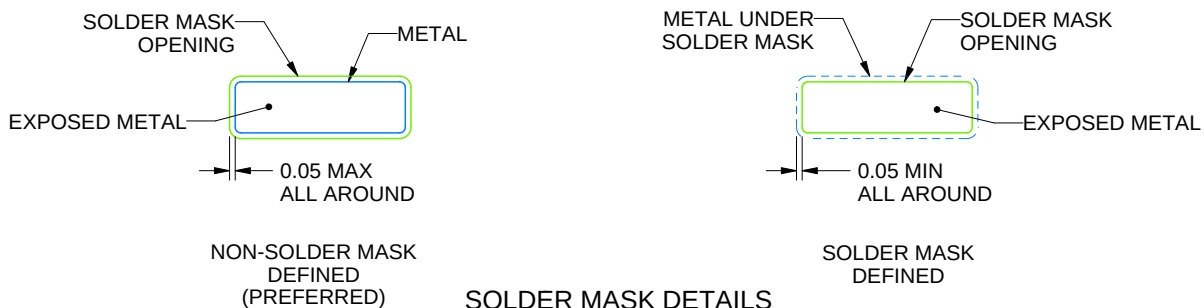
PW0016A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 10X



4220204/B 12/2023

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

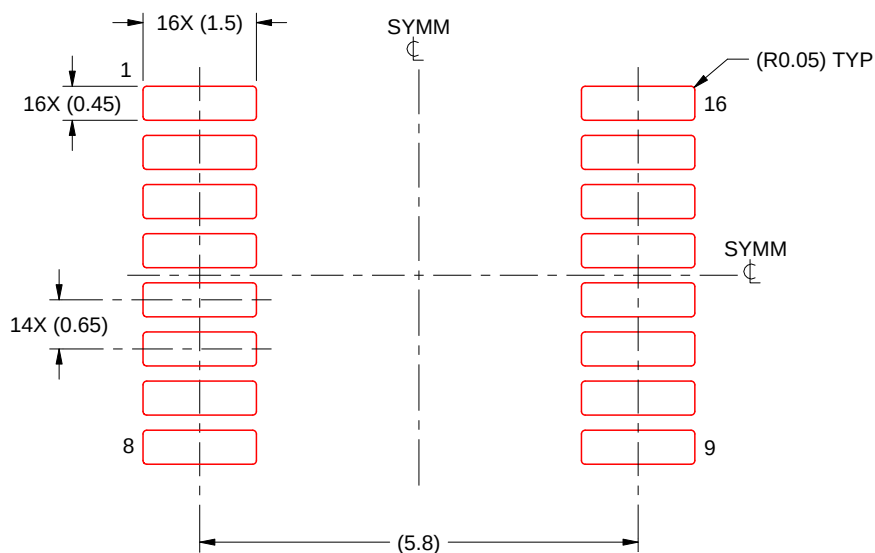
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

PW0016A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

4220204/B 12/2023

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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